Caste Ties and Firm Growth

India's Mining Industry

Public-data evidence, source inventory,
fieldwork design and causal identification

Detailed research report

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12 October 2026

Evidence boundary. I analyzed historical aggregate data from the Sixth Economic Census and Census 2011, fitted nine regional establishment specifications and four controlled worker-duration models, and prepared nineteen figures and a 65-entry source inventory. I did not observe caste ties or longitudinal firm growth and have not estimated a causal caste-network effect. Prospective designs and trial-power calculations are explicitly identified as proposals.

Abstract

I extracted mining and quarrying tables containing 85,176 establishments and 557,474 workers. Within the proprietary subset, workers per establishment are 2.63 for SC owners, 3.68 for ST owners, 3.77 for OBC owners and 5.62 for the residual Others category. These aggregate ratios establish a historical difference in establishment scale. They do not identify a network mechanism, an individual-firm size distribution or growth. A demographic merge supports regressions for 35 historical state/UT regions and 35 Maharashtra districts. Adjusted log-size associations with SC and ST population shares have wide HC3 intervals. One unadjusted state-density coefficient has $p = 0.036$, but its Benjamini–Hochberg value is 0.651 across the specified 18-coefficient family. A detailed prospective design combines consent-based relationship measurement, repeated firm outcomes, administrative linkage and a randomized business-introduction intervention. The added Census worker baseline contains 2,105,736 workers, 20.87 percent SC and 17.32 percent ST. Within state-division-sex-residence strata, employment-weighted SC and ST marginal-duration contrasts versus the residual group are +1.49 and +5.08 percentage points; these are descriptive adjusted associations. Local Rajasthan studies report job-entry routes, caste composition and working conditions; Coal India administrative snapshots add employee and vacancy information. I retained all observed-data tables, all nineteen figures, source access classifications and fieldwork modules in this reproducible package.

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Chapter 1

Research question and what this report establishes

I developed a dissertation research programme on caste relationships and enterprise growth in India's mining economy, and completed the empirical analyses supported by the public data I acquired. The central finding is a substantial descriptive difference in the employment scale of proprietary mining establishments across owners' broad social groups. The central identification finding is equally important: the available tabulations do not observe business relationships or repeated outcomes for the same firm. They therefore cannot identify the causal effect of caste ties on growth. The completed regressions describe geographic associations; the fieldwork and causal designs specify how to answer the original question credibly.

I used three source documents for my original establishment analysis: the government-authored All India Report of the Sixth Economic Census, Maharashtra's Sixth Economic Census report, and the Census of India 2011 Primary Census Abstract spreadsheet. I obtained the national report from an Ecensus mirror after official download attempts failed, and the Maharashtra report from NITI for States. I retained source files, acquisition routes, hashes, table locations, extracted data, transformations, statistical outputs and figures in my research package. These are historical data, principally 2011 and 2013-14, not a description of the mining economy in 2026. [S02] [S07] [S09]

There are 85,176 mining and quarrying establishments and 557,474 workers in the extracted national tables. The proprietary subset contains 73,170 establishments and 319,425 workers. Among that subset, workers per establishment are 2.63 for SC owners, 3.68 for ST owners, 3.77 for OBC owners and 5.62 for the residual Others category. These are ratios of employment totals to establishment totals, not medians of individual firms. The SC ratio is about 46.9 percent of the Others ratio. That difference motivates research on resources, market access and exclusion, but does not by itself establish any one mechanism.

I fitted nine regional OLS specifications. The national demographic merge contains 35 historical state/UT regions after pooling residual Andhra Pradesh and Telangana to match the 2011 undivided Andhra Pradesh geography. Log-size models have 33 observations because Delhi and Lakshadweep have no mining establishments in the extracted table. The Maharashtra replication contains 35 historical districts. Adjusted log-size associations with regional SC and ST population shares are imprecise. One SC coefficient in the state density model has an unadjusted p-value of 0.036, but the Benjamini-Hochberg adjusted value is 0.651 for the specified family of 18 social-composition coefficients. No causal conclusion follows from it.

I propose building a consent-based owner and business-relationship panel, linked where authorized to mine, lease, company and procurement identifiers. Fieldwork should cover operators and contractors separately, include both registered and less visible businesses, and follow surviving firms, exits and new entrants. A randomized introduction or procurement-brokerage intervention is the most defensible prospective route to a causal estimate of access to business relationships. Observational policy designs

remain useful, but must establish a credible comparison group and cannot simply label a policy coefficient a caste-network effect.

I extended the analysis with official Census B18, B18SC and B18ST counts, two Coal India employee snapshots, selected recruitment disclosures and published Rajasthan worker evidence. The worker baseline contains 2,105,736 people in NIC divisions 05-09. Four fitted aggregate duration models show how SC/ST differences change with sex, rural/urban residence, state and mining division controls. The expanded inventory has 65 entries, the report includes nineteen graphs, and the questionnaire now has 86 draft measures including 30 worker/job-entry items. The worker chapter specifies source-specific denominators and limitations. [S56] [S57] [S58] [S59] [S60] [S61] [S62]

1.1 Evidence status

I completed source discovery and classification; downloading the three actual inputs; extracting national state tables and 23 industry by four social-group cells; constructing state and district demographic joins; checking arithmetic; producing descriptive statistics, correlations, nine OLS models, HC3 intervals, influence diagnostics, leave-one-out ranges, multiple-testing corrections, aggregate industry fixed-effect contrasts, twelve establishment graphs and illustrative trial-power calculations; the worker extension adds seven graphs and four employment-weighted controlled comparisons.

I did not conduct interviews, invent respondents or access a proprietary firm panel. No actual caste-tie variable, firm sales-growth series, individual-firm survival history, randomized assignment, policy treatment panel, instrumental-variable first stage or valid regression-discontinuity running variable was acquired. Consequently, I have not estimated a causal effect of caste ties or regressed observed mining firm growth on observed caste ties. I developed a detailed research specification for those analyses once the required data exist.

I assembled an expanded source inventory containing 65 entries, including documents, databases, administrative systems and source families. It is deliberately broad, but cannot certify that every possible Indian record has been located. Entries distinguish downloaded inputs, verified public documentation, conditional or commercial access, candidate source families and primary data still to be collected. A portal's existence is not treated as proof that its transaction-level data are publicly downloadable.

Chapter 2

Conceptual framework

2.1 Caste identity and caste ties are different objects

An owner's broad category is an attribute. A caste tie is a relationship with another person or organization that may transmit information, trust, credit, labour, referrals, political access or obligations. A person can share a broad category with another entrepreneur without sharing a jati or any actual relationship. Conversely, a business tie can cross jati, category, religion or locality. Administrative SC, ST, OBC and residual classifications must not be treated as four internally uniform social networks.

I propose distinguishing at least four measurements in my future fieldwork. First, record self-reported broad social category and optional self-described jati. Second, observe named or coded business contacts and the respondent's own account of whether a relationship is connected to shared community identity. Third, record what each contact actually did, when it occurred and whether money or business information changed hands. Fourth, measure the eligible opportunity set: nearby businesses, procurement buyers, lenders, labour brokers and associations with which contact could plausibly have occurred.

Scheduled Tribe identity also deserves a distinct analytical treatment. Adivasi social organization, community claims, land access, displacement and the institutional circumstances of mining are not adequately represented by assuming they are identical to caste-based entrepreneur networks. SC and ST results should be presented separately before any pooled disadvantaged-group analysis. The EC residual Others category is a coding residual; it is not a validated indicator of upper-caste identity or dominance. [S01] [S13]

I will not infer individuals' caste from surnames, photographs, geography, directors' names or association labels. Identity and relationship information should come from voluntary self-report or an explicitly authorized administrative classification, with a separate unknown or declined-response code. This preserves construct validity and avoids turning uncertain social classification into a hidden measurement error.

2.2 Mechanisms and competing hypotheses

H1 is an information mechanism: community contacts reduce the cost of learning about leases, tenders, buyers, equipment and credit. H2 is an enforcement mechanism: repeated social interaction permits trade credit and informal insurance where legal collection is costly. H3 concerns labour recruitment and monitoring: referrals can ease hiring, but may also segment jobs and exclude outsiders. H4 concerns access to gatekeepers: connected firms may navigate administrative requirements or secure introductions more quickly. These hypotheses require process measures, not merely a coefficient on owner category.

H5 is a constraint mechanism. Dense within-community networks can lock firms into a narrow buyer

pool, impose transfers or obligations, and discourage relationships outside a familiar market. H6 is an exclusion mechanism: established networks may screen out entrepreneurs from historically disadvantaged groups even when those entrepreneurs have comparable productive capabilities. The effect of ties can therefore vary by type of contact, social location, mineral, scale and institutional setting. A single positive or negative average coefficient would conceal that heterogeneity.

Resources and selection produce competing explanations. Wealthier households may both enter capital-intensive mining and sustain influential contacts. Larger firms may attract buyers and lenders, creating ties as a consequence of success. Migration may alter both network composition and business opportunities. Technology, concession size, mineral quality, inherited land, local demand and formal-company organization can generate size differences even if relationships have no causal effect. The report treats these alternatives as hypotheses to discriminate, not as explanations already demonstrated.

The literature offers motivation rather than a transferable mining estimate. Iyer, Khanna and Varshney's work establishes the usefulness of Economic Census ownership information for studying entrepreneurship and social hierarchy. Munshi and Rosenzweig examine how traditional networks can influence choices and mobility in a different setting. Neither study licenses extrapolating a causal coefficient to Indian mining firms. The research contribution here would be direct relationship measurement in a sector with strong licensing, geology and buyer concentration, followed by longitudinal or experimental identification.

2.3 Outcomes and estimands

The preferred primary outcome is employment growth over a pre-specified interval, with real sales growth and survival as separate outcome families. A secondary outcome is production growth in mineral-specific physical units. Employment should separate owner work, unpaid family work, directly hired labour and contract labour. Sales should separate production sales, trading revenue and intercompany transactions. A firm with several mines requires both establishment-level and legal-entity-level accounts.

For continuing businesses with positive employment at both dates, annualized log employment growth is defined as follows. It should be supplemented by levels and by a symmetric growth measure that handles entry and exit, since conditioning only on survivors selects on an outcome affected by relationships.

$$g_i^E = \frac{\log E_{i,t_1} - \log E_{i,t_0}}{t_1 - t_0}, \qquad g_i^{DHS} = \frac{2(E_{i,t_1} - E_{i,t_0})}{E_{i,t_1} + E_{i,t_0}}.$$

For a prospectively randomized brokerage offer, the primary estimand is the intention-to-treat effect of being offered the programme among eligible baseline firms. A treatment-on-the-treated interpretation for actual relationship formation needs additional assumptions. It is inappropriate to describe the effect of an offer containing training, information and introductions as the pure effect of caste ties. Those components must be measured and, where feasible, separated by design.

$$\text{ITT} = \mathbb{E}[Y_i(1) - Y_i(0)], \qquad Y_i(z) = \text{outcome under assignment } z.$$

For the observational panel, the target must be stated as an association unless a defensible identification argument is supplied. A one-standard-deviation increase in a pre-specified relationship index, a new buyer introduction and a change in within-jati tie share are different exposures and estimands. The dissertation should not switch among them after inspecting which specification is significant.

Chapter 3

Mining sector scope and the empirical population

3.1 Operators and the mining supply chain

The core operator universe uses the relevant mining and quarrying classification. NIC 2008 Section B covers divisions 05 through 09, including coal and lignite, petroleum and natural gas, metal ores, other mining and quarrying, and support activities. The Sixth EC broad activity code 05 is a broad mining category; it is not NIC division 05, which specifically denotes coal and lignite. Confusing those two code systems would create a serious sample error. [S01] [S02]

Mining-dependent suppliers can belong to transport, machinery repair, construction, trading, professional services or other classifications. A supplier should be included on the basis of a documented mining revenue share or verified mining contracts, not on its industry code alone. A practical threshold is to pre-specify a mining sales-exposure measure and retain the continuous share for robustness. Operator estimates and supplier estimates should be reported separately even when both populations are sampled in the same district.

Major-mineral mines, small stone quarries, sand operations, coal enterprises and hydrocarbon operators face different reporting and concession regimes. The all-India public-data analysis is broad; the prospective fieldwork should concentrate on populations in which owner relationships can be measured reliably and enough independent businesses exist. Large public enterprises can serve as buyers and institutional context, but their own output cannot be attributed to an entrepreneur's caste network.

3.2 Geographic priorities

An initial core could combine Rajasthan's stone and quarrying businesses, Odisha's mineral and contractor ecosystem, and Maharashtra's quarrying and contractor population. The inclusion of Maharashtra is supported by an actually acquired district dataset, which permits frame checks and a transparent replication. Odisha has a verified public lessee register and an administrative system with stable codes. Rajasthan offers a potentially useful small-operator setting, but its current frame needs to be acquired and checked before district selection. [S07] [S33] [S34] [S35]

The final cluster list should be chosen after a short listing pilot rather than by declaring well-known mining districts representative. Stratify within states by mineral, formal operator versus contractor, approximate scale, rurality and lease or contract status. Include commercially active locations and areas with small or intermittent operations. Oversample groups with small entrepreneur counts, while retaining probabilities and survey weights so the target population remains explicit.

Geography must be historical when historical data are used. The national merge pools EC residual

Andhra Pradesh and Telangana into the 2011 state of Andhra Pradesh. The Maharashtra merge uses 35 districts in their historical configuration, including Thane before Palghar's later separation and the older names in the source. Present-day district maps should not be applied to these counts without a documented crosswalk. There is no spatial regression or geocoded mine map in the completed analysis.

Chapter 4

Data discovery and acquisition strategy

4.1 Four layers of evidence

Layer 1 is the establishment frame: Economic Censuses, state directories, lease registers, Udyam and a new area listing. Layer 2 is outcomes: authorized mine returns, company accounts, production and dispatch records, procurement contracts and repeated field measurements. Layer 3 is relationships: a consented owner and alter module, association participation and dated introductions. Layer 4 is context: geology, mineral prices, transport access, population composition and institutional events.

No single identified source provides all four layers. Economic Census data are particularly useful for broad coverage and owner category but weak for direct relationships and firm-level growth. Company panels provide financial trajectories but exclude many small businesses and do not supply validated owner caste. Administrative mining records may provide physical outcomes and concession events but require access verification and still need a relationship module. IHDS offers household social-capital questions and longitudinal household information, with mining sample size and business attribution still to be checked. [S01] [S12] [S20] [S37] [S39]

ASUSE and the related unincorporated manufacturing, trade and service sources need a coverage check before use. The checked official coverage excludes mining Section B. They may be valuable for eligible ancillary enterprises, but treating them as a mining-operator dataset would misstate the population. ASI likewise requires an explicit factory and industry eligibility check; mineral processing and manufacturing are not automatically the same population as mine operators. [S15] [S16]

4.2 Acquiring usable records

First request original EC5 and EC6 microdata through the official registered access route, schedules, record layouts and code lists. Obtain social group by mining by district tabulations if direct unit access is delayed. Retrieve the official distribution when possible to verify the mirrored national report byte-for-byte or reconcile content differences. Do not mix provisional and final census totals. The official FAQ checked during this work describes seventh-census finalization; an unverified final seventh-census file is not included. [S03] [S05]

Second request a deidentified mine-month or mine-year panel from IBM and the selected state departments, including mine code, lease holder, mineral, physical output, employment, dispatch, operational status, filing status and revisions. Ask for unsuccessful and non-operating records as well as operating mines. A returns system can omit precisely the businesses most likely to exit or remain informal. Administrative non-filing must not be silently recoded as zero production. [S20] [S21] [S34]

Third seek the university's Prowess licence or authorized MCA research access. Define entity identifiers and preserve standalone and consolidated financial statements. Add procurement bidder and award records for a clearly scoped set of buyers and services. Losing bidders and eligibility failures are

particularly valuable: a winners-only dataset cannot estimate the probability of obtaining a contract. Obtain the relevant portal terms before downloading or redistributing restricted extracts. [S38] [S39] [S45]

Fourth undertake field listing and an owner panel. Record permission for each linkage, maintain contact information outside the analytical files, and use a stable pseudonymous firm identifier. Mine codes, legal firms and operating establishments form a many-to-many system. The linkage table must preserve ownership transfers, separate subsidiaries and multiple mines, rather than collapsing them into a deceptively simple one-row-per-company file.

4.3 Access status and search limits

The archive includes an acquisition log with successful and failed attempts. The official national-report landing and some older MoSPI download routes did not yield a usable report during my acquisition attempts, while a government-authored mirror did. EC microdata metadata were discoverable, but the requested downloadable unit records were not acquired. Original microdata have not been replaced with invented establishment observations or surname-derived owner labels.

The 65-entry appendix records source-specific access and analytical roles. Families such as GST, payroll, bank credit, GeM, state-board consents, other state mining departments, exporter records and association lists remain conditional research leads. Their presence in the appendix is not a claim of access or an assessment of every file's schema. I have not sent institutional data requests or opened accounts to obtain restricted research records.

Chapter 5

Construction of the completed analytical data

5.1 Extraction and source traceability

The national report contributes Table 2.5 on mining establishments at PDF page 145, printed page 135, and Table 4.5 on mining employment at PDF page 169, printed page 159. Table 12 at PDF page 195, printed page 185, and Table 15 at PDF page 198, printed page 188, provide industry by owner-group establishment and employment totals. The Maharashtra report contributes Table 2.5 at PDF page 123, printed page 117, and Table 4.5 at PDF page 147, printed page 141. Page references are one-based.

I extracted the trailing numerical columns for each geography and checked every row against own-account plus hired-establishment totals, rural plus urban totals and combined counts. The extracted national totals reconcile to 85,176 establishments and 557,474 workers. The Maharashtra totals reconcile to 6,229 establishments and 29,527 workers and match the national Maharashtra row. The four mining owner-group counts reconcile to 73,170 proprietary establishments and 319,425 workers.

The numerical columns labelled employment in establishments with at least one hired worker include all persons working in those establishments. They are not counts of hired workers alone. Similarly, an own-account establishment has no hired worker under the census definition but may involve several household members. Misreading these columns would distort both employment shares and average size.

There are 92 aggregate industry-group observations, consisting of 23 broad industries crossed with four owner groups. These are not 92 sampled firms and are certainly not 52 million independent microdata rows. I therefore computed the industry contrasts as ratios of published counts and reported descriptive point estimates without manufacturing firm-level standard errors.

5.2 Demographic joins and transformations

The PCA input supplies total population, SC population, ST population, urban population and literate population at state and district levels. I joined state codes to historical geographic units after correcting Andhra Pradesh, and checked Maharashtra district names against both the source names and the historical district codes. The completed merge retains all 35 state/UT regions and all 35 Maharashtra districts; no unmatched geography is silently dropped.

$$\overline{E}_r = \frac{W_r}{N_r}, \qquad d_r = 100000\frac{N_r}{P_r}, \qquad h_r = 100\frac{N_r^H}{N_r}.$$

Here W is workers, N is establishments, P is the 2011 population and NH is the count of establishments with at least one hired worker. The population reference date differs from the establishment enumeration period. Density is a contextual historical measure with that timing limitation. SC and ST shares equal their population counts divided by total population, multiplied by 100. Literacy equals the literate population divided by population aged seven and above, not total population.

Size and hired shares are undefined when establishments are zero. They remain missing for Delhi and Lakshadweep; they are not filled with zero. Density includes the zeros. The density regression uses log of one plus establishments per 100,000 residents, making that dependent variable explicitly different from log density. The 50-establishment restriction is a sensitivity analysis intended to reduce instability in small district denominators, not a rule for deleting genuine large mines.

5.3 Quality and measurement limitations

Arithmetic reconciliation validates extraction, not census completeness. Under-enumeration, industry miscoding, intermittent activity and ownership classification can remain. The EC does not supply a verified longitudinal business identity in these tabulations. The national ownership categories are reported for proprietary establishments; incorporated operators, partnerships and public enterprises do not inherit a caste label from a table about proprietors.

The mineral composition of a region is not observed in the completed regressions. Major metal mines, smaller stone units and support activities can have very different labour intensity. This missing composition is a plausible explanation of geographic variation. Local demographics and infrastructure may also influence selection into mining. The report does not claim that basic literacy and urbanization controls remove these confounders.

The package preserves raw numerical counts and full-precision transformations. Rounded published percentages are retained for checking, while reported shares are recomputed from counts. The distinction matters because rounded percentages may sum to 100.1 or 99.9 without any extraction error. Unknown categories, non-applicable ownership and true zero establishments should be kept conceptually separate.

Chapter 6

Descriptive findings

6.1 Ownership and scale

The national average is 6.54 workers per mining establishment when employment and establishments are summed before division. About 73.74 percent of establishments are rural. About 52.03 percent have at least one hired worker, and those establishments account for 88.84 percent of workers in the mining table. Proprietary businesses comprise 85.90 percent of establishments but 57.30 percent of mining employment. This indicates that the establishment frame is dominated numerically by proprietors while employment is more concentrated in larger and other organizational forms.

Within proprietary mining, SC owners account for 12.03 percent of establishments but 7.25 percent of workers. ST owners account for 8.87 percent and 7.47 percent respectively; OBC owners account for 38.93 percent and 33.60 percent. Others account for 40.18 percent of establishments and 51.68 percent of workers. Employment in a social-group row is employment at establishments owned by that group. It is not employment of workers from that group.

The SC mean is about 53.1 percent lower than the Others mean. The ST and OBC means are about 34.5 and 32.9 percent lower respectively. This is meaningful for describing business scale in this historical proprietary population. It cannot establish whether differences arise at entry, through subsequent expansion, through survival, through mineral specialization or through access to finance and contracts. Those explanations require different data and empirical tests.

Comparing owner shares with the whole population can provide a rough context, but it is not a properly adjusted entrepreneurship participation rate. The eligible-owner denominator depends on age, location, labour-market attachment, household resources and business opportunity. A national OBC participation ratio cannot be calculated from the acquired PCA because that source has no comparable general OBC population denominator. Broad Others categories also differ across datasets.

6.2 Geography and heterogeneity

After combining Andhra Pradesh and Telangana, the five largest state/UT regions account for 49.41 percent of establishments and 51.25 percent of workers. The establishment Herfindahl index is 0.0722 and the employment index is 0.0768 across those historical regions. These indices describe geographic concentration, not firm market concentration or buyer competition. They are affected by region size and aggregation.

The unweighted mean of the 33 non-empty regional size ratios is 9.18 workers, while their median is 6.48. These differ from the national 6.54 because the national ratio weights each regional mean by its establishment count. An average of regional ratios answers a different question from the ratio of national totals. The discrepancy is particularly large when small regions contain a few large operations.

Maharashtra has 6,229 mining establishments and 29,527 workers, for a pooled mean of 4.74 workers. Across districts the unweighted mean is 5.73 and the median is 3.15. Chandrapur's ratio is 58.5 workers per establishment, reflecting 5,148 workers across 88 establishments in the table. This is not a reason to classify the observation as an error: it is a reason to inspect sector composition and influence. The 23-district restriction with at least 50 establishments yields another explicit target population.

The report provides raw regional counts, denominator-based indicators, distribution summaries, rank figures, social-group shares, cross-industry ratios and the state correlation matrix. Histograms describe distributions of regional mean sizes. They do not describe the individual-firm employment distribution. Individual medians, firm-size percentiles, inequality within owner groups, and the fraction of SC mining firms above a given size cannot be recovered from the acquired aggregates.

6.3 Cross-industry contrasts

An aggregate descriptive model relates log of the group-specific industry mean size to industry indicators and group indicators. With equal weight per industry-group cell, the fitted SC contrast is -0.261 log points, approximately 23.0 percent smaller than Others. The OBC and ST contrasts are approximately 16.6 and 15.3 percent smaller. With establishment-count weights, the SC contrast is about 17.5 percent smaller and the OBC and ST contrasts about 13.5 and 13.4 percent smaller.

These results are summaries of published cell means across industries. Changing the weights changes the target summary, so the two estimates are not robustness proofs for a causal parameter. The particularly large raw mining SC contrast is useful for prioritizing a sector-specific investigation. It cannot show that a mining mechanism is statistically different from an individual-firm mechanism in other industries. No sampling p-values are presented for the cell models.

Chapter 7

Regression methods and completed results

7.1 Specifications

The completed regional models use a baseline specification with SC and ST population shares, and an adjusted specification adding urban population share and literacy. All shares are scaled so a unit increase means ten percentage points. The source observations are regions, not individual owners. The dependent variables are log mean employment size, log of one plus establishment density, and the share of establishments with at least one hired worker. Each outcome has its own units.

$$\log \overline{E}_r = \alpha + \beta_{SC} \frac{s_{SC,r}}{10} + \beta_{ST} \frac{s_{ST,r}}{10} + \gamma_U \frac{u_r}{10} + \gamma_L \frac{\ell_r}{10} + \varepsilon_r.$$

S1 is the state log-size baseline. S2 adds the two controls. S3 repeats the adjusted size model among the 28 historical states excluding 2011 UTs. S4 uses transformed establishment density across all 35 regions, including zero-count regions. S5 uses the hired-establishment share in 33 non-empty regions. D1, D2 and D3 repeat adjusted size, density and hired-share models in the 35 Maharashtra districts. D4 restricts adjusted district size to the 23 districts with at least 50 establishments.

I used HC3 heteroskedasticity corrections and t-based intervals for the regional OLS models. The HC3 adjustment gives greater weight to the uncertainty introduced by high-leverage observations. These are model-based uncertainty summaries across a hypothetical population of comparable regions; the published census counts are not a random sample of these particular regions. Spatial dependence remains unaddressed because verified coordinates or a contemporaneous boundary adjacency file were not acquired. HC3 is not a correction for all spatial correlation.

7.2 Main coefficients and interpretation

In S2, the SC coefficient is 0.0306 with a 95 percent interval from -0.4932 to 0.5544; the ST coefficient is 0.0084 with an interval from -0.1421 to 0.1589. These associations are small at their point estimates but extremely imprecise. For example, the SC interval permits a substantial negative or positive relationship of local population composition with regional mean mining size. It does not permit a confident finding of no relationship.

In D1, the SC coefficient is -0.0406 with an interval from -0.6760 to 0.5948. The ST coefficient is 0.1273 with an interval from -0.2685 to 0.5230. The district data similarly do not identify a stable adjusted association. The ecological regressions should therefore be used as a description of context and a check against overconfident geographic narratives, not as an owner-level caste-effect analysis.

S4 yields an SC coefficient of -0.5284 on log of one plus establishment density, with an unadjusted

p-value of 0.0362. A ten-percentage-point difference in SC population share corresponds to a fitted factor exp(-0.5284), about 0.59, for one plus density, conditional on the included controls. This cannot simply be reported as a 41 percent reduction in density itself, particularly near zero. The association may reflect geology, mineral composition, administrative coverage, region size or other omitted factors.

I defined an exploratory testing family of eighteen SC/ST coefficients across the nine regional models and calculated Holm and Benjamini-Hochberg adjustments, which I retained in the CSV. The S4 SC adjusted value is approximately 0.651; no coefficient meets a 5 percent threshold after the corrections. The models and sample restrictions were selected for feasibility and sensitivity analysis rather than chosen after screening for significance. This remains exploratory work, not a preregistered confirmatory study.

7.3 Influence and robustness

Cook's distances and leverage are retained for every observation in every OLS model. Leave-one-out coefficient ranges show how a fitted relationship changes when each region is removed in turn. The report also contrasts inclusion and exclusion of UTs and removes very small district denominators in D4. Such checks can reveal instability, but they do not cure omitted geology or replace direct network measurement.

When a large mine creates a high district average, deleting it would change the question from all mining establishments to a selected small-scale segment. The appropriate response is to collect mineral, organizational form and operating-scale information, then stratify or model those features. Winsorization should be a pre-specified sensitivity exercise, not a method of forcing a convenient coefficient. I did not winsorize the completed model data.

The raw regression table includes all intercepts and control coefficients, sample sizes, R-squared values, HC3 standard errors, raw p-values and confidence intervals. The source package also contains the influence file, the correlation matrix and full descriptive output. I emphasize social-composition coefficients in my main discussion because they address the research question, and I retain the other fitted terms in the complete results.

Chapter 8

Mining workers: caste composition, job entry and employment conditions

8.1 What the additional evidence contributes

Worker composition is observable in several sources, even though caste-linked recruitment and firm growth remain incompletely observed. I added a national Census worker baseline, Coal India administrative employee snapshots, published recruitment flows and two Rajasthan stone-sector studies. They answer different questions. Census records industry-classified persons; Coal India records corporate employees; Rajasthan studies interview selected workers across extraction and processing. None supplies a linked national employer-worker network panel. I used actual published cells and retained their denominators, geographical coverage, dates and source locations. [S56] [S57] [S58] [S59] [S60] [S61] [S62]

The first distinction is between workers' own social identity and their employers' identity. EC6 employment classified by the proprietor's group does not reveal the group of the people employed. The B18 SC and ST tables directly classify workers by their own broad group. These series therefore substantially improve the labour-side description, but cannot be merged to EC6 to construct a worker-employer pairing. The second distinction is between a route into employment and the social identity of the person providing that route. A relative or colleague can provide a referral without sharing a caste; a same-category contact need not share a jati. Recruitment through relationships is evidence of relational hiring, rather than a direct measure of caste homophily.

8.2 National Census extraction and population definitions

I used three official Census 2011 Excel workbooks: B18 all persons, B18SC and B18ST. I selected NIC 2008 divisions 05, 06, 07, 08 and 09, with group code 000 and class code 0000. This avoids counting a division and then counting its constituent groups and classes again. The five divisions cover coal/lignite, petroleum/natural gas, metal ores, other mining/quarrying and mining-support services. Historical state codes 01-35 are retained; code 00 is the separate India total and never enters the regression. [S56] [S57] [S58]

The national total is 2,105,736 main plus marginal workers in the selected divisions. SC workers number 439,519, or 20.87 percent, and ST workers number 364,749, or 17.32 percent. The remaining 1,301,468 workers, or 61.81 percent, are derived by subtraction. That remainder contains OBC and other workers; it is not an upper-caste population. The source does not provide an OBC worker table. I did not manufacture a population-wide OBC denominator from the remainder.

Main workers worked for at least six months during the preceding reference year; marginal workers worked for less than six months. This is an annual-duration distinction. It does not classify

permanent versus temporary contracts, regular versus contractor payroll, skilled versus unskilled employment, or insured versus uninsured workers. It also includes industry-classified workers beyond wage employees. Thus a marginal-worker share is a defensible duration outcome, while calling it the informal-employment rate would be incorrect. The official definitions are also available at https://data.gov.in/catalog/main-workers-marginal-workers-non-workers-and-those-marginal-workers-non-workers .

I checked every extracted numeric column for male plus female equalling persons and rural plus urban equalling total. SC and ST counts must not exceed the corresponding all-worker cell when combined. Missing published zero division rows are padded only against the all-worker grid, and all state sums then reconcile to the national workbook entries. I applied these checks separately to main and marginal employment and retained original workbook row numbers. Published counts are not used as a substitute for access to individual Census records.

The worker total is much larger than the EC6 establishment employment total of 557,474. The series have different observation units, reference periods, industry classification procedures and enumeration coverage. This discrepancy must be investigated through the original schedules and producer clarification before any coverage adjustment. It cannot be interpreted as growth, a measured informal-work residual, or evidence that a specific fraction of miners was missed by EC6. Neither arithmetic difference nor a cross-source ratio creates a valid estimate of unregistered employment.

8.3 Descriptive segmentation across subsectors, places and sex

SC and ST participation varies sharply across the five divisions. In metal-ore mining the ST share is 34.83 percent, compared with 13.46 percent in coal/lignite and 8.31 percent in petroleum/natural gas. SC shares are 23.14 percent in other mining/quarrying, 18.29 percent in coal/lignite and 11.72 percent in metal ores. These are industry-worker denominators, rather than shares of the SC or ST population who work in mining. The distinction matters: an industry composition statistic cannot establish an individual's probability of getting a mining job.

The unadjusted marginal-worker shares are 15.59 percent among SC workers, 22.74 percent among ST workers and 11.75 percent among the remainder. Coal/lignite has relatively low marginal shares; other mining/quarrying has substantially higher values across groups. The appended sex-specific graph and all-state table show why pooling mining as a single industry can conceal composition differences. State charts include small workforces, and therefore retain counts alongside percentages. A high group share in a small island or northeastern workforce should not be mistaken for a large contribution to national mining employment.

Three descriptive targets are kept separate: national pooled shares, state-level shares and the average across state cells. I used pooled counts for the first target, each state's own count for the second, and employment weights in my regressions. These choices answer different questions; an unweighted average over states would overweight very small mining workforces. No worker is represented by a reconstructed individual record. The full rural/urban-by-sex-by-division-by-group cells are available for further descriptive work.

8.4 Controlled comparisons: four fitted models

I defined the added regression outcome as the percentage of workers in each observed cell who are marginal workers. Cells cross historical state, NIC division, male/female, rural/urban and SC/ST/remainder. Zero-employment cells have no defined percentage and are excluded from fitting. There are 1,275 nonempty cells in 35 historical state/UT regions, with denominator weights summing to 2,105,736. This is 1,275 aggregate observations, not 2.11 million independently observed survey records.

$$y_c = 100\frac{M_c}{N_c}, \qquad \underset{\theta}{\text{minimize}} \sum_c N_c(y_c - X_c\theta)^2.$$

W1 includes SC and ST indicators with the remainder as the reference group. W2 adds sex and rural/urban residence. W3 adds historical state and NIC-division fixed effects. W4 instead absorbs the complete state-by-division-by-sex-by-residence stratum, comparing social groups within the same joint context. The weights are the cell's main-plus-marginal count. W4 is a more demanding descriptive comparison because additive controls alone cannot account for every interaction between place, mineral and labour-market composition.

$$y_{g,s,d,k,a} = \alpha_{s,d,k,a} + \beta_{SC}\mathbf{1}\{g = SC\} + \beta_{ST}\mathbf{1}\{g = ST\} + u_{g,s,d,k,a}.$$

The unadjusted SC contrast is +3.83 percentage points, falling to +2.47 in W2, +1.89 in W3 and +1.49 in W4. The ST contrast falls from +10.99 to +7.58, +6.23 and +5.08 percentage points, respectively. Relative to the unadjusted contrasts, the W4 values are about 61 percent smaller for SC and 54 percent smaller for ST. This attenuation shows that the pooled differences are partly associated with where people work, the mining division and sex/residence composition. It is not a decomposition into causal contributions of those factors.

The W3 model-based 95 percent intervals are +0.35 to +3.43 points for SC and +5.16 to +7.30 for ST. W4 intervals are -0.46 to +3.45 and +3.58 to +6.58. Thus the more demanding within-stratum SC comparison is less conclusive, while the ST duration difference remains positive in these fitted specifications. All coefficients and controls are archived, including the absorbed stratum coefficients, and the regression table identifies the reference group and units. The four models form a stated sensitivity sequence; their overlapping intervals should not be read as independent replications.

I clustered model inference at the 35 historical states/UTs and used a finite-sample covariance correction and t reference distribution. These intervals describe a model across regional aggregates. They do not quantify Census enumeration errors, recover unavailable household survey design variance, or cure selection and omitted-variable bias. Thirty-five clusters give limited asymptotic reassurance; W4 uses many strata relative to the cell count, so its clustered intervals merit particular caution. The point estimates and their change with controls are the principal descriptive evidence. A policy or causal dissertation would need suitable design-based or small-cluster inference under its actual sampling/assignment scheme.

Age, education, occupation within the division, work experience, landholding, firm capital, contractor status and recruitment route are not jointly observed in these Census cells. I therefore did not add these unavailable controls. The remaining coefficients cannot be described as equal-productivity comparisons, discrimination effects or the effect of caste ties on hiring. Duration itself may depend on preferences, seasonality, secondary work and labour demand. Workers already employed in mining are selected from a wider population, so even a perfectly fitted employment-duration model cannot explain who gained access to mining employment in the first place.

8.5 Formal coal employment and observable recruitment flows

Coal Ministry chapter 16 reports Coal India's employment groups A-D and SC/ST/OBC composition. On 1 January 2024 the disclosed total is 231,058 employees: 44,601 SC, 33,143 ST and 58,296 OBC. On 1 January 2026 the total is 214,333: 35,351 SC, 26,755 ST and 40,395 OBC. Recomputed 2026 shares are 16.49, 12.48 and 18.85 percent. These are corporate employee stocks, including roles beyond mine-face work, and do not represent all Indian miners or the full contractor workforce. The 2026 residual includes the separately disclosed 510 EWS employees. Employment groups A-D are retained as administrative labels rather than automatically renamed as occupations. [S59] [S60]

The difference between snapshots is not a recruitment effect. Explaining a change in a group's stock requires opening stock plus hires and transfers minus retirements, deaths and other exits, together

with consistent coverage and classification. Those matched flows are absent. In addition, published recruitment reservation percentages describe eligibility rules for vacancies, not realised workforce shares. A disability table in the same chapter has the same total employee denominator but reports only employees with benchmark disabilities; it is deliberately excluded from the caste-workforce calculation.

An observable recruitment example is the SECL Mining Sirdar disclosure for FY2024-25: 350 advertised vacancies and 345 filled. The ST vacancy category has 97 advertised and 92 filled, with five non-joiners reported; UR, SC, EWS and OBC category counts fill their disclosed advertised numbers. The resulting 94.85 percent ST vacancy-fill rate has advertised reserved vacancies as its denominator. It is not the probability an ST applicant was hired. UR describes an unreserved vacancy category and cannot be relabelled as a general-caste employee identity. Recruitment through advertised examinations, GATE/CBT for specified management-trainee streams, and separately reported dependent-employment arrangements should be measured as distinct routes. [S60]

This distinction between public recruitment institutions and informal referral chains is analytically useful. It suggests separate dissertation strata rather than assuming a single Indian mining labour market. A comparison of company payroll groups with quarry interviewees can motivate questions about institutional access, but requires sampling and common outcomes before numerical gaps between those sources can be interpreted.

8.6 Rajasthan stone workers and job-entry evidence

Arisa/CLRA's 2024 report covers 97 individual interviews: 47 in Budhpura and 50 in Makrana; ten Budhpura home-workers used a different instrument. Published SC/ST/OBC shares are 36/19/45 percent in Budhpura and 82/0/18 percent in Makrana. The Budhpura composition paragraph leaves the precise instrument denominator unresolved. Of Makrana's 50 workers, 80 percent entered through relatives/co-workers, seven approached an employer and three used contractors. These are local, non-probability supply-chain samples, not nationally representative miners. No caste-by-route cross-tabulation is published. [S61]

Additional Makrana counts include 20 illiterate workers, seven with secondary education or higher, 48 local residents, 20 PPE users and no written employment contracts. An advance paragraph reports 18/50, while the conclusion reports 38 percent; the discrepancy is retained rather than silently corrected. Budhpura's main instrument reports 31/37 piece-rate workers. The report's wage averages are not used to reconstruct individual pay or an adjusted caste wage gap. [S61]

The ILO's 2021 Rajasthan study interviewed 115 workers and 43 owners, largely in the formal value chain across two clusters. Worker composition is reported as 34 percent SC, 30 ST, 30 OBC and 5 General; the 99-percent sum reflects rounded publication values and is not forced to 100. It describes informal connections and direct approaches as hiring routes. Other published indicators cover literacy, financial inclusion, local residence and Provident Fund access. Reported annual work duration differs by sex and production stage. Individual records and caste-by-recruitment tables were not obtained. [S62]

These observations motivate specific fieldwork questions. The recruitment agent, legal employer, wage payer and current supervisor may be different people. Being local does not imply recruitment without relationships, and a contractor is not necessarily the person who first supplied job information. An advance may affect the ability to leave a job, while the original referral may affect access to one. Each event needs its own date and relationship code. Safety equipment use also differs from effective protection; a survey should measure task, exposure, provision, training and use separately, without inferring disease from an interview response.

8.7 Additional usable sources and the controls they would permit

The acquired PLFS 2023-24 layout and interview instructions verify household social group and person-level industry, occupation, enterprise type, enterprise workforce band, contract and earnings fields. The first-visit layout uses sg at household level, ind_pas and ind_sas for usual principal/subsidiary industry, and job_pas/job_sas for job-contract measures. Records must be joined using the full survey household identifier and visit/quarter context, rather than row order. Documentation is acquired; the unit records are not. No PLFS wage regression is claimed. [S11] [S63]

A defensible next wage analysis would define wage-employed miners under one activity-status convention, separate monthly regular earnings from daily casual earnings, and estimate log earnings against group indicators, age and age squared, education, sex, state, rural/urban residence and occupation. Hours, days and tenure should be added only when genuinely measured for the same earnings period. Industry and establishment characteristics can enter subsequent specifications to reveal how the descriptive gap changes. Report weighted effective sample sizes by group and mining division before fitting; a small ST oil-sector cell may not sustain a precise estimate. Survey strata, PSUs, multipliers and annualization instructions must be applied from the exact release.

Occupational controls may themselves reflect the labour-market channel under study. Consequently, an occupation-adjusted pay contrast answers a different question from the total pay difference. The same is true of current contractor status, advances and job tenure if recruitment affects them. A controlled coefficient should therefore be accompanied by an estimand and a causal diagram, rather than a claim that adding every available variable always improves identification. Separate baseline determinants from possible mediators and report the sequence of specifications.

The Labour Bureau's Occupational Wage Survey offers occupation-level pay and employment information for selected mining industries; the identified seventh-round four-mining-industry report is a specific acquisition lead. Its caste field and unit-record access have not been verified. Separate Labour Bureau SC/ST working-and-living-condition, unorganised-worker and women-worker survey families should be searched by industry, locality and vintage, rather than presumed to contain a modern national mining panel. Company reservation returns, union records, contractor wage registers and consented site rosters can complement these sources but will have distinct coverage and access rules. [S64] [S65]

8.8 Matched worker-firm fieldwork and identified extensions

The additional W01-W30 instrument links a worker's broad self-reported group, job and recruitment event to an anonymous establishment identifier. It separately records who supplied information, who recommended the worker, who selected them, who pays wages, and whether a contact relationship existed before the vacancy. Optional jati and self-reported shared-community relationships require consent and declined/unknown codes; they are never inferred from surnames. Employer rosters should be reconciled with workers' accounts, including off-roster contractors and home workers where these fall inside the chosen supply-chain scope.

Sample workers through both site listing and area/household listing, with probabilities recorded. Employer-supplied lists alone may exclude mobile, indebted, dismissed or informal workers. Track incumbents, new recruits, applicants and rejected applicants separately. For hiring access, the relevant denominator is the eligible applicant/opportunity population; for retention it is the baseline recruited cohort; for productivity it is the matched firm over repeated periods. Recording those populations makes it possible to distinguish caste-related access from sorting among people already employed.

A future descriptive recruitment regression can use referral route as an outcome and pre-job age, education, prior experience, origin and recruitment cohort as controls, with firm or site effects where overlap permits. It needs observed joint records. I cannot identify a referral advantage from the published 80-percent Makrana referral share and 82-percent SC share: many joint distributions are

compatible with those two margins. An individual-route regression reconstructed from them would merely encode arbitrary assumptions.

For causal work, an explicitly randomized job-information or introduction offer could estimate whether widening information access changes applications, interview invitations, hiring and retention, with owner and worker category heterogeneity specified before analysis. A trial manipulating information should be described as an access intervention; it does not randomize caste. Any paired employer-worker design must protect workers from disclosure to employers and monitor interference, job displacement and substitution between applicants. Firm sales and employment follow-ups are then required to test whether recruitment and retention channels affect growth. My controlled Census results establish a labour-duration baseline and sharpen hypotheses; they do not substitute for that design.

Chapter 9

Why the completed results cannot identify caste ties and growth

9.1 Missing treatment and missing longitudinal outcomes

Identification requires an observed exposure and an outcome related to the desired causal contrast. The acquired caste tables observe owner category and total employment at a single enumeration. Regional tables observe population composition and aggregate mining scale. None observes the tie intensity Ti, the baseline firm size Ei0 and the subsequent outcome Ei1 jointly for a business. Repeated aggregate censuses would help describe net changes but would still mix entry, exit and incumbent expansion.

Many different business histories can produce the same aggregate table. One world could have no effect of community relationships and strong inherited-resource differences. Another could have powerful network advantages offset by other disadvantages. Both could reproduce the observed owner-group counts and employment totals. Because those worlds imply different causal effects while yielding the same acquired data, the desired effect is not identified by these tabulations.

This is not a problem resolved by adding more regression commands. Fixed effects, matching, propensity scores, a complex machine-learning model or an instrumental-variable label cannot reconstruct an unobserved treatment. Data construction and the assignment mechanism must lead the econometric choice. The observed disparity is an important fact about scale; it is not an estimate of discrimination or the economic return to same-caste contacts.

9.2 Causal pathways and adjustment

The conceptual graph has baseline household resources, geology, inherited land and business ability affecting both relationships and outcomes. Baseline social identity can affect treatment opportunities and resources. Relationships can affect finance, buyer access and hiring, which can affect growth. Previous firm success can also create relationships. Local institutions and commodity demand influence several pathways simultaneously.

If the goal is the total effect of relationships, controlling for credit or contract wins measured after the relationship was formed may block mechanisms the study is trying to estimate. If the goal is a direct effect holding finance constant, stronger assumptions are required and the estimand changes. Baseline predictors, mediators and outcome consequences need separate variable tags in the data dictionary.

A district fixed effect can compare firms facing a shared local environment, but does not remove owner-specific wealth or ability. A firm fixed effect can remove time-invariant characteristics, but not a new buyer relationship formed in anticipation of expansion. Lagging contacts improves temporal ordering but cannot guarantee exogeneity. Fixed effects do not randomize owner identity.

Chapter 10

Prospective causal inference designs

10.1 A randomized business introduction design

The most credible prospective intervention is a randomized offer of structured business introductions or brokerage among eligible firms. Randomize clusters of connected business environments when introductions can spill over. Stratify assignment by state, mineral, baseline scale and the sampled owner categories. Pre-specify an equal-probability treatment assignment and retain a control group receiving any ordinary public information required by the study's ethical commitments.

Measure whether the offer creates additional business relationships, whether those relationships cross existing community boundaries, and whether new contacts yield actual loans, quotations, contracts or orders. Primary outcomes can be real sales, employment and survival at 12 and 24 months. Programme delivery must not discriminate against participants based on caste. Subgroup analyses should examine heterogeneity in access and gains while presenting the overall programme effect first.

$$Y_{ic,1} = \alpha + \tau Z_c + \rho Y_{ic,0} + \lambda_{\text{randomization stratum}} + X'_{ic,0}\theta + e_{ic}.$$

Estimate intention-to-treat by assignment with inference at the randomization unit. Report take-up and a first stage for relationship formation. Instrumenting actual ties with assignment could target a local effect only if the exclusion, monotonicity and relevance conditions are credible. A package that independently teaches procurement rules or improves applications violates a pure relationship exclusion restriction unless those channels are isolated. The principal claim should remain the causal effect of the offered programme.

Networks create interference. A contact made by a treated business can become available to controls; contracts can also be displaced from controls to treated firms. Record contact sharing and define exposure to treated neighbours before analysis. A saturation or two-stage randomized design can estimate direct and spillover effects, but it requires enough clusters and a defensible interference map. Arbitrary post hoc network definitions would undermine the estimand. Aronow and Samii provide a formal framework for exposure-based inference; the specific exposure definitions here remain a proposed study decision.

10.2 Policy and auction difference in differences

A plausible observational programme would examine a verified change in auctions, procurement support, registration, portal access or licensing that varies across eligible units and time. Collect several outcome waves before the event and enough after it, including unaffected or not-yet-affected comparable units. Define exposure using eligibility and actual legal implementation, rather than whether a firm benefited or won a contract.

If treatment timing is staggered, estimate cohort-time effects using a method appropriate for heterogeneous treatment effects and state the comparison groups explicitly. Callaway and Sant'Anna offer one such framework. Avoid interpreting a conventional two-way fixed-effects coefficient mechanically when already-treated firms become controls. Show event-study estimates with simultaneous uncertainty, composition checks and robustness to comparison-group choice. A failure to reject pre-trends is not proof of parallel trends.

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_t(1) - Y_t(0) \mid G = g], \quad t \geq g.$$

An interaction with baseline self-reported network strength can test effect heterogeneity, but it identifies a policy effect differing by baseline connections, not necessarily the causal effect of those connections. Policy adoption can reflect local mining conditions, administrative capacity or political demand. The nationwide 2015 change is not automatically a staggered experiment. A dissertation must show the actual exposure variation and legal timing relevant to the selected population. [S24] [S25]

10.3 Regression discontinuity and close auctions

A verified support-programme cutoff could permit a discontinuity design if eligibility is determined by a predetermined score that firms cannot precisely manipulate, outcomes are smooth in the score absent treatment, and observations exist on both sides. Collect the complete running variable, the operational threshold, exceptions, application outcomes and programme take-up. Inspect bunching, balance and the local support. None of those inputs was acquired for this report.

A close-auction design needs the exact bidding mechanism, all bids, tie rules, eligibility conditions and reasons bids were rejected. Merely comparing winners with losers close in an endogenous bid does not create random assignment. Even a credible local allocation effect concerns acquiring a concession or contract. A caste-network interpretation additionally requires observed relationships and an argument separating allocation from other channels.

10.4 Instruments that should not be assumed valid

Historical caste concentration, distance to co-community settlements, geology, mineral prices, local political ties and family business ownership are tempting instruments. Each plausibly affects growth directly or proxies persistent resources and demand. Their exclusion restrictions are therefore weak without unusually strong evidence. Historical instruments can add a veneer of causality while magnifying measurement error and geographic confounding.

If a valid instrument emerges, report the institutional argument first, then relevance, weak-instrument diagnostics, reduced-form outcomes and the scope of the local effect. Use uncertainty methods that remain valid when the first stage is weak. Overidentification tests cannot establish that all excluded instruments are valid. The present report does not estimate an IV model because no defensible observed instrument-treatment-outcome dataset exists.

10.5 Observational matching and doubly robust estimation

Once the firm panel exists, matching or inverse-probability approaches can improve comparisons on observed baseline characteristics. They require conditional exchangeability, positivity, consistent treatment definition and adequate measurement. Diagnose common support and effective sample size; trim according to pre-specified rules and report the resulting population. Outcome and treatment nuisance models can be estimated with cross-fitting in sufficiently large samples.

These methods remain vulnerable to unobserved inherited resources, ability and anticipated growth. Report estimates as conditional associations unless the exchangeability argument is credible. A sensitivity analysis should ask how much omitted confounding would overturn the substantive conclusion, not merely whether the adjusted p-value is below a threshold. Network spillovers and endogenous tie formation are additional complications beyond ordinary binary treatment selection.

Chapter 11

Fieldwork design and questionnaire

11.1 Sampling and recruitment

Start with a frame merge of leases, registrations, firms, directory entries, buyer vendors and a new area listing. Deduplicate establishments while retaining multiple sites under one legal firm. List inactive concessions, intermittently operating units and known closed businesses. Do not rely exclusively on referrals from influential association members: such recruitment can build the treatment into the sample selection mechanism.

A proposed baseline target is 1,200 businesses across 80 clusters, with roughly 15 businesses per cluster. This is a design assumption, not an observed sample. Allocate separate operator and contractor strata and ensure multiple clusters within each selected state and mineral. A trial would need an explicit cluster assignment unit; a descriptive panel could use different sampling clusters. Final allocation should follow pilot response rates, subgroup incidence, costs and measured outcome variability.

Use probability sampling within completed listings, with documented oversampling for smaller entrepreneur groups. If a hard-to-reach segment requires a supplemental referral sample, label it separately and do not claim that inverse-probability weights solve unknown inclusion probabilities. Weights for population estimation should reflect listing and within-cluster selection; regression estimands should clarify whether they target the sampled firms or the frame population.

11.2 Owner and business module

Collect legal and operating names, pseudonymous business ID, organizational form, mines and sites, operating start, mineral, support activity, licence and registration status, and the respondent's role. Record all active owners and ownership shares where feasible. Ask who exercises effective control, separately from the legal proprietor. Do not automatically assign a multi-owner business the caste of the interviewer's main respondent.

The identity module asks voluntary self-reported broad category, optional jati name, state of category recognition, religion where relevant to the conceptual question, and whether the respondent wishes to decline. Retain original text in a protected coding environment and use a documented harmonized classification in the analytical file. Record ambiguity rather than forcing a category. Interview owners in a setting where colleagues or gatekeepers cannot overhear sensitive answers.

The baseline resource module covers education, earlier experience, inherited business or concession, initial capital source, household assets in broad bands, previous employment, land access, equipment and initial workforce. Sensitive wealth and revenue questions should offer ranges and a declined-response option. Distinguish inherited capability, family financing and commercial relationships in both questions and analysis.

11.3 Relationship name generators

Ask the respondent to identify coded contacts who helped obtain the most recent major buyer, loan, equipment, licence-related information, large contract or recruitment batch. For each contact record role, relationship origin, date first known, date of assistance, geographic location, frequency of interaction and whether the contact has business authority. Separate family relationships from explicitly reported community relationships and from ordinary commercial referrals.

Ask whether the respondent considers each contact part of the same jati or community, whether that fact mattered to the introduction, and whether the contact's identity is directly known. Do not force respondents to classify an alter they do not know. Relationship coding should include unknown and declined responses, as well as cross-community contacts. Contacts need not be named publicly: local identifiers and a protected linkage key can support panel tracking.

For each assistance event ask what changed: a buyer introduction, a quotation, a guarantee, a loan amount or range, deferred payment, information about a tender, a hiring recommendation, or no observable result. Record the sequence of dates. A contact merely known to the owner is different from a contact activated for business, and the latter is different from a completed transaction.

11.4 Financial and growth module

Collect monthly or quarterly employment and production where records exist, plus the preceding fiscal year's sales, costs, inventories, equipment purchases, borrowing and receivables. Retain evidence flags: audited statement, ledger, payslip or payroll, dispatch record, respondent recall, or bounded estimate. Align recall windows to the fiscal year or operating season and record downtime. Separate quantity, grade and price to distinguish production growth from commodity-price changes.

Measure buyer and supplier concentration, top-customer revenue share, payment delays, contract values, bidding participation, wins, reasons for nonparticipation and geographic market reach. Record credit applications and rejections as well as loans received. Ask about formal and informal finance separately. Contract workers require establishment and employer identifiers to avoid counting the same labour both at the contractor and mine operator.

Panel follow-ups at 12 and 24 months should verify operating status and collect growth outcomes. A closed business remains in the panel. Ask the owner or a permitted substitute about closure timing and reasons, and distinguish temporary suspension from exit. Track ownership transfers and mergers. New entrants can form a supplementary entry sample, but they should not be mixed with the original experimental cohort without explicit definitions.

11.5 Qualitative work

Conduct a nested qualitative sample selected for analytical contrasts: different minerals, owner categories, business scales, strong versus weak networks, successful versus unsuccessful applications and surviving versus exiting firms. A provisional target of 60 to 90 interviews is a planning range, not a saturation claim. Include lenders, buyer procurement officers, association representatives and workers where the research question requires them.

Interview guides should reconstruct events rather than solicit only general views about caste. Compare accounts of the same transaction when permission permits, triangulate documents, and record disagreement. Analyse negative cases in which relationships failed to help or cross-community ties became commercially important. Qualitative evidence can sharpen mechanisms and reveal mismeasurement; it does not substitute for the assignment assumptions of a causal estimate.

Chapter 12

Network measurement and analysis plan

12.1 Core indicators

Define the activated business-contact set before estimating relationships with growth. A within-jati share is the fraction of classified activated contacts the respondent reports as sharing jati. Record its denominator and the fraction whose identity is unknown. A categorical broad-group homophily index is a different measure and should not replace the jati-specific index without explanation.

$$H_i = \frac{\sum_j A_{ij} 1\{J_i = J_j\}}{\sum_j A_{ij}}, \qquad C_i = \frac{\text{activated contacts with classification}}{\text{all activated contacts}}.$$

H should be missing when no eligible classified contacts exist, while Ci records classification completeness. Report the number of contacts, within-community share, cross-community bridging ties, functional diversity, geographic reach and dependence on the strongest contact separately. A high share with only one contact is different from a high share across ten contacts. Fix the maximum number of elicited alters or record truncation so interview caps do not masquerade as network limits.

Adjust homophily for the plausible opportunity set where that set is measured. For example, local co-jati prevalence among listed eligible suppliers can benchmark observed contact shares. General village SC share is not a substitute for the prevalence of the owner's jati among eligible mining buyers. Availability-adjusted homophily is especially important when an apparent same-community preference is mechanically generated by who is nearby.

An ego-network survey does not identify complete network centrality, global betweenness or the entire mining industry's network structure. Such measures require a sufficiently complete bounded network and accurate alter linkage. Use degree and brokerage measures appropriate to the observed design, preserve anonymous unlinked alters, and test sensitivity to missing links. Any reported network visualization should avoid making protected identities reidentifiable.

12.2 Firm panel specifications

The descriptive baseline panel can regress later growth on baseline tie measures, owner identity and baseline resources, with location by mineral effects. Separate a total-association model from models including potentially mediating credit and buyer access. Interactions with SC, ST and OBC identity should be pre-specified and interpreted within their support. Report group-specific sample sizes and tie distributions.

$$g_{i,t+1} = \alpha_{d\times m} + \beta T_{i,t} + \sum_c \delta_c 1\{C_i = c\} + \sum_c \phi_c T_{i,t} 1\{C_i = c\} + X'_{i,0}\theta + u_{i,t+1}.$$

Repeated employment levels can support firm fixed effects and mineral-by-time controls. Dynamic specifications need enough waves and careful treatment of endogenous lagged outcomes. A two-wave observational panel cannot justify an elaborate dynamic GMM procedure merely because such software exists. With small samples, simple transparent specifications and credible assignment are preferable to a long list of fragile estimators.

Analyze employment, real sales and production in separate outcome families. Use appropriate deflators and price exposure. Report survival separately and present a growth measure that includes exits. Conditional-survivor sales growth answers a selected question; it should not be the sole headline result if the programme changes survival. Quantile or distributional analysis needs adequate firm-level data and cluster-aware inference.

Chapter 13

Power, inference and pre-analysis commitments

13.1 Power assumptions

The supplied power table is illustrative. It assumes equal treatment assignment, 15 baseline businesses per cluster, 20 percent attrition, 80 percent power, a two-sided 5 percent test, and intracluster correlations of 0.02, 0.05 and 0.10. Calculations use a cluster-based t critical value plus the normal 80-percent power quantile and a design effect evaluated at the retained average cluster size. They do not incorporate any verified pilot residual variance or covariate gain.

$$\mathrm{DEFF} = 1 + (m_{\mathrm{retained}} - 1)\rho, \qquad \mathrm{MDE}_{SD} \approx (t_{0.975,G-2} + z_{0.80})\sqrt{\frac{4\,\mathrm{DEFF}}{n_{\mathrm{retained}}}}.$$

The 1,200-business, 80-cluster proposal with ICC 0.05 yields an MDE of roughly 0.23 outcome standard deviations under these assumptions. If the residual standard deviation of log growth were 0.5, that corresponds to roughly 0.114 log points. That standard deviation is assumed, not estimated from this study. Unequal clusters, rare group incidence, noncompliance, spillovers and multiple outcomes may require more clusters or larger samples.

Heterogeneous effects by owner group require their own power calculations. A design adequately powered for the overall programme effect can be badly underpowered for an SC-versus-ST interaction. Do not promise precise subgroup estimates without pilot incidence and cluster allocation. Compute detectable first-stage changes in relationships as well as detectable outcome changes before funding the intervention.

13.2 Pre-analysis plan

Register the primary estimand, primary outcome window, assignment unit, strata, sample exclusions and treatment coding before outcomes are observed. Fix rules for permanent exit, temporary shutdown, multiple sites, ownership transfer and missing accounts. Pre-specify baseline adjustment and the distinction between total programme effects, mechanisms and exploratory heterogeneity.

Declare outcome families and multiplicity corrections. For an experimental study, use randomization inference when appropriate and cluster inference at the assignment level. For few clusters, use methods suited to that design rather than relying only on asymptotic robust standard errors. Report confidence intervals and substantively meaningful magnitudes; absence of statistical significance does not establish absence of an economically meaningful effect.

Track missingness by treatment assignment, owner category, mineral and baseline size. Report attri-

tion rates and reasons. Weighting under a missing-at-random assumption should be accompanied by sensitivity analysis. Bound outcomes under transparent scenarios when missingness is plausibly nonignorable; monotonicity-based bounds require a defended direction of selection. No attrition adjustment was needed in the completed census-table analysis because it is not a respondent panel.

13.3 Placebos and robustness

For the observational policy design, inspect pre-treatment outcomes, alternative unaffected minerals or service categories, eligibility placebo definitions, policy anticipation and concurrent changes. A convincing placebo test cannot rescue an implausible assignment mechanism, but contradictory placebo evidence can undermine it. Collect the same information for treated and comparison groups to prevent differential outcome measurement.

For the firm panel, vary credible baseline resource controls, geographic and mineral strata, outcome definitions and treatment measures. Show sensitivity to network classification completeness, ties that predate business establishment, self-reported versus documentary outcomes, and the presence of family firms. Use a principled limited set of checks; do not search every transformation and report the subset with favourable signs.

Chapter 14

Ethics, governance and field implementation

14.1 Consent and disclosure control

The relationship module concerns potentially sensitive social identity and commercial arrangements. Explain the research purpose, voluntary participation, right to skip questions, intended use and linkage options in the respondent's preferred language. Seek the applicable institutional ethics review before fieldwork. Separate permission to be interviewed from permission to link records, contact alters or conduct follow-ups. Participation should not affect licensing or procurement opportunities.

Store names, phone numbers, exact locations and linkage keys separately from analytical data. Limit access to the minimum research team, encrypt retained identifiers and define a deletion schedule. Public releases should suppress small cells and remove reidentifiable relationship diagrams. A public GitHub repository should contain code and properly licensed aggregates; it should not contain raw owner identities, classified alter identities or unauthorized commercial histories.

Sensitive statements about influence, exclusion or administrative contact should be recorded without assuming illegality. Qualitative claims need corroboration and appropriate anonymization. Enumerators must not pressure respondents to name officials or discuss actions they consider risky. This study is designed to measure business relationships and economic outcomes, not to classify people from external clues or make unsupported allegations.

14.2 Enumerator and data quality protocol

Pilot the identity and relationship questions in the selected local languages. Use neutral examples, separate the question about knowing a contact from the question about receiving assistance, and test whether respondents distinguish caste, family and business association. Rotate or standardize module order to reduce priming. Capture interview duration, skipped modules and reasons for ambiguous classifications.

Use checks that flag inconsistent units, duplicated sites, improbable employment jumps and mismatched fiscal years. Confirm a subset of nonsensitive numerical responses and document enumerator corrections. Back-checking should not disclose one respondent's caste or commercial relationships to another. Record provenance for each correction rather than overwriting the original response invisibly.

14.3 Proposed timetable

Months 1-3 should cover original-data access, administrative negotiations, instrument translation and ethics review. Months 4-5 should conduct listing and a small pilot, finalize power and cluster selection, and freeze the pre-analysis plan. Months 6-9 should complete baseline fieldwork and any randomized rollout. Short verification contacts can monitor operating status, while outcome follow-ups occur at 12 and 24 months after baseline. Analysis and dissertation writing follow validated data freezes rather than an arbitrary promise of immediate causal results.

A formal budget should be built from local quotes for listing, enumerators, travel, translation, data management, follow-up and record acquisition. No fabricated rupee budget is presented. The scope can be scaled by reducing minerals or states while preserving enough independent clusters; expanding geographic coverage without sufficient cluster depth can weaken the actual identification design.

Chapter 15

Interpretation and dissertation contribution

The acquired data demonstrate historical inequality in the scale of proprietary mining businesses across broad owner categories. The strongest current result is the gap in ratios of employment to establishments, supported by exact published counts and reproducible arithmetic. The findings do not identify the effects of caste ties, discrimination, ability or inherited resources. The regional regressions reinforce that caution: local population shares are weak and imprecise proxies for the actual relationships the question concerns.

The relevance is practical. A fieldwork project can use the Economic Census as a coverage benchmark, administrative mine and lease systems as a current operator frame, financial or production records as growth outcomes, and a new relationship panel as the essential treatment measurement. A study that combines those sources can ask whether community relationships facilitate entry and expansion, constrain diversification, or mediate exclusion, with distinct evidence for each mechanism.

A credible dissertation contribution would be threefold: direct measurement of functional business relationships in a differentiated mining population; longitudinal separation of entry, survival and incumbent growth; and an identified intervention or policy comparison showing how access to relationships changes economic outcomes. I assembled the source map, historical statistical baseline and research architecture needed to pursue that contribution. It should be cited as a preliminary public-data analysis and research design, not as a completed causal dissertation.

Policy implications remain conditional. The observed size gap supports investigating whether entrepreneurs face unequal financing, buyer access, equipment or licensing constraints. It does not yet establish that a particular caste-based programme will improve growth. A brokerage trial could test whether opening business access helps and whom it helps; complementary resource and discrimination measures are needed before attributing any observed advantage to networks alone.

Chapter 16

Reproducibility and reading the research package

The package contains the self-contained LaTeX report, source inventory, raw acquired reports and spreadsheet, extracted text, derived CSV files, full statistical results, all nineteen PNG figures, scripts, requirements and a README. Hashes identify the downloaded files. The national mirror and official producer are both recorded. Download failures and unavailable unit records are documented rather than hidden.

The analysis script reproduces the source extraction, joins, ratios, regressions, diagnostics, descriptive cell contrasts and power assumptions. An additional plotting script produces the extended figure set. Code does not require a commercial licence or a hidden respondent dataset. The raw-source hashes and totals provide meaningful integrity checks; exact coefficient reproduction also depends on numerical-library versions, which are specified.

The standalone LaTeX document embeds plotting coordinates and tables, so it is designed to compile without external figure assets. The PNG versions are supplied for reuse in slides or papers and correspond to the information reproduced in the report; panel arrangement may differ. The source is opened in the built-in editor. The archive records the actual compilation outcome and any environment limitations. I prepared a public GitHub research deposit containing the compiled report, editable source, public inputs, analysis outputs and reproducible scripts. The deposit does not contain confidential respondent records.

Historical tables can be reused for context, but a current dissertation should update the frame and validate the original releases before making current-policy claims. New microdata, administrative panels or fieldwork responses should be added as separately versioned inputs, with their own terms and access log. Re-running the public-data scripts will not magically add the unobserved relationships or growth outcomes.

Chapter 17

Selected methodological and substantive references

Iyer, Lakshmi, Tarun Khanna and Ashutosh Varshney. Caste and Entrepreneurship in India. Harvard Business School working paper 12-028, 2011; related article in Economic and Political Weekly 48(6), 2013. Primary working paper: https://www.hbs.edu/ris/Publication%20Files/12-028_0e6e69ba-da4d-4abc-b66a-9611dc07c53b.pdf . Used to motivate ownership and enterprise research, not to supply a mining causal estimate.

Munshi, Kaivan and Mark Rosenzweig. Traditional Institutions Meet the Modern World Caste Gender and Schooling Choice in a Globalizing Economy. American Economic Review 96(4), 2006. https://pubs.aeaweb.org/doi/abs/10.1257/aer.96.4.1225 . Used for conceptual motivation about networks and mobility in a different population.

Callaway, Brantly and Pedro H C Sant'Anna. Difference in Differences with Multiple Time Periods. Journal of Econometrics 225(2), 2021. Author manuscript: https://psantanna.com/files/Callaway_SantAnna_2020.pdf . Used to specify a potential staggered-policy design; no such design was estimated with the acquired cross-section.

Aronow, Peter M and Cyrus Samii. Estimating Average Causal Effects under General Interference with Application to a Social Network Experiment. Annals of Applied Statistics, 2017. https://arxiv.org/abs/1305.6156 . Used for a proposed exposure and spillover framework; no observed network experiment is available here.

Central Statistics Office. All India Report of Sixth Economic Census. Ministry of Statistics and Programme Implementation, 2016. Official landing: https://www.mospi.gov.in/all-india-report-sixth-economic-census . Acquired mirror and exact empirical tables are documented in source S02.

Directorate of Economics and Statistics Government of Maharashtra. Sixth Economic Census 2013 Maharashtra. Acquired official repository copy: https://www.nitiforstates.gov.in/public-assets/Policy/policy-repo/urbanization/RSS71M000226.pdf . The exact mining district tables are documented in source S07.

Office of the Registrar General and Census Commissioner India. Primary Census Abstract India and States State and District Level 2011. https://censusindia.gov.in/nada/index.php/catalog/61191 . The downloaded workbook supplies the demographic denominators and controls.

Chapter 18

All graphical results

Every graph generated for this study is reproduced below using inline vector coordinates. The archive also contains nineteen PNG versions. State geography is harmonized to Census 2011; district geography uses the 35 historical Maharashtra districts. Captions state the population and units. Proposed trial power is a design calculation, not an observed treatment effect.

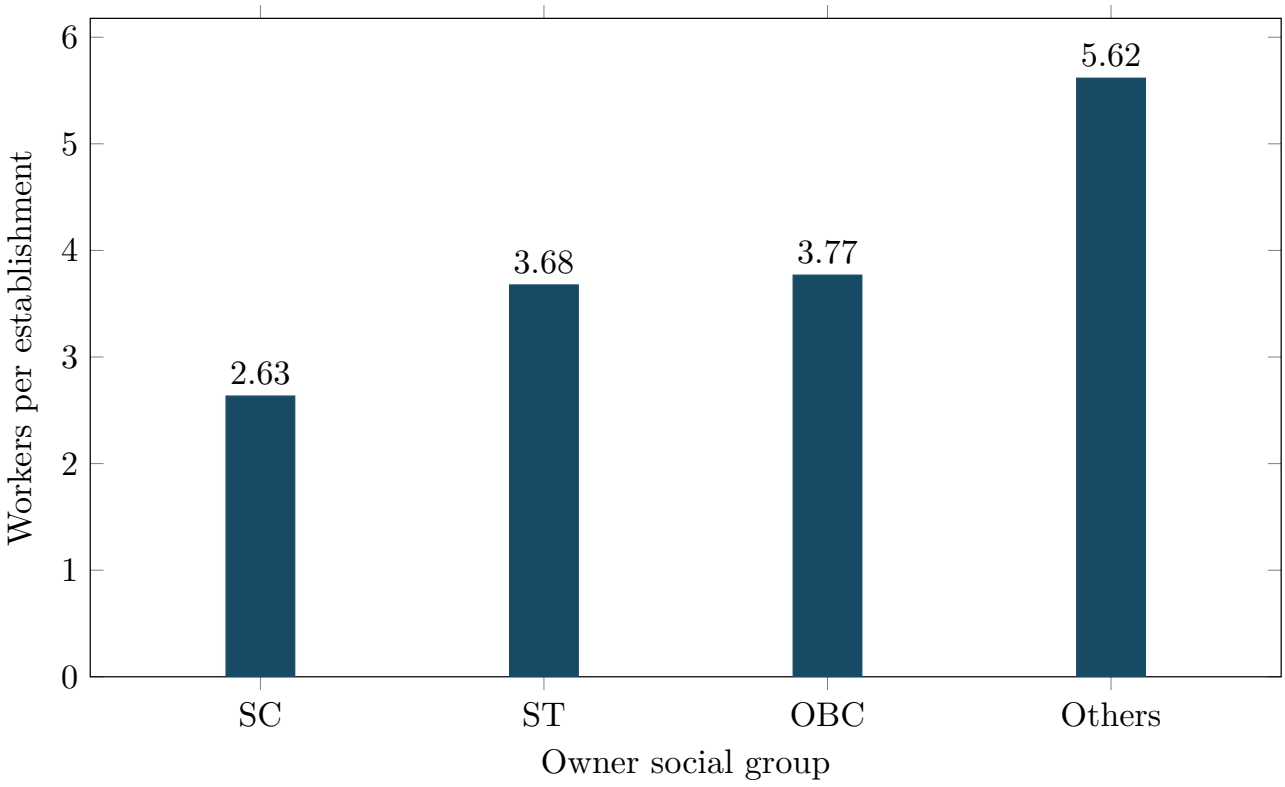


Figure 18.1: Mean employment scale by owner social group. Proprietary mining and quarrying establishments only. Source: EC6 Tables 12 and 15, PDF pages 195 and 198. Ratios of totals, not firm medians. Others is a residual category.

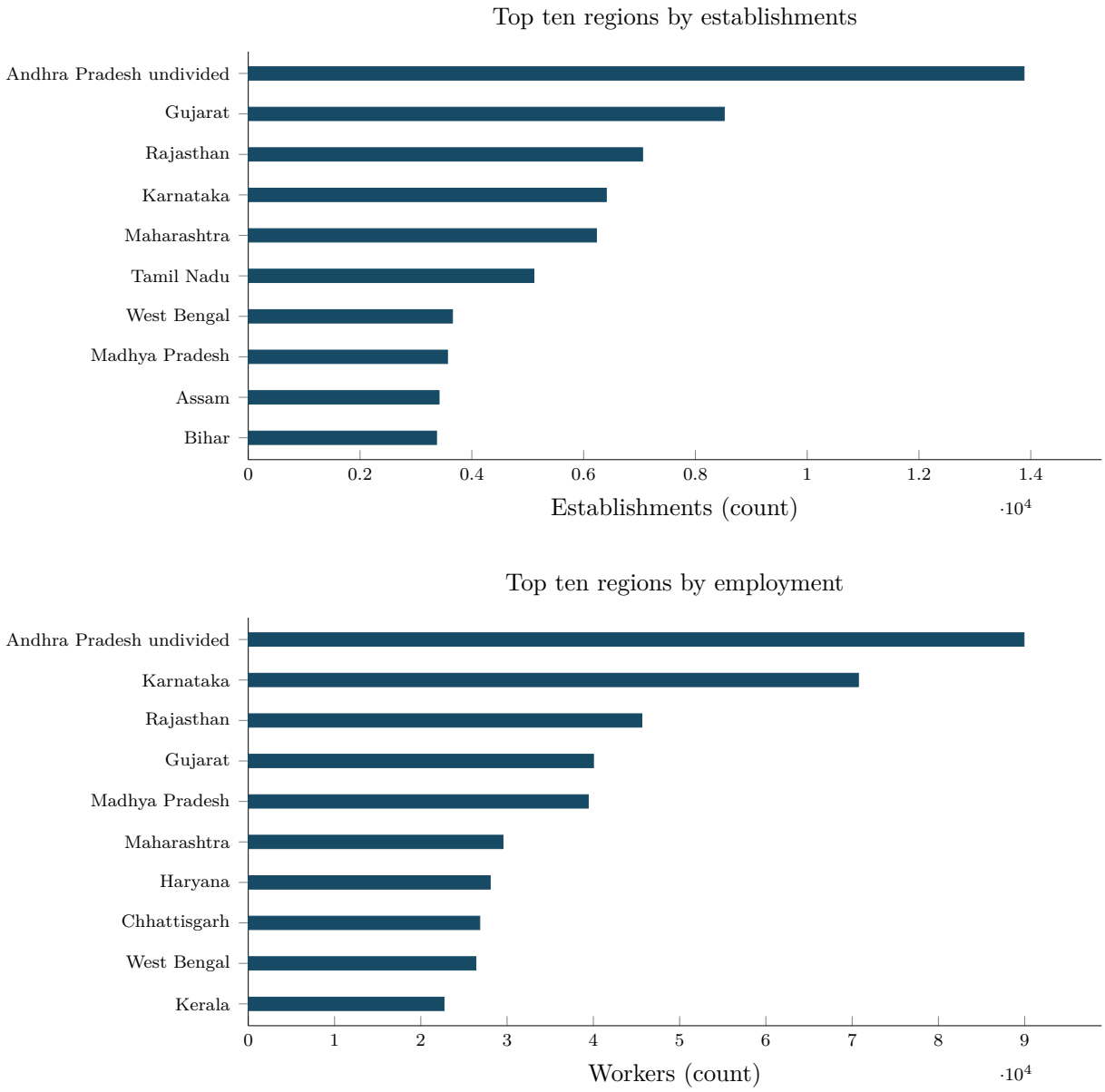


Figure 18.2: Geographic concentration in establishments and workers. Mining and quarrying, all ownership forms. Sources: EC6 Tables 2.5 and 4.5. Undivided Andhra Pradesh combines residual Andhra Pradesh and Telangana for comparability with Census 2011.

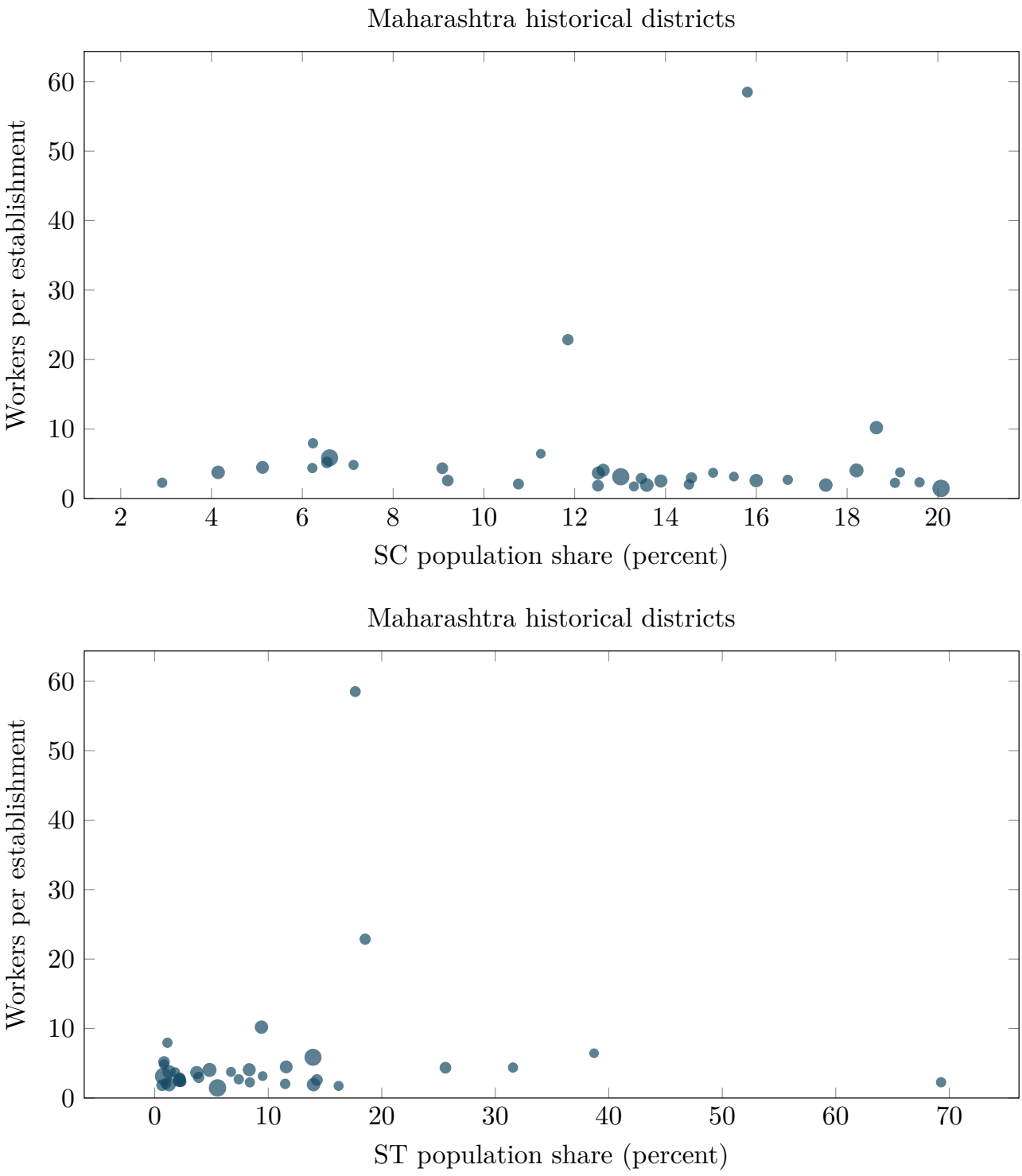


Figure 18.3: District population composition and mining establishment scale. Each point is one of 35 historical Maharashtra districts; marker area increases with establishment count. Sources: Maharashtra EC6 and Census 2011 PCA. Population shares are contextual variables, not owner identities or caste relationships. Chandrapur has mean size 58.5.

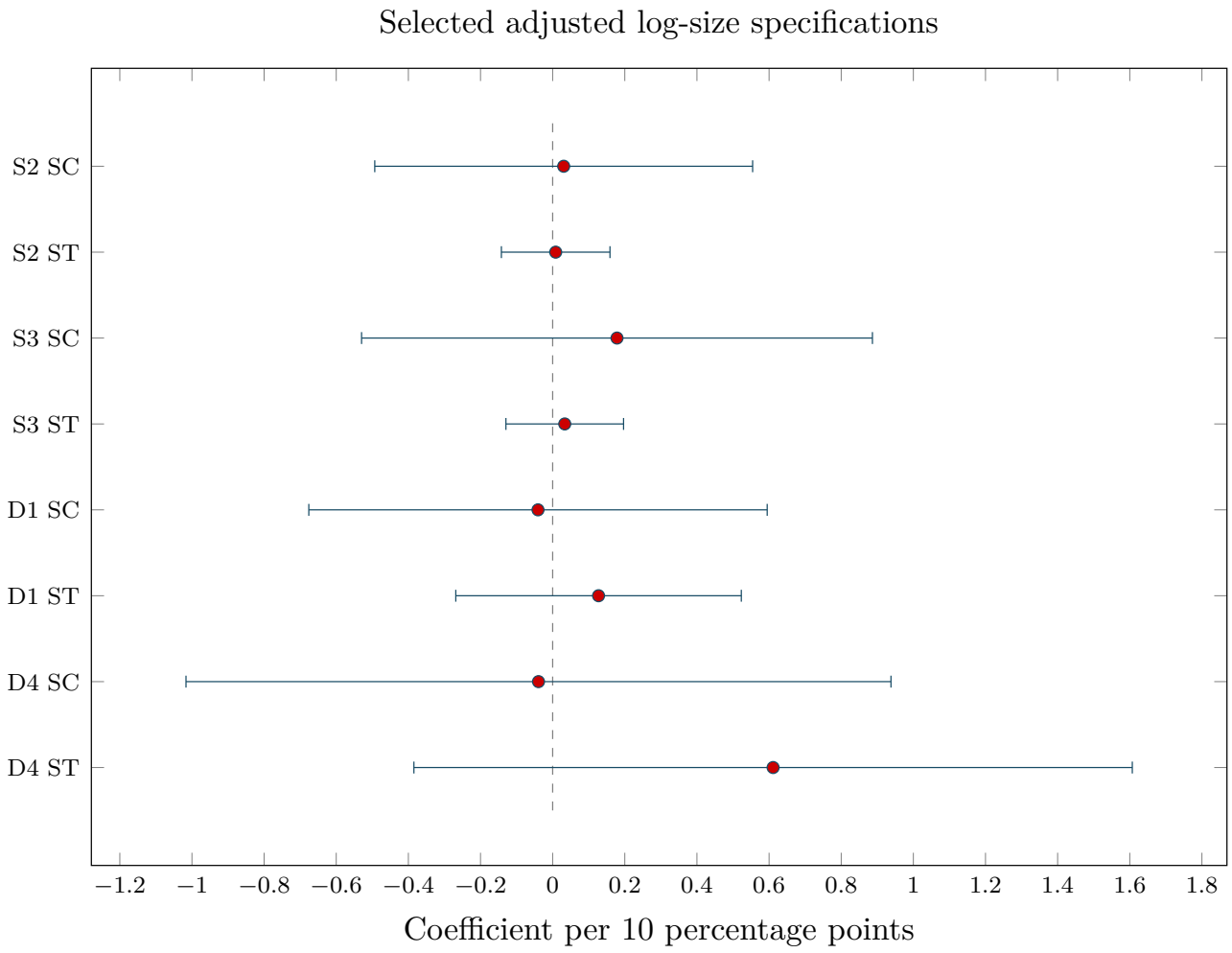


Figure 18.4: Selected SC and ST population-composition coefficients with 95 percent HC3 intervals. S2 all nonempty state/UT regions; S3 excludes 2011 UTs; D1 all Maharashtra districts; D4 districts with at least 50 mining establishments. Outcome is log workers per establishment. Intervals are model-based, not census sampling errors.

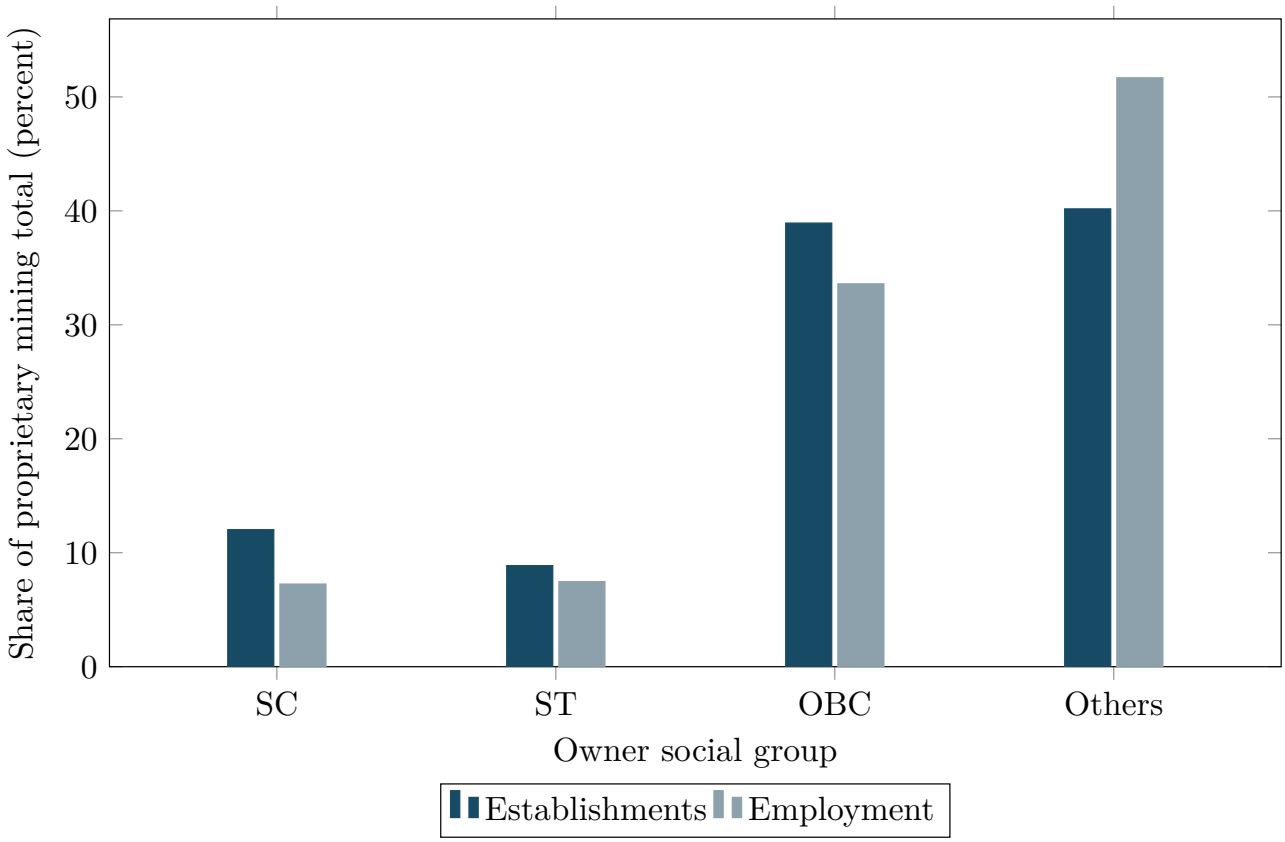


Figure 18.5: Owner-group shares of establishments and employment. Proprietary mining only, recomputed from exact counts. Sources: EC6 Tables 12 and 15. The lower SC employment share relative to establishment share reflects smaller aggregate employment scale.

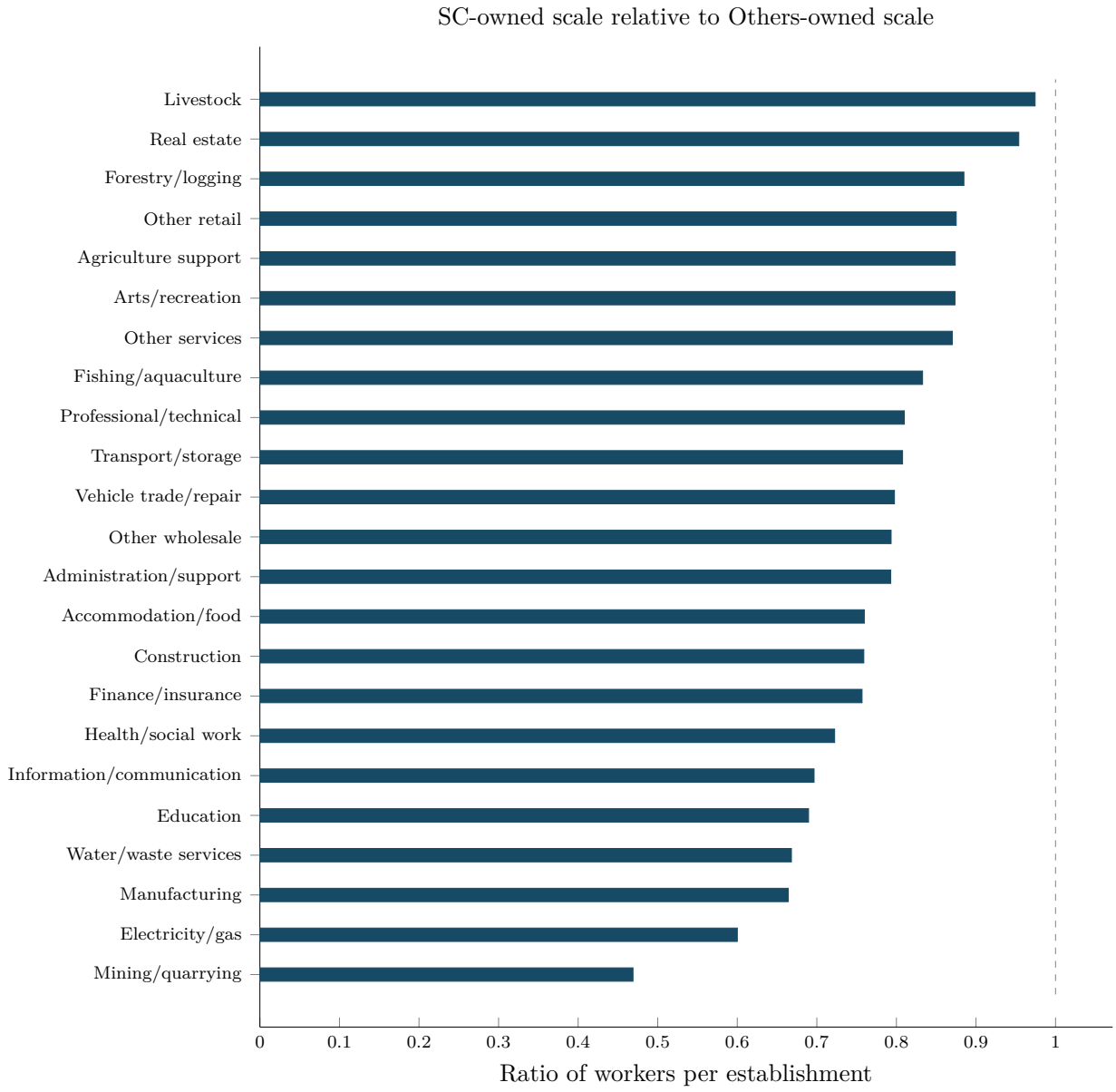


Figure 18.6: Cross-industry SC to Others mean-size ratios for 23 EC6 broad industries. Proprietary establishments only. Sources: Tables 12 and 15. A ratio of one is equal aggregate scale. Sector labels are shortened; full names appear in the complete industry-cell table. These are broad-sector cells, not adjusted firm-level treatment effects; mining is the Mining/quarrying row.

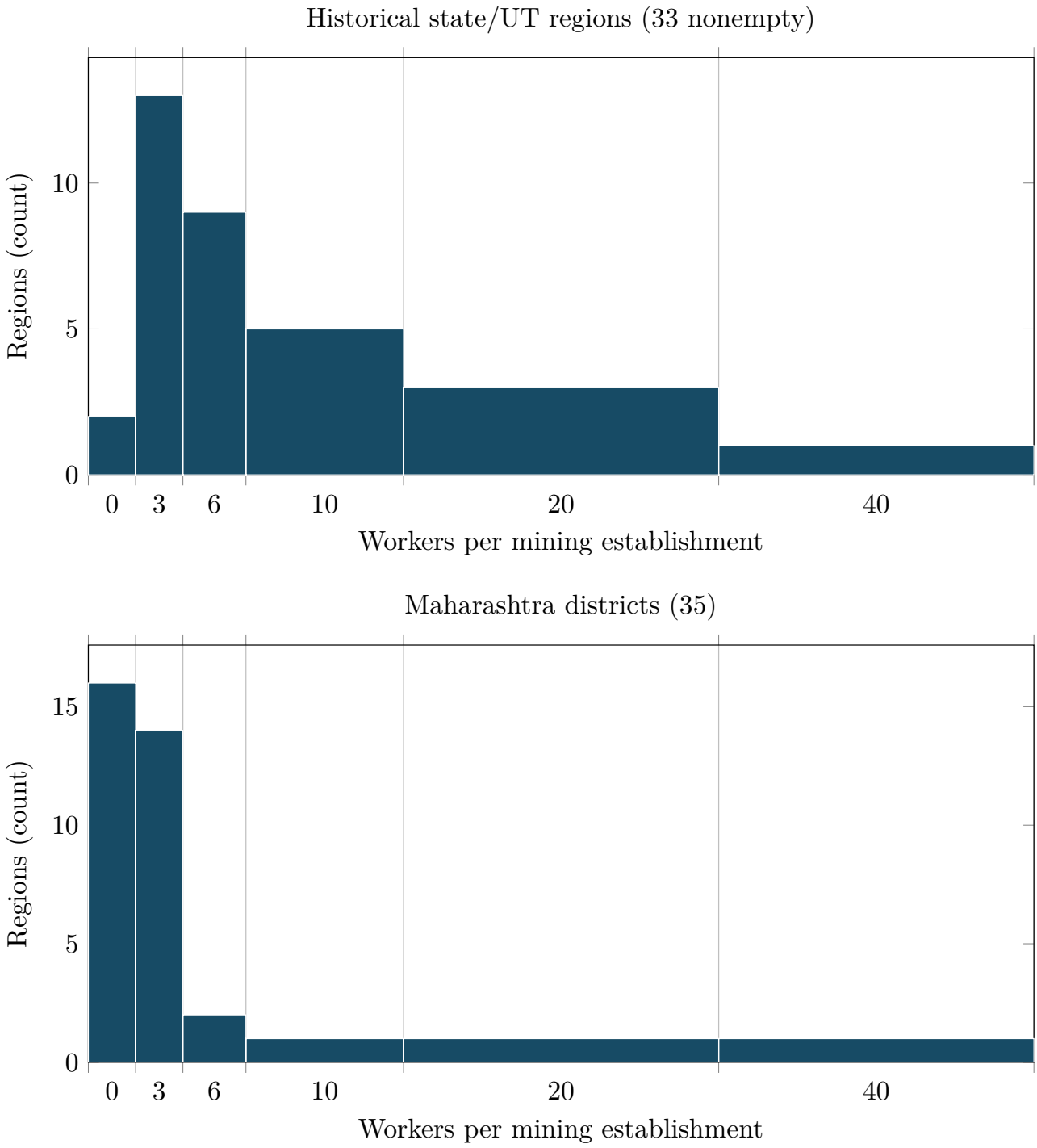


Figure 18.7: Distribution of regional mean sizes. EC6 mining counts and employment. These histograms describe variation in regional ratios, not a size distribution of individual firms. Bin boundaries are 0, 3, 6, 10, 20, 40 and 60 workers per establishment.

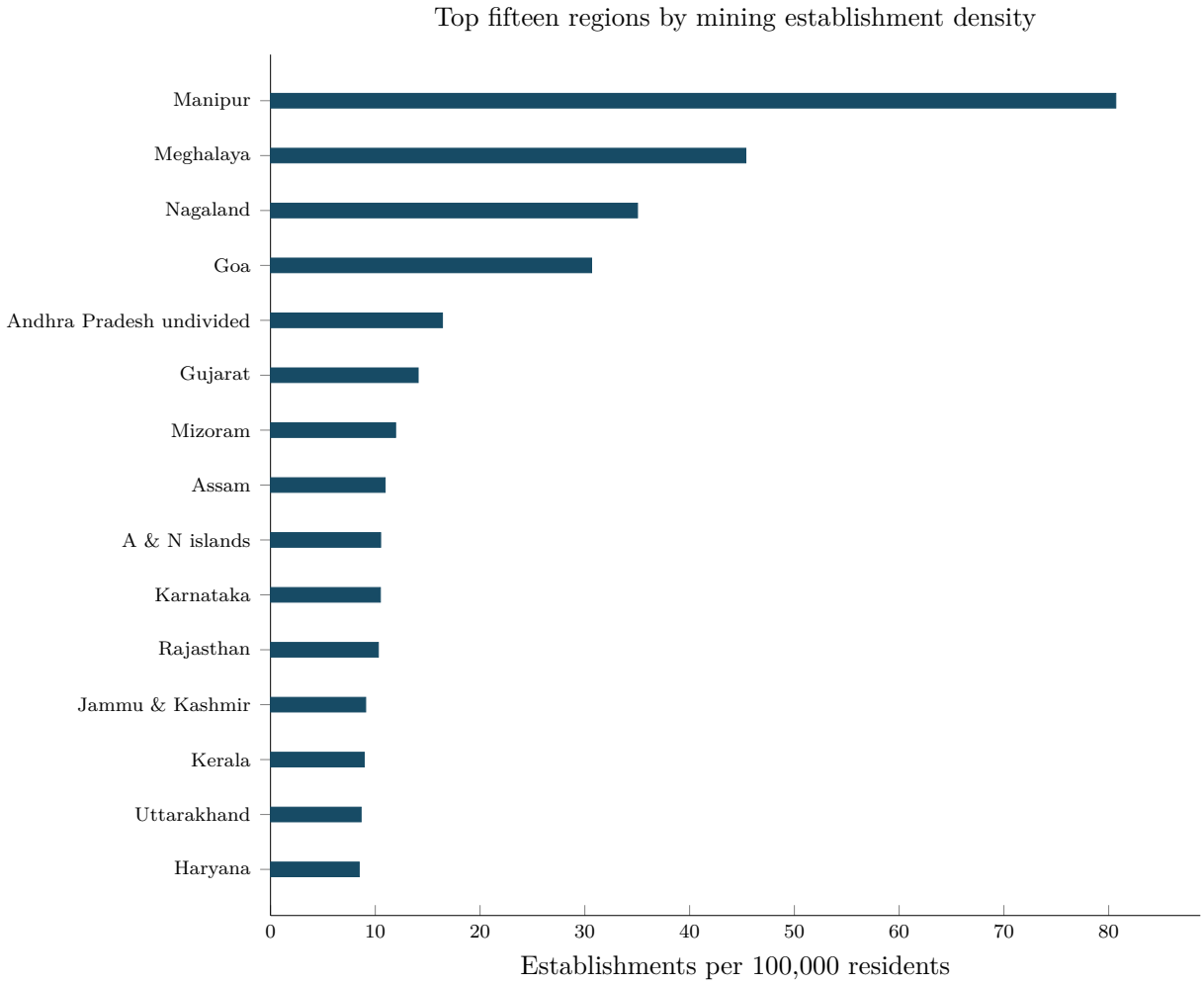


Figure 18.8: Historical mining establishment density. Sources: EC6 establishment counts and Census 2011 total population. Counts include all ownership forms; the population denominator predates EC6. Undivided Andhra Pradesh includes Telangana.

	Size	Hired share	Density	SC share	ST share	Urban	Literacy
Size	1.00	0.30	-0.27	0.01	-0.01	0.23	0.15
Hired share	0.30	1.00	-0.35	-0.24	-0.02	0.24	0.16
Density	-0.27	-0.35	1.00	-0.38	0.31	-0.15	-0.02
SC share	0.01	-0.24	-0.38	1.00	-0.67	-0.12	-0.23
ST share	-0.01	-0.02	0.31	-0.67	1.00	-0.06	0.11
Urban	0.23	0.24	-0.15	-0.12	-0.06	1.00	0.65
Literacy	0.15	0.16	-0.02	-0.23	0.11	0.65	1.00

Figure 18.9: State-level Pearson correlation heatmap. Pairwise-complete observations: 33 for pairs involving size or hired share, 35 otherwise. Blue indicates positive correlation and orange negative correlation; intensity reflects absolute magnitude. Hired share means the share of establishments with hired workers, not the share of workers who are hired. Sources: EC6 and Census 2011.

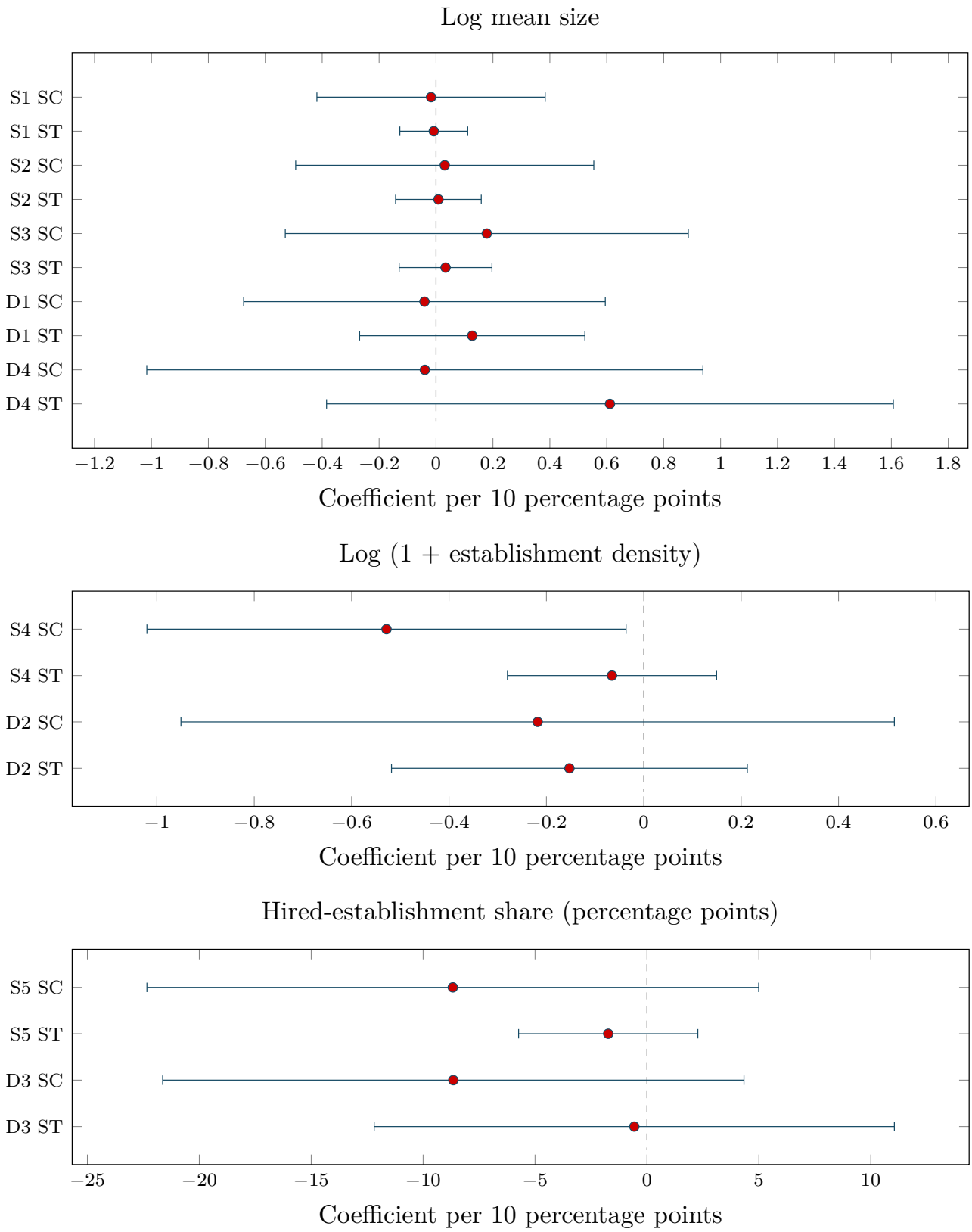


Figure 18.10: All 18 social-composition coefficients in nine models, with 95 percent HC3 intervals. Panels use separate outcome units; compare coefficients within an outcome, not across panels. SC/ST shares are scaled in ten percentage points. None passes the stated Holm or Benjamini–Hochberg adjustment over the 18-coefficient family.

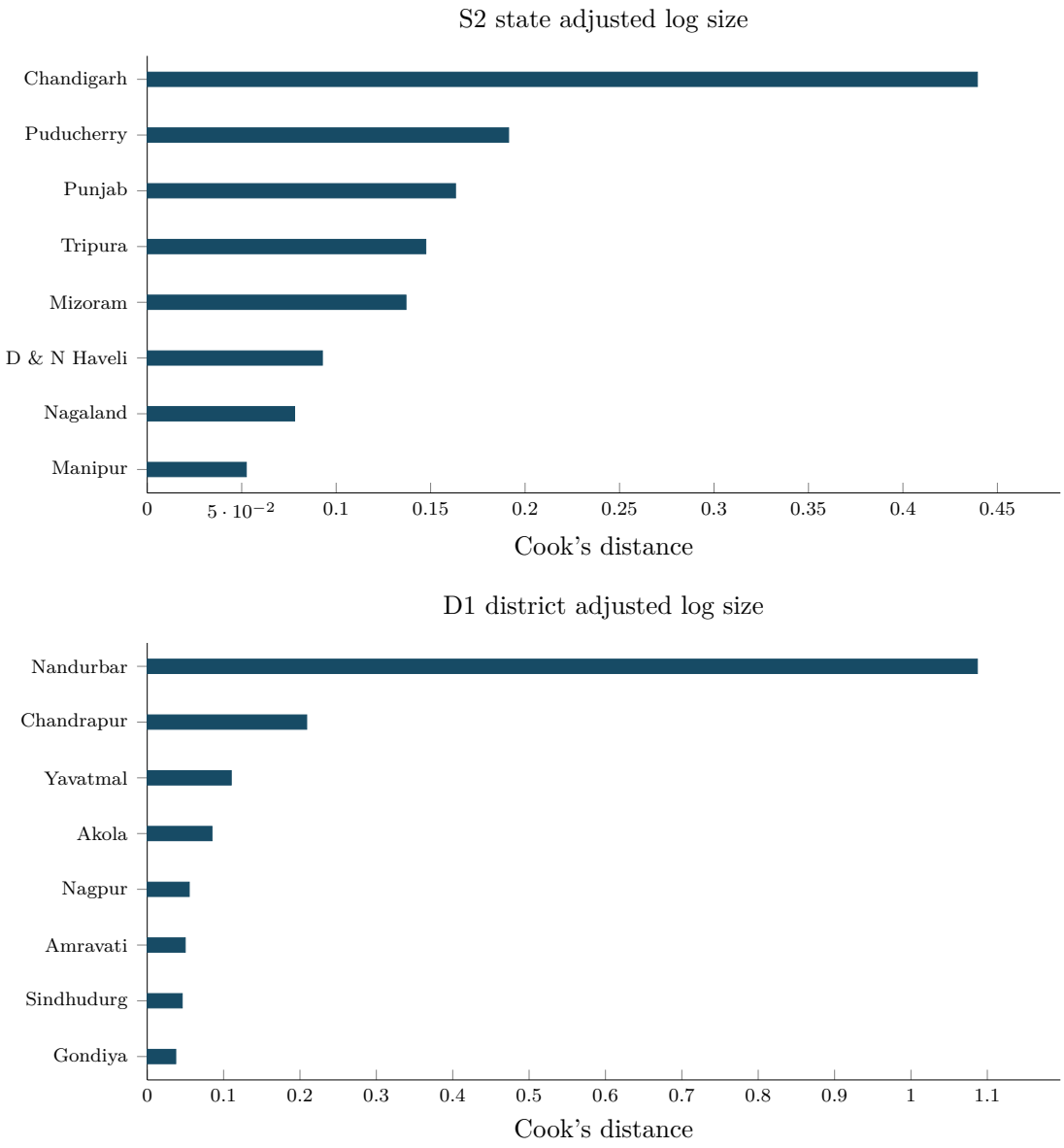


Figure 18.11: The eight most influential regions in each adjusted log-size regression. Cook's distance measures OLS sensitivity, not a probability that an observation is wrong. Full leverage, residual and leave-one-out coefficient ranges are in the statistical appendix and CSV archive.

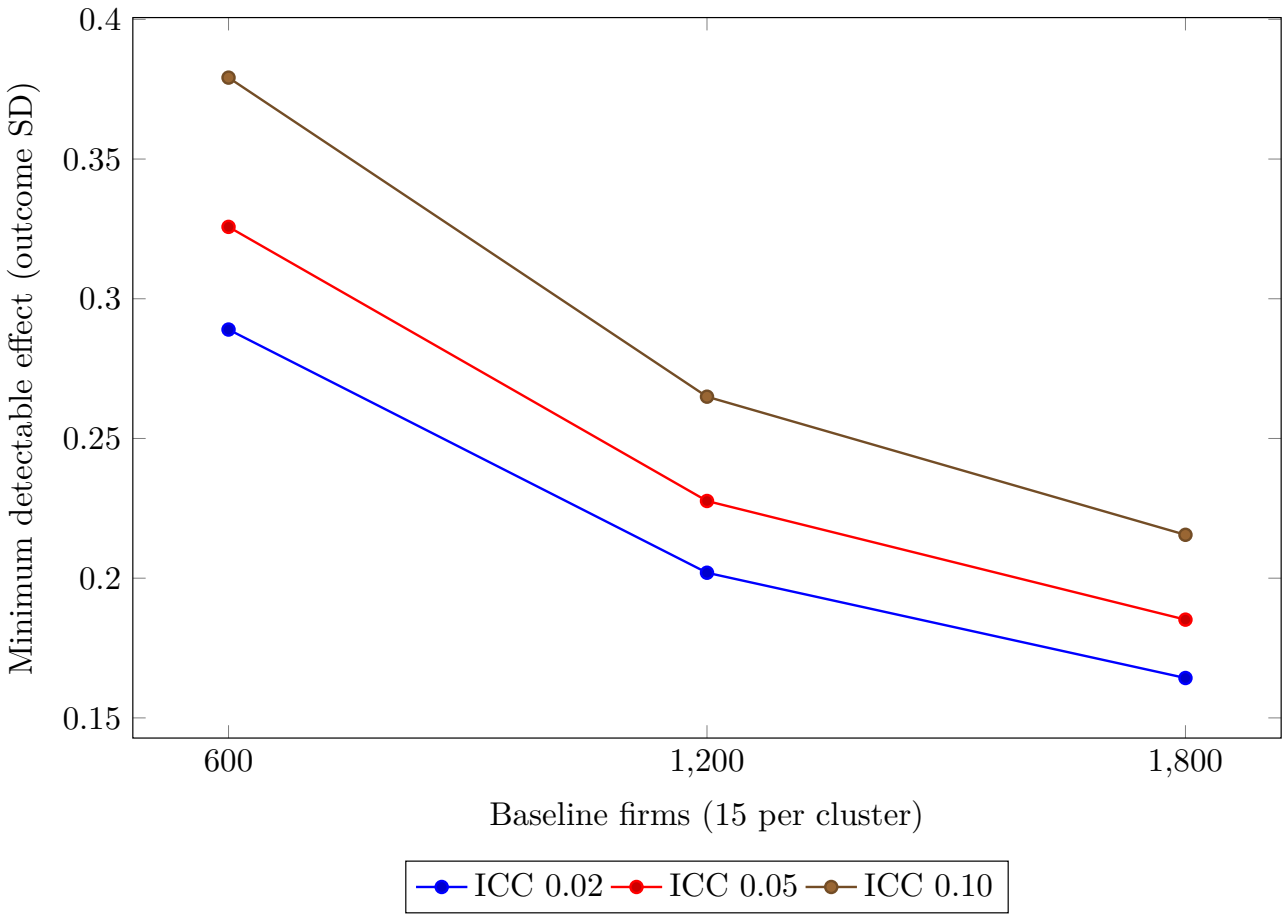


Figure 18.12: Illustrative power for a proposed cluster-randomized introduction intervention. Assumptions: 80 percent power, five percent two-sided test, equal assignment, 15 baseline firms per cluster, 20 percent attrition and the stated ICC. Retained cluster size is 12. No intervention was conducted; these are prospective design assumptions.

Chapter 19

Complete statistical tables

All tables below are produced from the archived CSV files. Totals are published census tabulations; HC3 inference concerns a fitted regional model. Missing size values arise where establishment counts are zero. Rounded displays do not replace full precision in the CSV outputs.

Table 19.1: National proprietary mining ownership and employment scale

Owner group	Establishments	Workers	Mean size	Est. share %	Emp. share %
SC	8,799	23,171	2.63	12.03	7.25
ST	6,489	23,853	3.68	8.87	7.47
OBC	28,485	107,323	3.77	38.93	33.60
Others	29,397	165,078	5.62	40.18	51.68

Table 19.2: Mining establishments and employment by historical state/ UT region

Historical region	Estab.	Workers	Mean size	Hired %	Density
Jammu & Kashmir	1,139	4,783	4.20	69.97	9.08
Himachal Pradesh	238	1,910	8.03	67.23	3.47
Punjab	1,207	2,971	2.46	49.30	4.35
Chandigarh	60	231	3.85	46.67	5.68
Uttarakhand	872	4,345	4.98	70.41	8.65
Haryana	2,146	28,045	13.07	48.56	8.46
Delhi	0	0	–	–	0.00
Rajasthan	7,055	45,628	6.47	66.68	10.29
Uttar Pradesh	3,182	13,959	4.39	52.51	1.59
Bihar	3,368	13,378	3.97	40.32	3.24
Sikkim	14	57	4.07	42.86	2.29
Arunachal Pradesh	46	507	11.02	65.22	3.32
Nagaland	693	2,119	3.06	36.36	35.03
Manipur	2,304	3,795	1.65	6.68	80.68
Mizoram	131	551	4.21	61.07	11.94
Tripura	51	1,089	21.35	31.37	1.39
Meghalaya	1,346	11,179	8.31	72.36	45.37
Assam	3,414	12,118	3.55	28.15	10.94
West Bengal	3,654	26,386	7.22	25.86	4.00
Jharkhand	2,599	18,067	6.95	69.22	7.88
Odisha	3,234	21,451	6.63	39.58	7.70
Chhattisgarh	1,188	26,834	22.59	72.47	4.65
Madhya Pradesh	3,564	39,410	11.06	65.88	4.91
Gujarat	8,516	40,015	4.70	72.49	14.09
Daman & Diu	13	159	12.23	92.31	5.34
D & N Haveli	11	303	27.55	81.82	3.20
Maharashtra	6,229	29,527	4.74	42.05	5.54

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Historical region	Estab.	Workers	Mean size	Hired %	Density
Karnataka	6,406	70,725	11.04	53.15	10.49
Goa	447	2,657	5.94	56.60	30.65
Lakshadweep	0	0	–	–	0.00
Kerala	2,988	22,686	7.59	81.96	8.94
Tamil Nadu	5,110	20,873	4.08	57.30	7.08
Puducherry	31	1,453	46.87	67.74	2.48
A & N islands	40	340	8.50	92.50	10.51
Andhra Pradesh undivided	13,880	89,923	6.48	41.33	16.41

Table 19.3: State demographic controls and population denominators

Historical region	Population	SC %	ST %	Urban %	Literacy %
Jammu & Kashmir	12,541,302	7.38	11.91	27.38	67.16
Himachal Pradesh	6,864,602	25.19	5.71	10.03	82.80
Punjab	27,743,338	31.94	0.00	37.48	75.84
Chandigarh	1,055,450	18.86	0.00	97.25	86.05
Uttarakhand	10,086,292	18.76	2.89	30.23	78.82
Haryana	25,351,462	20.17	0.00	34.88	75.55
Delhi	16,787,941	16.75	0.00	97.50	86.21
Rajasthan	68,548,437	17.83	13.48	24.87	66.11
Uttar Pradesh	199,812,341	20.70	0.57	22.27	67.68
Bihar	104,099,452	15.91	1.28	11.29	61.80
Sikkim	610,577	4.63	33.80	25.15	81.42
Arunachal Pradesh	1,383,727	0.00	68.79	22.94	65.38
Nagaland	1,978,502	0.00	86.48	28.86	79.55
Manipur	2,855,794	3.41	40.88	29.21	76.94
Mizoram	1,097,206	0.11	94.43	52.11	91.33
Tripura	3,673,917	17.83	31.76	26.17	87.22
Meghalaya	2,966,889	0.58	86.15	20.07	74.43
Assam	31,205,576	7.15	12.45	14.10	72.19
West Bengal	91,276,115	23.51	5.80	31.87	76.26
Jharkhand	32,988,134	12.08	26.21	24.05	66.41
Odisha	41,974,218	17.13	22.85	16.69	72.87
Chhattisgarh	25,545,198	12.82	30.62	23.24	70.28
Madhya Pradesh	72,626,809	15.62	21.09	27.63	69.32
Gujarat	60,439,692	6.74	14.75	42.60	78.03
Daman & Diu	243,247	2.52	6.32	75.17	87.10
D & N Haveli	343,709	1.80	51.95	46.72	76.24
Maharashtra	112,374,333	11.81	9.35	45.22	82.34
Karnataka	61,095,297	17.15	6.95	38.67	75.36
Goa	1,458,545	1.74	10.23	62.17	88.70
Lakshadweep	64,473	0.00	94.80	78.07	91.85
Kerala	33,406,061	9.10	1.45	47.70	94.00
Tamil Nadu	72,147,030	20.01	1.10	48.40	80.09
Puducherry	1,247,953	15.73	0.00	68.33	85.85
A & N islands	380,581	0.00	7.50	37.70	86.63
Andhra Pradesh undivided	84,580,777	16.41	7.00	33.36	67.02

Table 19.4: Maharashtra historical district mining establishments and employment

District	Estab.	Workers	Mean size	Hired %	Density
Nandurbar	48	109	2.27	16.67	2.91
Dhule	44	193	4.39	59.09	2.15
Jalgaon	141	366	2.60	24.82	3.33
Buldana	332	1,343	4.05	25.00	12.84
Akola	669	974	1.46	18.09	36.88
Washim	29	109	3.76	55.17	2.42

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District	Estab.	Workers	Mean size	Hired %	Density
Amravati	288	558	1.94	21.18	9.97
Wardha	56	114	2.04	30.36	4.31
Nagpur	278	2,836	10.20	55.40	5.97
Bhandara	46	124	2.70	32.61	3.83
Gondiya	31	54	1.74	9.68	2.34
Gadchiroli	9	58	6.44	55.56	0.84
Chandrapur	88	5,148	58.50	62.50	3.99
Yavatmal	112	2,561	22.87	68.75	4.04
Nanded	43	97	2.26	30.23	1.28
Hingoli	13	41	3.15	38.46	1.10
Parbhani	114	328	2.88	45.61	6.21
Jalna	253	638	2.52	29.25	12.91
Aurangabad	109	324	2.97	44.04	2.94
Nashik	144	628	4.36	56.94	2.36
Thane	628	3,682	5.86	81.53	5.68
Mumbai Suburban	46	366	7.96	56.52	0.49
Mumbai	36	174	4.83	80.56	1.17
Raigarh	230	1,031	4.48	50.00	8.73
Pune	254	932	3.67	46.46	2.69
Ahmadnagar	242	986	4.07	57.85	5.33
Bid	296	574	1.94	33.78	11.45
Latur	45	105	2.33	31.11	1.83
Osmanabad	275	710	2.58	43.64	16.59
Solapur	27	100	3.70	44.44	0.63
Satara	90	188	2.09	14.44	3.00
Ratnagiri	288	1,085	3.77	43.40	17.83
Sindhudurg	134	696	5.19	54.48	15.77
Kolhapur	649	2,033	3.13	36.36	16.74
Sangli	142	262	1.85	25.35	5.03

Table 19.5: Maharashtra historical district demographic controls

District	Population	SC %	ST %	Urban %	Literacy %
Nandurbar	1,648,295	2.91	69.28	16.71	64.38
Dhule	2,050,862	6.22	31.56	27.84	72.80
Jalgaon	4,229,917	9.20	14.29	31.74	78.20
Buldana	2,586,258	18.21	4.83	21.22	83.40
Akola	1,813,906	20.07	5.53	39.68	88.05
Washim	1,197,160	19.17	6.72	17.66	83.25
Amravati	2,888,445	17.53	13.99	35.91	87.38
Wardha	1,300,774	14.52	11.49	32.54	86.99
Nagpur	4,653,570	18.65	9.40	68.31	88.39
Bhandara	1,200,334	16.69	7.41	19.48	83.76
Gondiya	1,322,507	13.31	16.20	17.08	84.95
Gadchiroli	1,072,942	11.25	38.71	11.00	74.36
Chandrapur	2,204,307	15.80	17.67	35.18	80.01
Yavatmal	2,772,348	11.85	18.54	21.58	82.82
Nanded	3,361,292	19.05	8.38	27.19	75.45
Hingoli	1,177,345	15.51	9.51	15.18	78.17
Parbhani	1,836,086	13.47	2.21	31.03	73.34
Jalna	1,959,046	13.90	2.16	19.27	71.52
Aurangabad	3,701,282	14.57	3.87	43.77	79.02
Nashik	6,107,187	9.08	25.62	42.53	82.31
Thane	11,060,148	6.60	13.95	76.99	84.53
Mumbai Suburban	9,356,962	6.23	1.12	100.00	89.91
Mumbai	3,085,411	7.13	0.81	100.00	89.21
Raigarh	2,634,200	5.12	11.58	36.83	83.14
Pune	9,429,408	12.52	3.70	60.99	86.15
Ahmadnagar	4,543,159	12.63	8.33	20.09	79.05
Bid	2,585,049	13.59	1.27	19.90	76.99

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District	Population	SC %	ST %	Urban %	Literacy %
Latur	2,454,196	19.60	2.34	25.47	77.26
Osmanabad	1,657,576	16.00	2.17	16.96	78.44
Solapur	4,317,756	15.05	1.80	32.40	77.02
Satara	3,003,741	10.76	0.99	18.99	82.87
Ratnagiri	1,615,069	4.15	1.26	16.33	82.18
Sindhudurg	849,651	6.54	0.82	12.59	85.56
Kolhapur	3,876,001	13.01	0.78	31.73	81.51
Sangli	2,822,143	12.51	0.65	25.49	81.48

Table 19.6: State/UT regions: descriptive moments and quartiles

Variable	N	Missing	Mean	SD	Min	Median	Max
Establishments	35	0	2,433.60	3,050.32	0.00	1,207.00	13,880.00
Workers	35	0	15,927.83	20,994.82	0.00	4,783.00	89,923.00
Mean size	33	2	9.18	8.97	1.65	6.48	46.87
Hired estab. share %	33	2	56.61	19.66	6.68	57.30	92.50
Estab. per 100k	35	0	11.13	15.50	0.00	7.08	80.68
SC population %	35	0	11.75	8.68	0.00	12.82	31.94
ST population %	35	0	23.10	29.27	0.00	10.23	94.80
Urban population %	35	0	38.84	22.22	10.03	31.87	97.50
Literacy %	35	0	77.85	8.59	61.80	76.94	94.00

Table 19.7: State/UT regions: distribution percentiles

Variable	P10	P25	P50	P75	P90
Establishments	13.40	55.50	1,207.00	3,391.00	6,335.20
Workers	187.80	820.00	4,783.00	24,536.00	39,773.00
Mean size	3.61	4.20	6.48	11.02	19.70
Hired estab. share %	32.37	42.05	57.30	69.97	79.95
Estab. per 100k	1.87	3.40	7.08	10.50	24.95
SC population %	0.04	2.96	12.82	17.83	20.49
ST population %	0.00	2.17	10.23	31.19	79.20
Urban population %	18.04	24.46	31.87	47.21	72.44
Literacy %	66.65	71.23	76.94	85.95	88.11

Table 19.8: Maharashtra districts: descriptive moments and quartiles

Variable	N	Missing	Mean	SD	Min	Median	Max
Establishments	35	0	177.97	176.80	9.00	114.00	669.00
Workers	35	0	843.63	1,136.18	41.00	366.00	5,148.00
Mean size	35	0	5.73	9.91	1.46	3.15	58.50
Hired estab. share %	35	0	42.27	18.01	9.68	43.64	81.53
Estab. per 100k	35	0	6.73	7.35	0.49	3.99	36.88
SC population %	35	0	12.64	4.79	2.91	13.31	20.07
ST population %	35	0	10.54	13.69	0.65	6.72	69.28
Urban population %	35	0	33.42	22.41	11.00	27.19	100.00
Literacy %	35	0	80.97	5.64	64.38	82.18	89.91

Table 19.9: Maharashtra districts: distribution percentiles

Variable	P10	P25	P50	P75	P90
Establishments	29.80	45.50	114.00	264.50	317.60
Workers	98.20	119.00	366.00	980.00	2,349.80

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Variable	P10	P25	P50	P75	P90
Mean size	1.94	2.30	3.15	4.43	7.35
Hired estab. share %	19.32	29.74	43.64	55.48	61.14
Estab. per 100k	1.13	2.35	3.99	9.35	16.26
SC population %	6.23	9.14	13.31	15.90	18.89
ST population %	0.89	1.98	6.72	13.97	22.79
Urban population %	16.48	19.13	27.19	36.37	65.38
Literacy %	73.74	77.72	82.18	84.74	87.78

19.1 Regression specifications and all coefficients

S1: state log size, SC/ST shares only. S2: adds urban and literacy shares. S3: S2 excluding historical UTs. S4: state log(1+density), adjusted. S5: state hired-establishment share, adjusted. D1: district log size, adjusted. D2: district log(1+density), adjusted. D3: district hired-establishment share, adjusted. D4: D1 among districts with at least 50 establishments. All share regressors use ten-percentage-point units; const is the intercept.

Table 19.10: Model sample sizes and fit

Model	Outcome	N	R squared
S1	log_size	33	0.0005
S2	log_size	33	0.0200
S3	log_size	28	0.0465
S4	log_density	35	0.2323
S5	hired_est_pct	33	0.1375
D1	log_size	35	0.1155
D2	log_density	35	0.2149
D3	hired_est_pct	35	0.2945
D4	log_size	23	0.2612

Table 19.11: All fitted regression coefficients and HC3 inference

Model	Term	Beta	HC3 SE	t-based p	CI low	CI high
S1	const	1.9681	0.3250	0.0000	1.3044	2.6318
S1	sc_10pp	-0.0175	0.1964	0.9296	-0.4186	0.3836
S1	st_10pp	-0.0079	0.0584	0.8939	-0.1271	0.1114
S2	const	1.5438	1.5336	0.3227	-1.5977	4.6852
S2	sc_10pp	0.0306	0.2557	0.9056	-0.4932	0.5544
S2	st_10pp	0.0084	0.0735	0.9099	-0.1421	0.1589
S2	urban_10pp	0.0501	0.1515	0.7432	-0.2602	0.3604
S2	literacy_10pp	0.0197	0.2080	0.9253	-0.4063	0.4457
S3	const	1.3326	1.4368	0.3633	-1.6396	4.3048
S3	sc_10pp	0.1784	0.3423	0.6073	-0.5297	0.8865
S3	st_10pp	0.0335	0.0788	0.6746	-0.1296	0.1966
S3	urban_10pp	-0.0459	0.1389	0.7437	-0.3332	0.2413
S3	literacy_10pp	0.0407	0.2056	0.8447	-0.3845	0.4660
S4	const	3.2009	1.8138	0.0878	-0.5034	6.9052
S4	sc_10pp	-0.5284	0.2409	0.0362	-1.0204	-0.0364
S4	st_10pp	-0.0652	0.1052	0.5397	-0.2800	0.1495
S4	urban_10pp	-0.1592	0.1273	0.2207	-0.4191	0.1008
S4	literacy_10pp	0.0290	0.2402	0.9047	-0.4615	0.5195
S5	const	67.2899	39.5409	0.0999	-13.7059	148.2857
S5	sc_10pp	-8.6736	6.6712	0.2041	-22.3389	4.9918
S5	st_10pp	-1.7298	1.9546	0.3837	-5.7337	2.2741
S5	urban_10pp	1.7362	3.5835	0.6318	-5.6043	9.0768
S5	literacy_10pp	-0.3628	6.1550	0.9534	-12.9708	12.2453
D1	const	-0.5228	2.4547	0.8328	-5.5359	4.4903

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Model	Term	Beta	HC3 SE	t-based p	CI low	CI high
D1	sc_10pp	-0.0406	0.3111	0.8970	-0.6760	0.5948
D1	st_10pp	0.1273	0.1938	0.5164	-0.2685	0.5230
D1	urban_10pp	0.0735	0.0600	0.2305	-0.0491	0.1960
D1	literacy_10pp	0.1864	0.3110	0.5534	-0.4487	0.8215
D2	const	-0.4305	3.2044	0.8940	-6.9748	6.1139
D2	sc_10pp	-0.2179	0.3587	0.5480	-0.9505	0.5146
D2	st_10pp	-0.1528	0.1789	0.4000	-0.5181	0.2126
D2	urban_10pp	-0.1683	0.0745	0.0313	-0.3205	-0.0161
D2	literacy_10pp	0.3886	0.3818	0.3170	-0.3912	1.1684
D3	const	53.7536	62.8033	0.3988	-74.5079	182.0151
D3	sc_10pp	-8.6509	6.3584	0.1838	-21.6366	4.3347
D3	st_10pp	-0.5663	5.6897	0.9214	-12.1863	11.0537
D3	urban_10pp	3.7347	1.7898	0.0455	0.0795	7.3900
D3	literacy_10pp	-1.5360	8.1623	0.8520	-18.2056	15.1336
D4	const	1.8579	3.4415	0.5959	-5.3725	9.0883
D4	sc_10pp	-0.0393	0.4651	0.9337	-1.0165	0.9380
D4	st_10pp	0.6112	0.4740	0.2136	-0.3847	1.6070
D4	urban_10pp	0.0131	0.1148	0.9106	-0.2281	0.2542
D4	literacy_10pp	-0.1172	0.4480	0.7966	-1.0585	0.8241

Table 19.12: Multiple-testing adjustments: 18 SC/ST coefficients

Model	Term	Unadj. p	Holm p	BH q
S1	sc_10pp	0.9296	1.0000	0.9337
S1	st_10pp	0.8939	1.0000	0.9337
S2	sc_10pp	0.9056	1.0000	0.9337
S2	st_10pp	0.9099	1.0000	0.9337
S3	sc_10pp	0.6073	1.0000	0.9337
S3	st_10pp	0.6746	1.0000	0.9337
S4	sc_10pp	0.0362	0.6513	0.6513
S4	st_10pp	0.5397	1.0000	0.9337
S5	sc_10pp	0.2041	1.0000	0.9337
S5	st_10pp	0.3837	1.0000	0.9337
D1	sc_10pp	0.8970	1.0000	0.9337
D1	st_10pp	0.5164	1.0000	0.9337
D2	sc_10pp	0.5480	1.0000	0.9337
D2	st_10pp	0.4000	1.0000	0.9337
D3	sc_10pp	0.1838	1.0000	0.9337
D3	st_10pp	0.9214	1.0000	0.9337
D4	sc_10pp	0.9337	1.0000	0.9337
D4	st_10pp	0.2136	1.0000	0.9337

Table 19.13: Leave-one-region-out coefficient ranges

Model	Term	Minimum beta	Maximum beta
S1	sc_10pp	-0.0875	0.1250
S1	st_10pp	-0.0276	0.0216
S2	sc_10pp	-0.0595	0.1993
S2	st_10pp	-0.0106	0.0518
S3	sc_10pp	0.0940	0.4719
S3	st_10pp	0.0123	0.0829
S4	sc_10pp	-0.6080	-0.4198
S4	st_10pp	-0.1009	0.0146
S5	sc_10pp	-11.1949	-5.2436
S5	st_10pp	-2.5249	-0.7149
D1	sc_10pp	-0.2425	0.0889
D1	st_10pp	0.0666	0.2918
D2	sc_10pp	-0.4500	-0.0806

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Model	Term	Minimum beta	Maximum beta
D2	st_10pp	-0.2875	-0.1022
D3	sc_10pp	-10.7489	-5.5807
D3	st_10pp	-1.7449	4.6616
D4	sc_10pp	-0.3069	0.1442
D4	st_10pp	0.3116	0.8831

Table 19.14: Three highest-influence regions per model

Model	Region	Cook D	Leverage	Residual
D1	Nandurbar	1.0870	0.6108	-0.8497
D1	Yavatmal	0.1100	0.0803	1.7624
D1	Chandrapur	0.2088	0.0676	2.6814
D2	Nandurbar	0.6920	0.6108	0.6960
D2	Mumbai Suburban	0.1590	0.3114	-0.8268
D2	Akola	0.2582	0.1549	1.8334
D3	Nandurbar	2.2666	0.6108	-26.9983
D3	Mumbai Suburban	0.1187	0.3114	-15.3139
D3	Sindhudurg	0.0840	0.2657	14.8687
D4	Nashik	0.2862	0.3665	-1.0060
D4	Chandrapur	0.4377	0.2053	2.0851
D4	Nagpur	0.1955	0.3362	0.9098
S1	Punjab	0.2283	0.2209	-1.0114
S1	Nagaland	0.1276	0.2114	-0.7825
S1	Puducherry	0.1252	0.0506	1.9069
S2	Puducherry	0.1913	0.1345	1.7441
S2	Punjab	0.1632	0.2348	-1.0779
S2	Chandigarh	0.4394	0.4531	-0.9101
S3	Nagaland	0.0943	0.2292	-0.6962
S3	Punjab	0.4928	0.3344	-1.1382
S3	Tripura	0.2138	0.2237	1.0691
S4	Chandigarh	0.1496	0.2897	0.9942
S4	Delhi	0.1434	0.2769	-1.0136
S4	Lakshadweep	0.5124	0.3342	-1.6060
S5	Himachal Pradesh	0.1855	0.2999	24.0368
S5	Chandigarh	0.2584	0.4531	-18.0264
S5	A & N islands	0.1335	0.2605	23.1034

Table 19.15: Complete regional influence diagnostics for all nine models

Model	Region	Cook D	Leverage	Residual
S1	Jammu & Kashmir	0.0119	0.0649	-0.5109
S1	Himachal Pradesh	0.0023	0.1113	0.1631
S1	Punjab	0.2283	0.2209	-1.0114
S1	Chandigarh	0.0129	0.0546	-0.5870
S1	Uttarakhand	0.0038	0.0515	-0.3270
S1	Haryana	0.0170	0.0604	0.6374
S1	Rajasthan	0.0001	0.0456	-0.0595
S1	Uttar Pradesh	0.0090	0.0631	-0.4528
S1	Bihar	0.0102	0.0480	-0.5599
S1	Sikkim	0.0102	0.0531	-0.5294
S1	Arunachal Pradesh	0.0246	0.1291	0.4858
S1	Nagaland	0.1276	0.2114	-0.7825
S1	Manipur	0.0870	0.0611	-1.4310
S1	Mizoram	0.0623	0.2636	-0.4572
S1	Tripura	0.0757	0.0792	1.1493
S1	Meghalaya	0.0098	0.2107	0.2175
S1	Assam	0.0212	0.0655	-0.6790
S1	West Bengal	0.0002	0.0898	0.0546

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Model	Region	Cook D	Leverage	Residual
S1	Jharkhand	0.0000	0.0320	0.0126
S1	Odisha	0.0000	0.0511	-0.0281
S1	Chhattisgarh	0.0369	0.0389	1.1958
S1	Madhya Pradesh	0.0060	0.0392	0.4790
S1	Gujarat	0.0070	0.0633	-0.3974
S1	Daman & Diu	0.0385	0.1520	0.5452
S1	D & N Haveli	0.1132	0.0806	1.3917
S1	Maharashtra	0.0041	0.0417	-0.3840
S1	Karnataka	0.0063	0.0429	0.4690
S1	Goa	0.0038	0.1470	-0.1746
S1	Kerala	0.0003	0.0822	0.0761
S1	Tamil Nadu	0.0112	0.0588	-0.5249
S1	Puducherry	0.1252	0.0506	1.9069
S1	A & N islands	0.0058	0.1946	0.1779
S1	Andhra Pradesh undivided	0.0001	0.0409	-0.0654
S2	Jammu & Kashmir	0.0106	0.1345	-0.4108
S2	Himachal Pradesh	0.0127	0.2999	0.2437
S2	Punjab	0.1632	0.2348	-1.0779
S2	Chandigarh	0.4394	0.4531	-0.9101
S2	Uttarakhand	0.0025	0.0676	-0.3042
S2	Haryana	0.0099	0.0605	0.6412
S2	Rajasthan	0.0000	0.0937	0.0024
S2	Uttar Pradesh	0.0056	0.0945	-0.3738
S2	Bihar	0.0132	0.1688	-0.3925
S2	Sikkim	0.0095	0.1001	-0.4686
S2	Arunachal Pradesh	0.0378	0.2158	0.5548
S2	Nagaland	0.0780	0.2148	-0.7998
S2	Manipur	0.0524	0.0677	-1.3872
S2	Mizoram	0.1370	0.3807	-0.6277
S2	Tripura	0.1475	0.2064	1.1334
S2	Meghalaya	0.0077	0.2146	0.2520
S2	Assam	0.0192	0.1463	-0.5220
S2	West Bengal	0.0001	0.0916	0.0466
S2	Jharkhand	0.0003	0.0861	0.0850
S2	Odisha	0.0001	0.0732	0.0497
S2	Chhattisgarh	0.0351	0.0568	1.2539
S2	Madhya Pradesh	0.0068	0.0629	0.5190
S2	Gujarat	0.0041	0.0646	-0.3965
S2	Daman & Diu	0.0208	0.2242	0.3991
S2	D & N Haveli	0.0927	0.1156	1.3388
S2	Maharashtra	0.0035	0.0509	-0.4203
S2	Karnataka	0.0037	0.0461	0.4574
S2	Goa	0.0063	0.1775	-0.2614
S2	Kerala	0.0001	0.2256	0.0303
S2	Tamil Nadu	0.0112	0.0760	-0.5988
S2	Puducherry	0.1913	0.1345	1.7441
S2	A & N islands	0.0089	0.2605	0.2306
S2	Andhra Pradesh undivided	0.0000	0.0999	-0.0304
S3	Jammu & Kashmir	0.0051	0.1541	-0.2169
S3	Himachal Pradesh	0.0001	0.4142	-0.0097
S3	Punjab	0.4928	0.3344	-1.1382
S3	Uttarakhand	0.0027	0.0721	-0.2531
S3	Haryana	0.0237	0.0755	0.7303
S3	Rajasthan	0.0000	0.1013	0.0160
S3	Uttar Pradesh	0.0094	0.0960	-0.3985
S3	Bihar	0.0289	0.1922	-0.4413
S3	Sikkim	0.0125	0.1533	-0.3406
S3	Arunachal Pradesh	0.0941	0.2376	0.6758
S3	Nagaland	0.0943	0.2292	-0.6962
S3	Manipur	0.0775	0.0874	-1.2106
S3	Mizoram	0.0911	0.4513	-0.3471
S3	Tripura	0.2138	0.2237	1.0691

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Model	Region	Cook D	Leverage	Residual
S3	Meghalaya	0.0162	0.2446	0.2742
S3	Assam	0.0469	0.2456	-0.4643
S3	West Bengal	0.0001	0.1071	0.0413
S3	Jharkhand	0.0011	0.0900	0.1430
S3	Odisha	0.0001	0.0935	-0.0428
S3	Chhattisgarh	0.0552	0.0597	1.2740
S3	Madhya Pradesh	0.0131	0.0701	0.5659
S3	Gujarat	0.0004	0.1161	-0.0771
S3	Maharashtra	0.0013	0.0981	-0.1462
S3	Karnataka	0.0167	0.0760	0.6105
S3	Goa	0.0454	0.3725	0.3087
S3	Kerala	0.0445	0.3141	0.3637
S3	Tamil Nadu	0.0176	0.1617	-0.3899
S3	Andhra Pradesh undivided	0.0009	0.1286	0.1001
S4	Jammu & Kashmir	0.0015	0.1281	-0.1817
S4	Himachal Pradesh	0.0251	0.2829	-0.4163
S4	Punjab	0.0286	0.2237	0.5405
S4	Chandigarh	0.1496	0.2897	0.9942
S4	Uttarakhand	0.0022	0.0660	0.3286
S4	Haryana	0.0034	0.0576	0.4486
S4	Delhi	0.1434	0.2769	-1.0136
S4	Rajasthan	0.0062	0.0921	0.4574
S4	Uttar Pradesh	0.0276	0.0884	-0.9927
S4	Bihar	0.0435	0.1462	-0.9075
S4	Sikkim	0.0575	0.0941	-1.3796
S4	Arunachal Pradesh	0.1062	0.2056	-1.1123
S4	Nagaland	0.0908	0.1711	1.1763
S4	Manipur	0.0704	0.0653	1.8906
S4	Mizoram	0.0393	0.2680	0.5461
S4	Tripura	0.0849	0.1994	-1.0176
S4	Meghalaya	0.1264	0.1812	1.3324
S4	Assam	0.0025	0.1210	-0.2469
S4	West Bengal	0.0000	0.0902	-0.0242
S4	Jharkhand	0.0000	0.0860	-0.0176
S4	Odisha	0.0001	0.0704	0.0713
S4	Chhattisgarh	0.0030	0.0565	-0.4259
S4	Madhya Pradesh	0.0009	0.0624	-0.2230
S4	Gujarat	0.0033	0.0632	0.4174
S4	Daman & Diu	0.0043	0.1927	-0.2350
S4	D & N Haveli	0.0194	0.0927	-0.8089
S4	Maharashtra	0.0003	0.0485	-0.1561
S4	Karnataka	0.0042	0.0426	0.5885
S4	Goa	0.0851	0.1698	1.1452
S4	Lakshadweep	0.5124	0.3342	-1.6060
S4	Kerala	0.0005	0.2252	0.0731
S4	Tamil Nadu	0.0047	0.0650	0.4916
S4	Puducherry	0.0025	0.0979	-0.2827
S4	A & N islands	0.0152	0.2507	-0.3598
S4	Andhra Pradesh undivided	0.0250	0.0949	0.9056
S5	Jammu & Kashmir	0.0073	0.1345	8.8242
S5	Himachal Pradesh	0.1855	0.2999	24.0368
S5	Punjab	0.0075	0.2348	5.9492
S5	Chandigarh	0.2584	0.4531	-18.0264
S5	Uttarakhand	0.0125	0.0676	17.5081
S5	Haryana	0.0007	0.0605	-4.5540
S5	Rajasthan	0.0140	0.0937	15.2619
S5	Uttar Pradesh	0.0002	0.0945	1.8642
S5	Bihar	0.0206	0.1688	-12.6626
S5	Sikkim	0.0166	0.1001	-15.9832
S5	Arunachal Pradesh	0.0124	0.2158	8.2160
S5	Nagaland	0.0598	0.2148	-18.0917
S5	Manipur	0.1141	0.0677	-52.8588

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Model	Region	Cook D	Leverage	Residual
S5	Mizoram	0.0104	0.3807	4.4754
S5	Tripura	0.0459	0.2064	-16.3417
S5	Meghalaya	0.0709	0.2146	19.6970
S5	Assam	0.0988	0.1463	-30.6145
S5	West Bengal	0.0303	0.0916	-22.7959
S5	Jharkhand	0.0125	0.0861	15.1755
S5	Odisha	0.0038	0.0732	-9.1573
S5	Chhattisgarh	0.0135	0.0568	20.1138
S5	Madhya Pradesh	0.0069	0.0629	13.5018
S5	Gujarat	0.0032	0.0646	9.0315
S5	Daman & Diu	0.0662	0.2242	18.4022
S5	D & N Haveli	0.0302	0.1156	19.7292
S5	Maharashtra	0.0099	0.0509	-18.2445
S5	Karnataka	0.0001	0.0461	-2.0426
S5	Goa	0.0309	0.1775	-14.9834
S5	Kerala	0.0636	0.2256	17.9422
S5	Tamil Nadu	0.0002	0.0760	2.0605
S5	Puducherry	0.0027	0.1345	5.3474
S5	A & N islands	0.1335	0.2605	23.1034
S5	Andhra Pradesh undivided	0.0125	0.0999	-13.8837
D1	Nandurbar	1.0870	0.6108	-0.8497
D1	Dhule	0.0003	0.1419	0.0634
D1	Jalgaon	0.0027	0.0499	-0.3587
D1	Buldana	0.0018	0.0817	0.2224
D1	Akola	0.0848	0.1549	-1.0231
D1	Washim	0.0012	0.1038	0.1577
D1	Amravati	0.0498	0.1462	-0.8152
D1	Wardha	0.0230	0.0982	-0.7143
D1	Nagpur	0.0550	0.2141	0.6521
D1	Bhandara	0.0016	0.0777	-0.2164
D1	Gondiya	0.0373	0.1249	-0.7833
D1	Gadchiroli	0.0203	0.1685	0.4723
D1	Chandrapur	0.2088	0.0676	2.6814
D1	Yavatmal	0.1100	0.0803	1.7624
D1	Nanded	0.0063	0.1404	-0.2991
D1	Hingoli	0.0000	0.0556	0.0449
D1	Parbhani	0.0000	0.1709	0.0113
D1	Jalna	0.0000	0.2009	0.0022
D1	Aurangabad	0.0010	0.0771	-0.1724
D1	Nashik	0.0008	0.0915	-0.1402
D1	Thane	0.0000	0.1650	-0.0003
D1	Mumbai Suburban	0.0096	0.3114	0.1975
D1	Mumbai	0.0184	0.3035	-0.2804
D1	Raigarh	0.0004	0.1243	0.0761
D1	Pune	0.0017	0.0770	-0.2273
D1	Ahmadnagar	0.0011	0.0441	0.2518
D1	Bid	0.0049	0.0865	-0.3571
D1	Latur	0.0027	0.1282	-0.2074
D1	Osmanabad	0.0002	0.0716	-0.0781
D1	Solapur	0.0016	0.0933	0.1968
D1	Satara	0.0069	0.0977	-0.3935
D1	Ratnagiri	0.0074	0.2688	0.1982
D1	Sindhudurg	0.0458	0.2657	0.4990
D1	Kolhapur	0.0000	0.0481	-0.0448
D1	Sangli	0.0068	0.0579	-0.5281
D2	Nandurbar	0.6920	0.6108	0.6960
D2	Dhule	0.0019	0.1419	-0.1659
D2	Jalgaon	0.0007	0.0499	-0.1887
D2	Buldana	0.0143	0.0817	0.6451
D2	Akola	0.2582	0.1549	1.8334
D2	Washim	0.0262	0.1038	-0.7562
D2	Amravati	0.0283	0.1462	0.6307

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Model	Region	Cook D	Leverage	Residual
D2	Wardha	0.0025	0.0982	-0.2414
D2	Nagpur	0.0500	0.2141	0.6381
D2	Bhandara	0.0064	0.0777	-0.4441
D2	Gondiya	0.0406	0.1249	-0.8383
D2	Gadchiroli	0.0592	0.1685	-0.8279
D2	Chandrapur	0.0005	0.0676	0.1358
D2	Yavatmal	0.0024	0.0803	-0.2655
D2	Nanded	0.0308	0.1404	-0.6765
D2	Hingoli	0.0279	0.0556	-1.1241
D2	Parbhani	0.0145	0.1709	0.4058
D2	Jalna	0.0994	0.2009	0.9447
D2	Aurangabad	0.0008	0.0771	-0.1539
D2	Nashik	0.0025	0.0915	-0.2514
D2	Thane	0.0408	0.1650	0.6976
D2	Mumbai Suburban	0.1590	0.3114	-0.8268
D2	Mumbai	0.0375	0.3035	-0.4115
D2	Raigarh	0.0085	0.1243	0.3838
D2	Pune	0.0021	0.0770	-0.2544
D2	Ahmadnagar	0.0001	0.0441	-0.0559
D2	Bid	0.0137	0.0865	0.6111
D2	Latur	0.0244	0.1282	-0.6387
D2	Osmanabad	0.0247	0.0716	0.9171
D2	Solapur	0.0556	0.0933	-1.1755
D2	Satara	0.0296	0.0977	-0.8349
D2	Ratnagiri	0.0553	0.2688	0.5574
D2	Sindhudurg	0.0149	0.2657	0.2926
D2	Kolhapur	0.0177	0.0481	0.9692
D2	Sangli	0.0012	0.0579	-0.2268
D3	Nandurbar	2.2666	0.6108	-26.9983
D3	Dhule	0.0262	0.1419	13.2890
D3	Jalgaon	0.0171	0.0499	-20.0039
D3	Buldana	0.0046	0.0817	-7.8454
D3	Akola	0.0622	0.1549	-19.2859
D3	Washim	0.0601	0.1038	24.5722
D3	Amravati	0.0426	0.1462	-16.6052
D3	Wardha	0.0075	0.0982	-8.9788
D3	Nagpur	0.0108	0.2141	6.3701
D3	Bhandara	0.0000	0.0777	-0.6944
D3	Gondiya	0.0784	0.1249	-24.9805
D3	Gadchiroli	0.0832	0.1685	21.0418
D3	Chandrapur	0.0306	0.0676	22.5711
D3	Yavatmal	0.0701	0.0803	30.9605
D3	Nanded	0.0039	0.1404	-5.1274
D3	Hingoli	0.0012	0.0556	4.9973
D3	Parbhani	0.0021	0.1709	3.3116
D3	Jalna	0.0178	0.2009	-8.5702
D3	Aurangabad	0.0001	0.0771	-1.1019
D3	Nashik	0.0073	0.0915	9.2576
D3	Thane	0.0625	0.1650	18.5068
D3	Mumbai Suburban	0.1187	0.3114	-15.3139
D3	Mumbai	0.0423	0.3035	9.3694
D3	Raigarh	0.0000	0.1243	0.3492
D3	Pune	0.0023	0.0770	-5.8012
D3	Ahmadnagar	0.0151	0.0441	20.1331
D3	Bid	0.0011	0.0865	-3.7484
D3	Latur	0.0013	0.1282	-3.2010
D3	Osmanabad	0.0059	0.0716	9.5621
D3	Solapur	0.0011	0.0933	3.5385
D3	Satara	0.0547	0.0977	-24.3076
D3	Ratnagiri	0.0000	0.2688	-0.1695
D3	Sindhudurg	0.0840	0.2657	14.8687
D3	Kolhapur	0.0012	0.0481	-5.4194

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Model	Region	Cook D	Leverage	Residual
D3	Sangli	0.0106	0.0579	-14.5460
D4	Jalgaon	0.0500	0.1539	-0.8661
D4	Buldana	0.0052	0.1666	0.2658
D4	Akola	0.0852	0.2601	-0.7613
D4	Amravati	0.1035	0.2076	-1.0056
D4	Wardha	0.0379	0.1371	-0.8155
D4	Nagpur	0.1955	0.3362	0.9098
D4	Chandrapur	0.4377	0.2053	2.0851
D4	Yavatmal	0.1618	0.2384	1.1274
D4	Parbhani	0.0006	0.2585	-0.0641
D4	Jalna	0.0078	0.3089	-0.1973
D4	Aurangabad	0.0004	0.1389	-0.0791
D4	Nashik	0.2862	0.3665	-1.0060
D4	Thane	0.0004	0.4808	-0.0256
D4	Raigarh	0.0013	0.1956	-0.1193
D4	Pune	0.0053	0.2506	0.1951
D4	Ahmadnagar	0.0000	0.0855	-0.0122
D4	Bid	0.0058	0.1232	-0.3434
D4	Osmanabad	0.0004	0.1331	-0.0823
D4	Satara	0.0020	0.1302	-0.1930
D4	Ratnagiri	0.0239	0.3055	0.3494
D4	Sindhudurg	0.1345	0.3375	0.7514
D4	Kolhapur	0.0014	0.0899	0.2012
D4	Sangli	0.0033	0.0902	-0.3144

Table 19.16: Descriptive industry fixed-effect owner-group contrasts

Weighting	Group vs Others	Log contrast	Percent contrast
equal industry cells	OBC	-0.1818	-16.63
equal industry cells	SC	-0.2610	-22.97
equal industry cells	ST	-0.1663	-15.32
establishment weighted cells	OBC	-0.1445	-13.45
establishment weighted cells	SC	-0.1925	-17.51
establishment weighted cells	ST	-0.1439	-13.41

These fixed-effect contrasts summarize 92 aggregate industry by owner-group cells. They do not supply microdata sampling errors or causal firm-level effects. Establishment weighting changes the descriptive target.

Table 19.17: Complete proprietary establishment and employment cells: 23 industries by four groups

Industry	Group	Estab.	Workers	Mean size	vs Others
Activities relating to agriculture other than crop production & plantation	SC	63,933	119,425	1.87	0.874
Activities relating to agriculture other than crop production & plantation	ST	39,003	81,085	2.08	0.973
Activities relating to agriculture other than crop production & plantation	OBC	291,719	575,993	1.97	0.924
Activities relating to agriculture other than crop production & plantation	Others	216,720	463,214	2.14	1.000
Livestock	SC	1,340,188	2,205,371	1.65	0.974
Livestock	ST	708,291	1,220,681	1.72	1.020
Livestock	OBC	5,212,297	8,898,499	1.71	1.011

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Industry	Group	Estab.	Workers	Mean size	vs Others
Livestock	Others	3,930,413	6,638,214	1.69	1.000
Forestry and Logging	SC	62,259	117,292	1.88	0.885
Forestry and Logging	ST	320,166	635,496	1.98	0.932
Forestry and Logging	OBC	113,199	226,448	2.00	0.940
Forestry and Logging	Others	72,394	154,140	2.13	1.000
Fishing and aqua culture	SC	95,747	141,497	1.48	0.833
Fishing and aqua culture	ST	33,320	52,149	1.57	0.882
Fishing and aqua culture	OBC	238,416	432,923	1.82	1.023
Fishing and aqua culture	Others	115,443	204,834	1.77	1.000
Mining and quarrying	SC	8,799	23,171	2.63	0.469
Mining and quarrying	ST	6,489	23,853	3.68	0.655
Mining and quarrying	OBC	28,485	107,323	3.77	0.671
Mining and quarrying	Others	29,397	165,078	5.62	1.000
Manufacturing	SC	1,125,759	2,316,709	2.06	0.664
Manufacturing	ST	466,592	957,450	2.05	0.662
Manufacturing	OBC	4,415,553	9,482,534	2.15	0.693
Manufacturing	Others	3,766,411	11,670,630	3.10	1.000
Electricity, gas, steam and air conditioning supply	SC	2,864	8,398	2.93	0.600
Electricity, gas, steam and air conditioning supply	ST	945	4,929	5.22	1.067
Electricity, gas, steam and air conditioning supply	OBC	8,942	28,170	3.15	0.645
Electricity, gas, steam and air conditioning supply	Others	18,479	90,300	4.89	1.000
Water supply, sewerage, waste management and remediation activities	SC	20,109	31,526	1.57	0.668
Water supply, sewerage, waste management and remediation activities	ST	3,898	7,332	1.88	0.802
Water supply, sewerage, waste management and remediation activities	OBC	44,389	77,678	1.75	0.746
Water supply, sewerage, waste management and remediation activities	Others	60,989	143,126	2.35	1.000
Construction	SC	183,022	317,974	1.74	0.759
Construction	ST	47,920	91,583	1.91	0.835
Construction	OBC	310,930	711,605	2.29	1.000
Construction	Others	391,744	896,657	2.29	1.000
Whole sale trade, retail trade & repair of motor vehicles & motor cycles	SC	80,494	171,289	2.13	0.798
Whole sale trade, retail trade & repair of motor vehicles & motor cycles	ST	29,518	69,231	2.35	0.879
Whole sale trade, retail trade & repair of motor vehicles & motor cycles	OBC	377,779	812,902	2.15	0.807
Whole sale trade, retail trade & repair of motor vehicles & motor cycles	Others	444,683	1,186,442	2.67	1.000
Whole sale trade (not covered in item-10 above)	SC	67,109	138,208	2.06	0.793
Whole sale trade (not covered in item-10 above)	ST	32,099	66,614	2.08	0.799
Whole sale trade (not covered in item-10 above)	OBC	253,983	575,103	2.26	0.872
Whole sale trade (not covered in item-10 above)	Others	505,034	1,311,017	2.60	1.000

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Industry	Group	Estab.	Workers	Mean size	vs Others
Retail trade (not covered in item-10 above)	SC	1,642,154	2,526,494	1.54	0.875
Retail trade (not covered in item-10 above)	ST	693,525	1,118,225	1.61	0.917
Retail trade (not covered in item-10 above)	OBC	5,779,080	9,162,114	1.59	0.902
Retail trade (not covered in item-10 above)	Others	6,977,154	12,265,443	1.76	1.000
Transportation and storage	SC	388,181	507,413	1.31	0.808
Transportation and storage	ST	135,410	183,258	1.35	0.836
Transportation and storage	OBC	1,016,466	1,422,987	1.40	0.865
Transportation and storage	Others	1,288,047	2,084,399	1.62	1.000
Accommodation and Food service activities	SC	220,667	440,461	2.00	0.760
Accommodation and Food service activities	ST	99,251	214,682	2.16	0.823
Accommodation and Food service activities	OBC	904,079	1,956,679	2.16	0.824
Accommodation and Food service activities	Others	980,819	2,577,149	2.63	1.000
Information & communication	SC	23,735	59,215	2.49	0.697
Information & communication	ST	8,664	19,891	2.30	0.641
Information & communication	OBC	91,401	213,841	2.34	0.653
Information & communication	Others	149,934	537,036	3.58	1.000
Financial and insurance activities	SC	27,188	52,541	1.93	0.757
Financial and insurance activities	ST	10,251	21,029	2.05	0.803
Financial and insurance activities	OBC	125,964	236,278	1.88	0.735
Financial and insurance activities	Others	219,071	559,384	2.55	1.000
Real estate activities	SC	31,076	45,611	1.47	0.954
Real estate activities	ST	14,978	20,832	1.39	0.904
Real estate activities	OBC	141,308	197,372	1.40	0.908
Real estate activities	Others	228,906	352,243	1.54	1.000
Professional, scientific & technical activities	SC	39,337	79,301	2.02	0.810
Professional, scientific & technical activities	ST	11,316	24,164	2.14	0.858
Professional, scientific & technical activities	OBC	144,779	295,745	2.04	0.821
Professional, scientific & technical activities	Others	276,855	689,060	2.49	1.000
Administrative and support service activities	SC	58,214	122,511	2.10	0.793
Administrative and support service activities	ST	21,986	46,121	2.10	0.790
Administrative and support service activities	OBC	248,720	504,526	2.03	0.764
Administrative and support service activities	Others	305,694	811,355	2.65	1.000
Education	SC	49,386	156,046	3.16	0.690
Education	ST	17,915	71,993	4.02	0.877
Education	OBC	169,572	770,772	4.55	0.992
Education	Others	498,529	2,284,284	4.58	1.000
Human health & social work activities	SC	55,507	118,472	2.13	0.722
Human health & social work activities	ST	15,008	38,999	2.60	0.879
Human health & social work activities	OBC	207,036	463,078	2.24	0.757
Human health & social work activities	Others	395,580	1,168,771	2.95	1.000
Arts entertainment, sports & amusement and recreation	SC	25,968	56,935	2.19	0.874

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Industry	Group	Estab.	Workers	Mean size	vs Others
Arts entertainment, sports & amusement and recreation	ST	5,542	12,107	2.18	0.870
Arts entertainment, sports & amusement and recreation	OBC	51,541	118,389	2.30	0.915
Arts entertainment, sports & amusement and recreation	Others	75,382	189,183	2.51	1.000
Other service activities not else where classified	SC	360,975	544,650	1.51	0.870
Other service activities not else where classified	ST	90,765	144,480	1.59	0.918
Other service activities not else where classified	OBC	1,152,873	1,790,577	1.55	0.896
Other service activities not else where classified	Others	1,228,811	2,130,517	1.73	1.000

Table 19.18: Prospective trial-power assumptions and calculations

Clusters	Baseline N	Retained N	ICC	DEFF	MDE SD	MDE log
40.0	600.0	480.0	0.02	1.22	0.289	0.144
40.0	600.0	480.0	0.05	1.55	0.326	0.163
40.0	600.0	480.0	0.10	2.10	0.379	0.190
80.0	1200.0	960.0	0.02	1.22	0.202	0.101
80.0	1200.0	960.0	0.05	1.55	0.228	0.114
80.0	1200.0	960.0	0.10	2.10	0.265	0.132
120.0	1800.0	1440.0	0.02	1.22	0.164	0.082
120.0	1800.0	1440.0	0.05	1.55	0.185	0.093
120.0	1800.0	1440.0	0.10	2.10	0.216	0.108

Assumed growth standard deviation is 0.5, not measured in the acquired data. The formula uses retained cluster size 12, a t critical value with G-2 degrees of freedom and a normal 80-percent-power quantile. Actual trial planning requires pilot estimates, unequal cluster-size checks, compliance and spillover analysis.

19.2 Complete state correlation matrix

Table 19.19: State Pearson correlations: all nine analytical variables

Variable	Est.	Workers	Size	Hired	Density	SC	ST	Urban	Literacy
Est.	1.00	0.91	-0.25	-0.20	0.06	0.26	-0.30	-0.23	-0.42
Workers	0.91	1.00	-0.09	-0.10	-0.02	0.30	-0.27	-0.22	-0.41
Size	-0.25	-0.09	1.00	0.30	-0.27	0.01	-0.01	0.23	0.15
Hired	-0.20	-0.10	0.30	1.00	-0.35	-0.24	-0.02	0.24	0.16
Density	0.06	-0.02	-0.27	-0.35	1.00	-0.38	0.31	-0.15	-0.02
SC	0.26	0.30	0.01	-0.24	-0.38	1.00	-0.67	-0.12	-0.23
ST	-0.30	-0.27	-0.01	-0.02	0.31	-0.67	1.00	-0.06	0.11
Urban	-0.23	-0.22	0.23	0.24	-0.15	-0.12	-0.06	1.00	0.65
Literacy	-0.42	-0.41	0.15	0.16	-0.02	-0.23	0.11	0.65	1.00

Correlations use pairwise-complete observations. Hired is the percentage of establishments with hired workers. Pairs involving size or hired share have 33 observations; other pairs have 35. These are geographic associations.

Chapter 20

Additional worker composition and recruitment graphs

All seven worker graphs use observed published cells or fitted aggregate models. Remaining means All minus SC minus ST in Census graphs, but total minus SC minus ST minus OBC in Coal India graphs; their definitions are source-specific. No individual observations were reconstructed. PNG counterparts are in the archive.

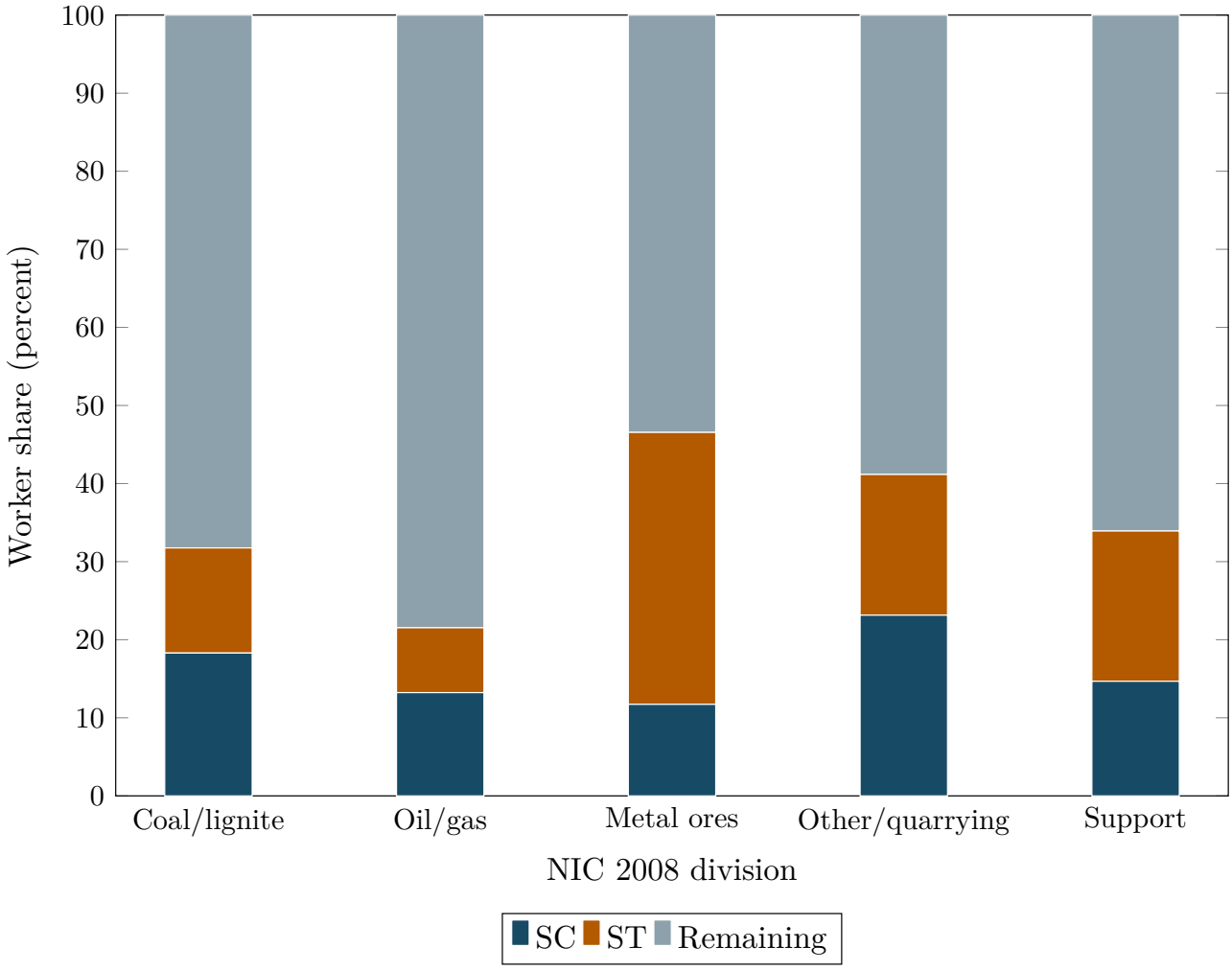


Figure 20.1: National SC/ST mining worker composition, main plus marginal, by NIC divisions 05-09. Source: Census 2011 B18, B18SC and B18ST. Remaining contains OBC and other workers. This directly classifies worker identity, unlike EC employment classified by owner identity.

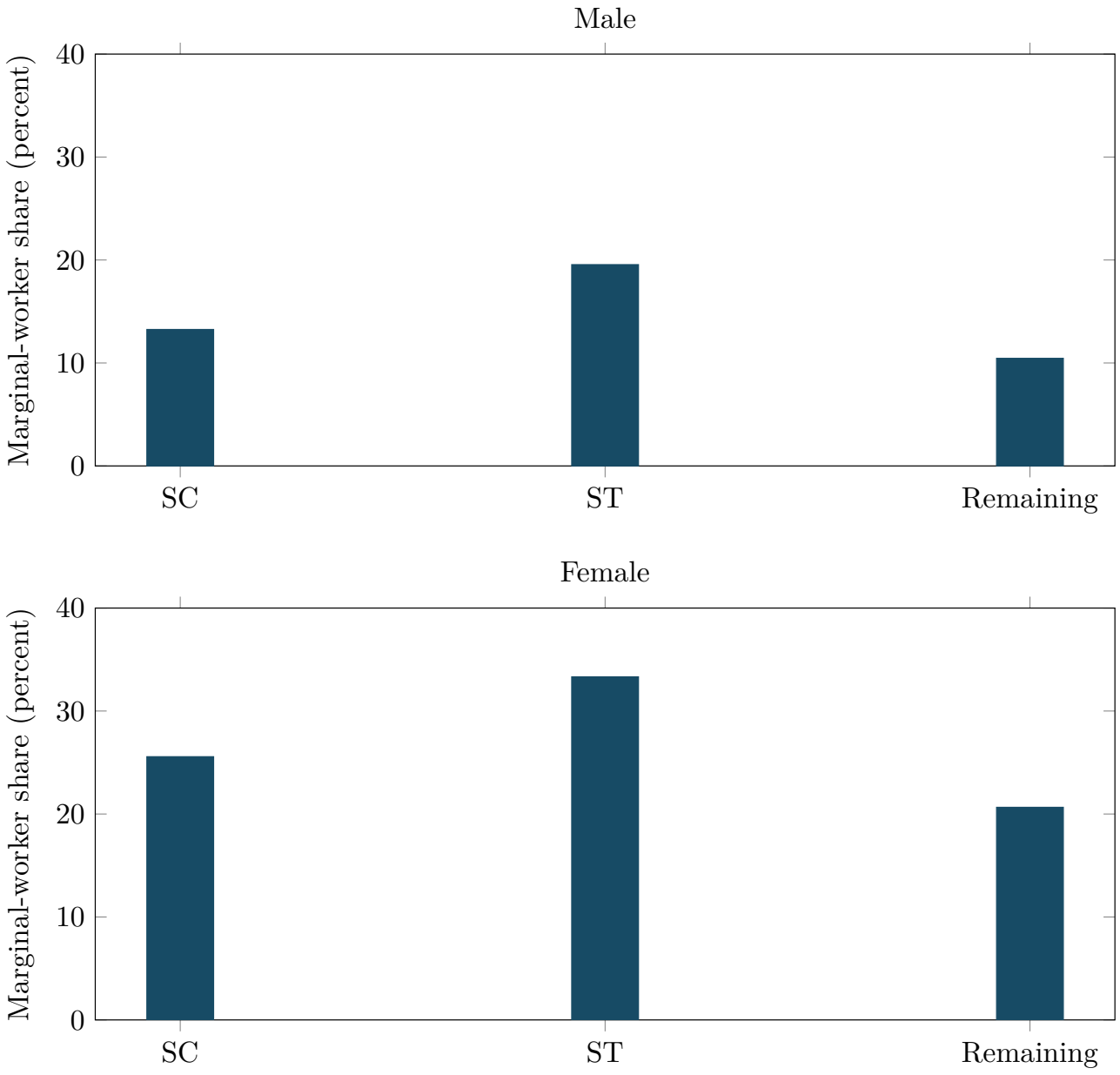


Figure 20.2: Sex-specific marginal-worker shares across NIC 05-09, Census 2011. Marginal denotes work for less than six months during the preceding reference year. It is not a measure of contractor employment, informality or a particular occupation.

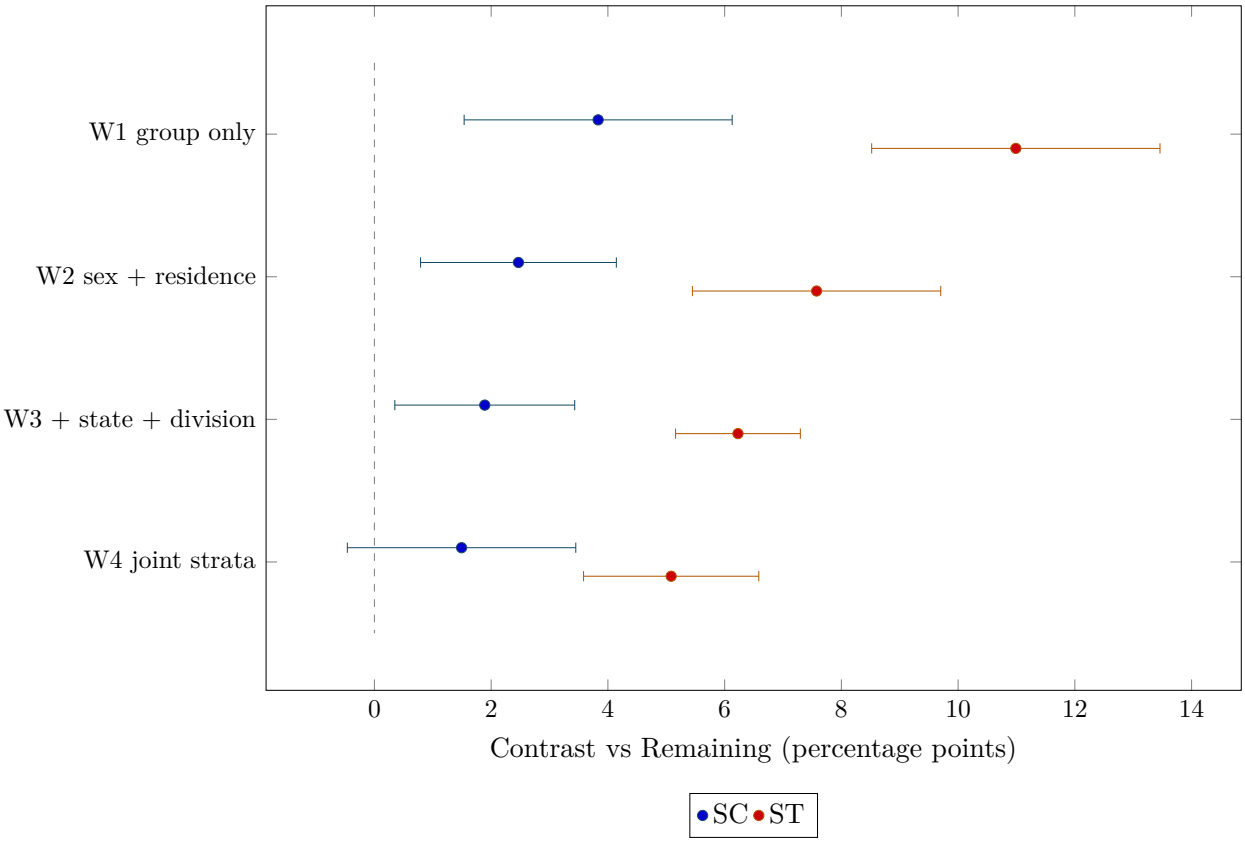


Figure 20.3: Observed aggregate duration differences before and after controls. Employment-weighted WLS on 1,275 cells. W4 absorbs state-by-division-by-sex-by-residence strata. Intervals cluster at 35 historical states and are model-based, not Census enumeration errors or causal caste effects. Missing age, education and occupation controls remain relevant.

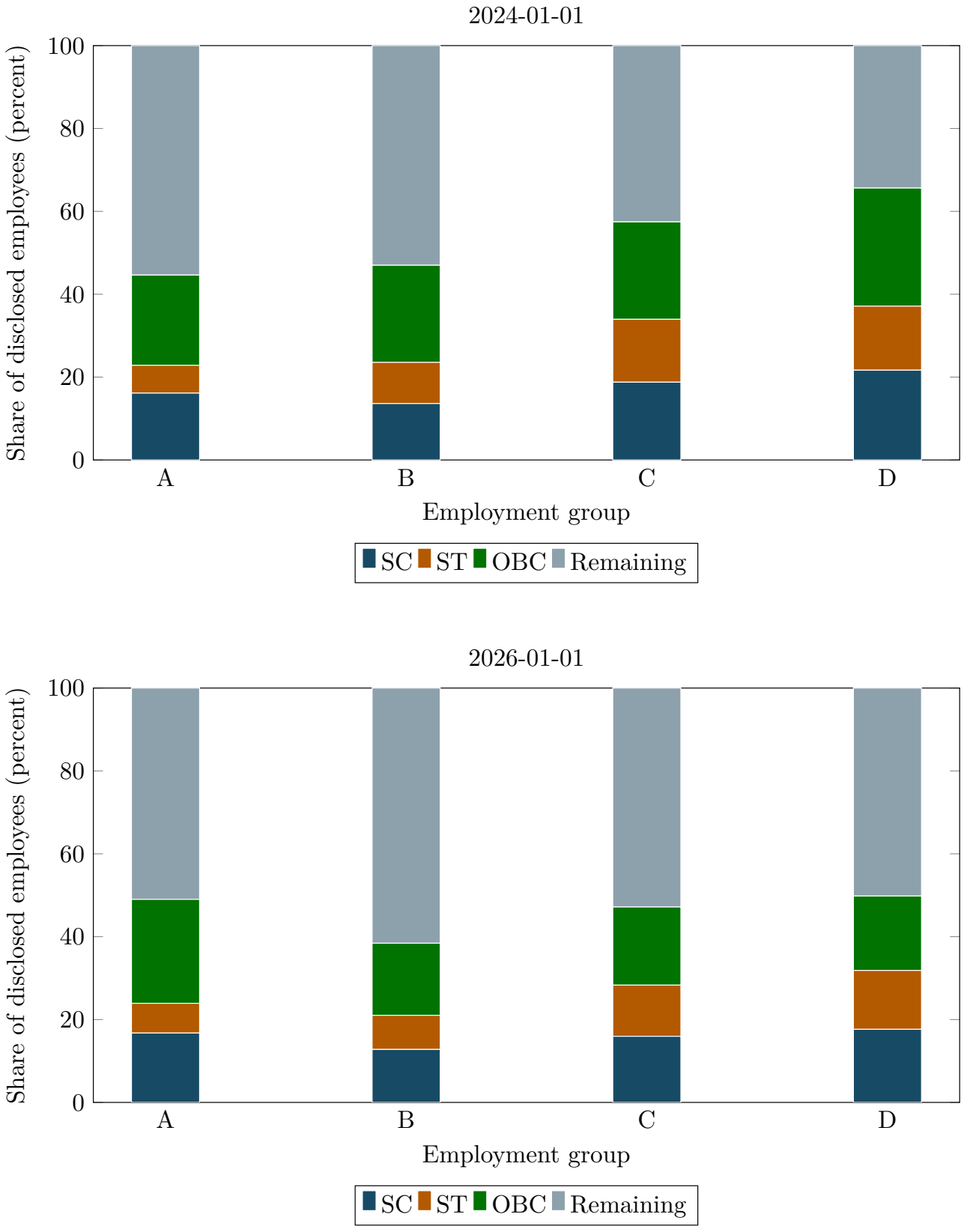


Figure 20.4: Coal India corporate employee composition by administrative employment group. Source: Coal Ministry annual reports 2023-24 and 2025-26, chapter 16, PDF pages 5 and 4. The 2026 residual contains EWS employees. Snapshots do not identify recruitment probabilities or the reasons for changes in employee stocks.

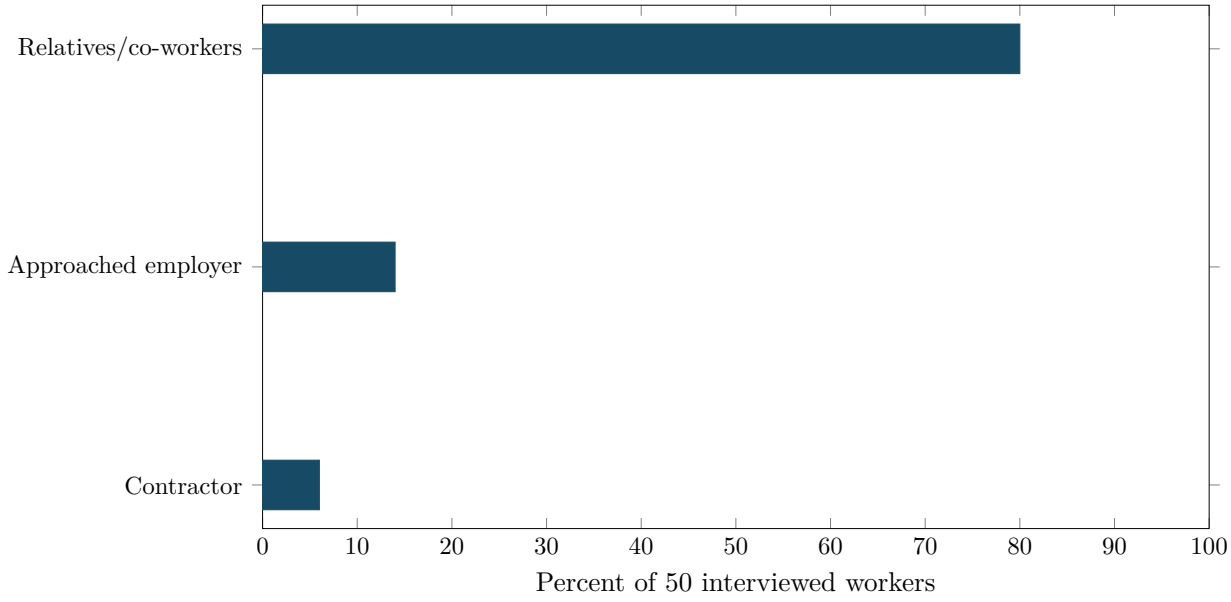


Figure 20.5: Reported Makrana job-entry routes, Arisa/CLRA 2024, PDF page 29. Published counts are seven direct approaches and three contractor recruits; 40 referral entrants are derived from the published 80 percent of N=50. This is a local supply-chain sample and referrals do not establish shared-caste relationships.

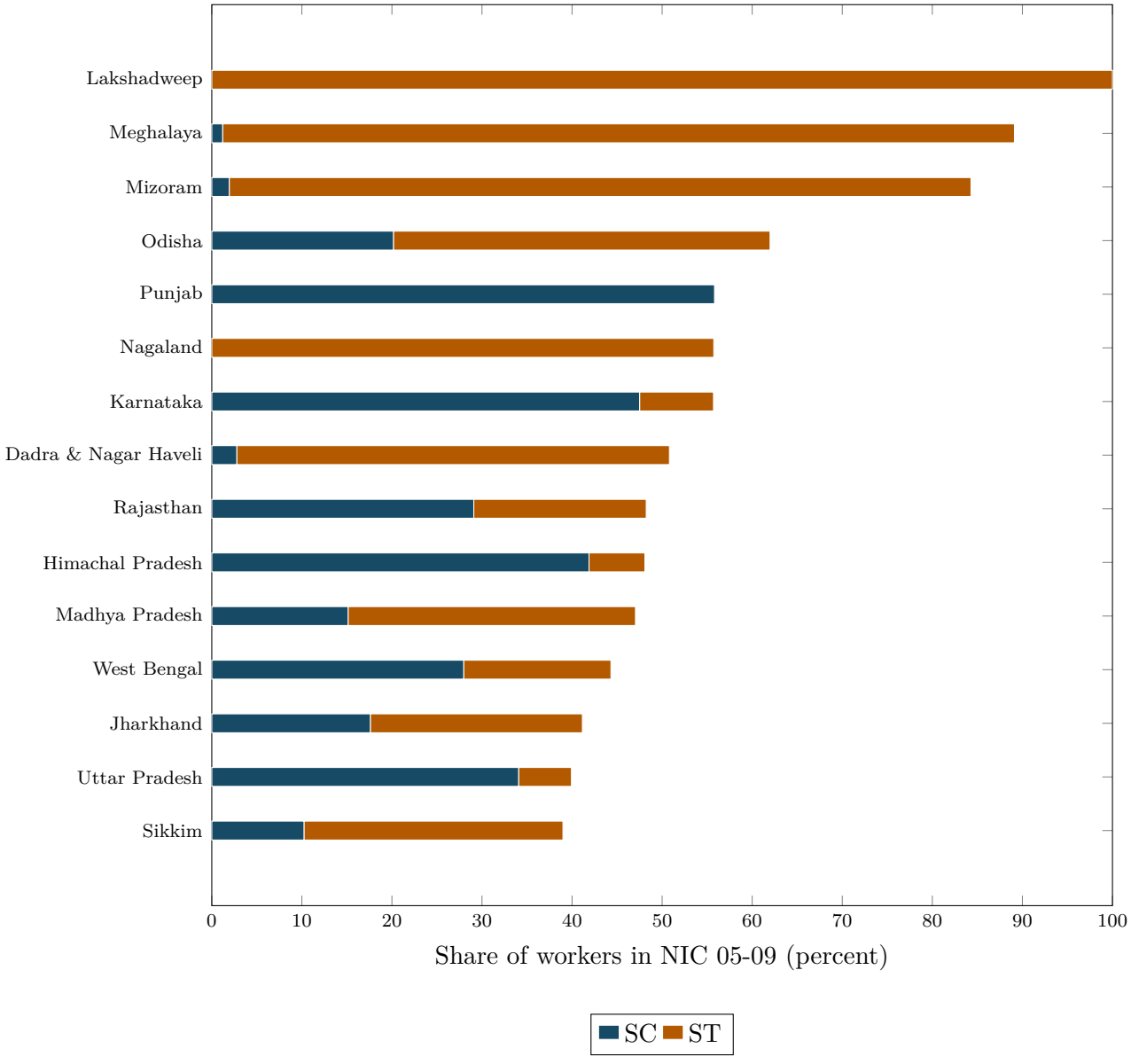


Figure 20.6: Fifteen highest SC plus ST worker shares, Census 2011 historical state/UT geography. Some regions have very small workforces; the full count table should accompany comparisons. These are workforce composition shares, not mining participation rates among the regional SC/ST population.

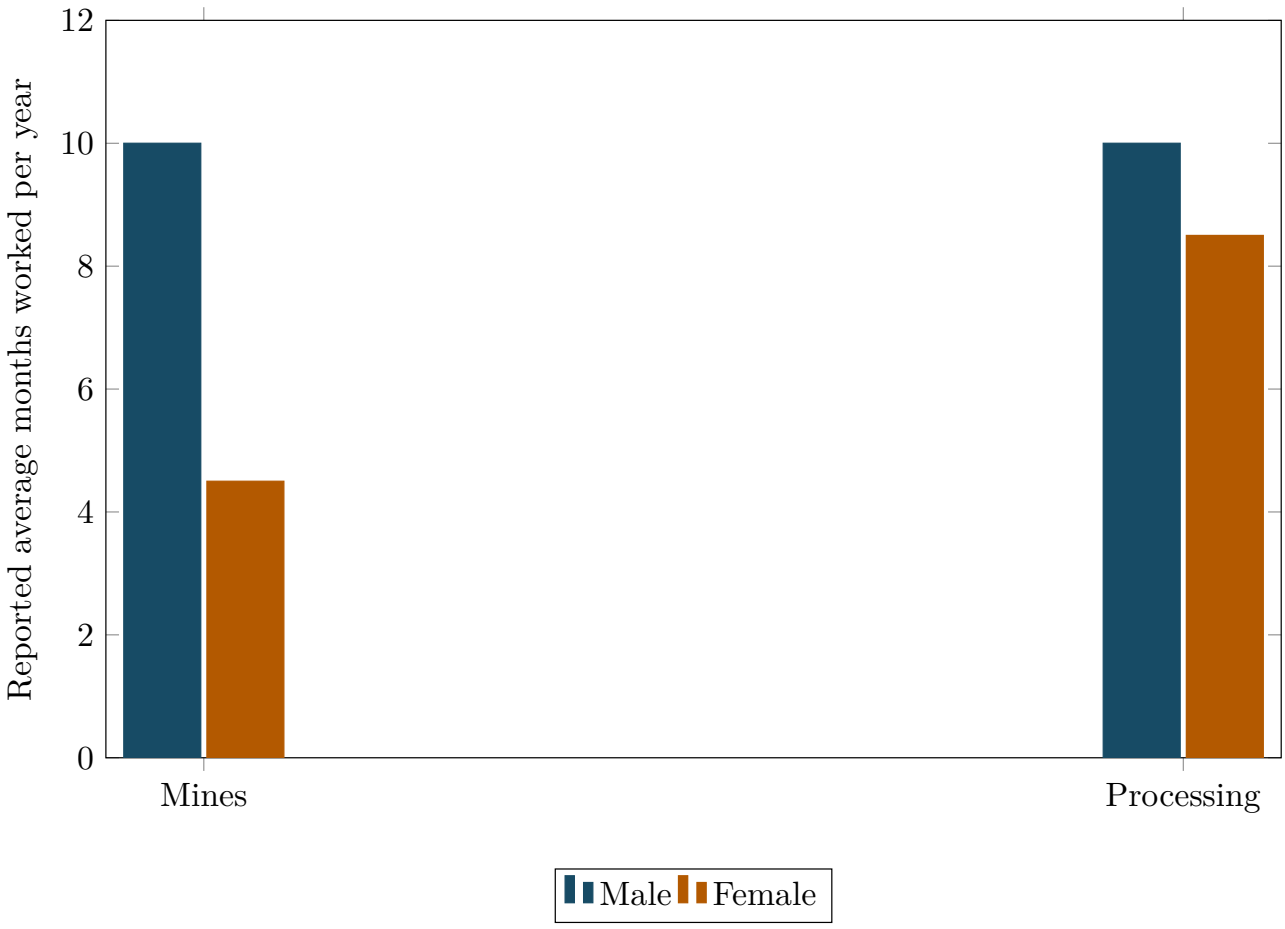


Figure 20.7: Published local working-year averages by sex and stone supply-chain stage. ILO Rajasthan study 2021, PDF page 59. These unadjusted averages do not control for occupation, caste or worker characteristics and should not be read as a causal gender effect.

Chapter 21

Worker statistics and controlled-result tables

All national and state values are derived from actual B18 cells. Census categories and company reservation classifications are kept separate. Model coefficients use percentage points. Full-precision values, source coordinates and all fitted coefficients are archived in CSV files.

Table 21.1: Census 2011 national mining workforce, NIC 05-09

Group	Main	Marginal	All workers	Worker %	Marginal %
SC	371,016	68,503	439,519	20.87	15.59
ST	281,799	82,950	364,749	17.32	22.74
Remaining	1,148,511	152,957	1,301,468	61.81	11.75

Table 21.2: National workforce composition by mining division

NIC	Division	Group	Workers	Worker %	Marginal %
05	Coal and lignite	Remaining	398,831	68.24	4.46
05	Coal and lignite	SC	106,906	18.29	6.04
05	Coal and lignite	ST	78,692	13.46	13.76
06	Oil and natural gas	Remaining	41,525	78.47	7.02
06	Oil and natural gas	SC	6,998	13.22	13.79
06	Oil and natural gas	ST	4,395	8.31	16.68
07	Metal ores	Remaining	52,952	53.45	9.09
07	Metal ores	SC	11,614	11.72	11.19
07	Metal ores	ST	34,504	34.83	17.88
08	Other mining / quarrying	Remaining	785,315	58.84	15.97
08	Other mining / quarrying	SC	308,926	23.14	19.08
08	Other mining / quarrying	ST	240,504	18.02	26.56
09	Mining support	Remaining	22,845	66.08	8.66
09	Mining support	SC	5,075	14.68	16.33
09	Mining support	ST	6,654	19.25	20.26

Table 21.3: All 35 historical regions: workforce counts and social composition

Region	Workers	SC count	ST count	SC %	ST %	Marg. %
Jammu & Kashmir	6,384	309	217	4.84	3.40	25.44
Himachal Pradesh	2,999	1,256	186	41.88	6.20	25.44
Punjab	4,239	2,366	0	55.82	0.00	10.43
Chandigarh	89	6	0	6.74	0.00	0.00
Uttarakhand	8,091	1,387	276	17.14	3.41	32.75

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Region	Workers	SC count	ST count	SC %	ST %	Marg. %
Haryana	9,392	1,802	0	19.19	0.00	19.73
Nct Of Delhi	777	136	0	17.50	0.00	3.09
Rajasthan	235,903	68,619	45,163	29.09	19.14	16.38
Uttar Pradesh	65,954	22,465	3,872	34.06	5.87	23.23
Bihar	27,736	8,478	964	30.57	3.48	23.33
Sikkim	1,026	105	295	10.23	28.75	18.13
Arunachal Pradesh	1,038	0	243	0.00	23.41	11.66
Nagaland	3,052	0	1,701	0.00	55.73	40.20
Manipur	1,858	47	134	2.53	7.21	17.98
Mizoram	1,394	27	1,148	1.94	82.35	16.57
Tripura	2,707	308	392	11.38	14.48	18.06
Meghalaya	23,367	280	20,544	1.20	87.92	29.66
Assam	75,909	4,578	7,291	6.03	9.60	19.31
West Bengal	155,250	43,413	25,386	27.96	16.35	14.68
Jharkhand	286,430	50,414	67,438	17.60	23.54	13.11
Odisha	166,609	33,595	69,678	20.16	41.82	21.55
Chhattisgarh	78,687	10,076	17,974	12.81	22.84	5.14
Madhya Pradesh	157,392	23,815	50,223	15.13	31.91	13.71
Gujarat	55,447	4,275	7,387	7.71	13.32	11.44
Daman & Diu	451	17	84	3.77	18.63	2.00
Dadra & Nagar Haveli	433	12	208	2.77	48.04	16.63
Maharashtra	126,535	16,055	17,257	12.69	13.64	9.96
Andhra Pradesh	223,499	37,379	9,354	16.72	4.19	8.77
Karnataka	141,530	67,227	11,610	47.50	8.20	14.03
Goa	18,321	261	2,178	1.42	11.89	14.88
Lakshadweep	3	0	3	0.00	100.00	33.33
Kerala	77,142	14,371	2,084	18.63	2.70	18.43
Tamil Nadu	144,416	26,314	1,456	18.22	1.01	10.37
Puducherry	822	126	0	15.33	0.00	6.08
Andaman & Nicobar Islands	854	0	3	0.00	0.35	12.41

Table 21.4: National worker sex and duration cells

Sex	Group	Main	Marginal	Workers	Marginal %
Female	Remaining	131,829	34,306	166,135	20.65
Female	SC	61,998	21,293	83,291	25.56
Female	ST	56,362	28,165	84,527	33.32
Male	Remaining	1,016,682	118,651	1,135,333	10.45
Male	SC	309,018	47,210	356,228	13.25
Male	ST	225,437	54,785	280,222	19.55

Table 21.5: Worker-cell specifications: weighting and control sets

Model	Controls	Cells	States	Weight sum	Rank	R sq.
W1	Group only	1275	35	2,105,736	3	0.1599
W2	Sex + rural/urban	1275	35	2,105,736	5	0.5201
W3	Sex + area + state + division	1275	35	2,105,736	43	0.7856
W4	State-division-sex-area strata	1275	35	2,105,736	552	0.9349

Table 21.6: Unadjusted and controlled SC/ST duration contrasts versus Remaining

Model	Group	Beta pp	Cluster SE	95% low	95% high	Model p
W1	SC	3.833	1.130	1.537	6.129	0.0018
W1	ST	10.989	1.215	8.519	13.459	0.0000

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Model	Group	Beta pp	Cluster SE	95% low	95% high	Model p
W2	SC	2.468	0.825	0.791	4.145	0.0052
W2	ST	7.575	1.047	5.448	9.702	0.0000
W3	SC	1.891	0.758	0.350	3.432	0.0177
W3	ST	6.228	0.526	5.160	7.297	0.0000
W4	SC	1.493	0.963	-0.464	3.450	0.1302
W4	ST	5.084	0.739	3.583	6.585	0.0000

W1 group-only; W2 sex/residence; W3 additionally state and division effects; W4 joint state-division-sex-residence strata. All models have 1,275 nonempty aggregate cells and 35 state clusters. Point estimates describe the acquired cells; p-values are exploratory model summaries. They are not part of the earlier 18-coefficient EC multiple-testing family and are not used for a causal claim.

Table 21.7: All coefficients including controls in W1-W3

Model	Term	Beta pp	Cluster SE	95% low	95% high
W1	Intercept	11.753	0.936	9.851	13.654
W1	C(group)[T.SC]	3.833	1.130	1.537	6.129
W1	C(group)[T.ST]	10.989	1.215	8.519	13.459
W2	Intercept	24.968	2.093	20.714	29.221
W2	C(group)[T.SC]	2.468	0.825	0.791	4.145
W2	C(group)[T.ST]	7.575	1.047	5.448	9.702
W2	C(sex)[T.Male]	-10.559	1.451	-13.508	-7.610
W2	C(area)[T.Urban]	-9.491	1.176	-11.880	-7.101
W3	Intercept	28.525	1.739	24.991	32.059
W3	C(group)[T.SC]	1.891	0.758	0.350	3.432
W3	C(group)[T.ST]	6.228	0.526	5.160	7.297
W3	C(sex)[T.Male]	-10.111	1.331	-12.816	-7.406
W3	C(area)[T.Urban]	-5.504	0.889	-7.311	-3.697
W3	C(state_code)[T.02]	-2.867	0.480	-3.843	-1.891
W3	C(state_code)[T.03]	-16.750	0.462	-17.689	-15.810
W3	C(state_code)[T.04]	-20.420	0.655	-21.750	-19.090
W3	C(state_code)[T.05]	6.286	0.181	5.919	6.654
W3	C(state_code)[T.06]	-6.897	0.219	-7.342	-6.453
W3	C(state_code)[T.07]	-15.449	1.093	-17.671	-13.228
W3	C(state_code)[T.08]	-12.533	0.371	-13.288	-11.778
W3	C(state_code)[T.09]	-3.505	0.309	-4.132	-2.878
W3	C(state_code)[T.10]	-2.416	0.340	-3.107	-1.725
W3	C(state_code)[T.11]	-13.550	0.635	-14.841	-12.259
W3	C(state_code)[T.12]	-16.212	0.510	-17.247	-15.176
W3	C(state_code)[T.13]	12.940	0.378	12.172	13.708
W3	C(state_code)[T.14]	-13.406	0.770	-14.972	-11.841
W3	C(state_code)[T.15]	-11.923	0.791	-13.531	-10.315
W3	C(state_code)[T.16]	-9.294	1.115	-11.560	-7.028
W3	C(state_code)[T.17]	1.728	0.881	-0.063	3.518
W3	C(state_code)[T.18]	-4.967	0.756	-6.503	-3.431
W3	C(state_code)[T.19]	-8.440	0.415	-9.284	-7.596
W3	C(state_code)[T.20]	-10.086	0.548	-11.200	-8.973
W3	C(state_code)[T.21]	-6.813	0.375	-7.574	-6.051
W3	C(state_code)[T.22]	-14.106	0.908	-15.950	-12.261
W3	C(state_code)[T.23]	-10.968	0.439	-11.860	-10.076
W3	C(state_code)[T.24]	-15.007	0.527	-16.077	-13.937
W3	C(state_code)[T.25]	-22.960	0.424	-23.822	-22.097
W3	C(state_code)[T.26]	-11.549	0.316	-12.190	-10.907
W3	C(state_code)[T.27]	-13.885	0.337	-14.571	-13.199
W3	C(state_code)[T.28]	-15.900	0.327	-16.564	-15.235
W3	C(state_code)[T.29]	-14.799	0.465	-15.745	-13.854
W3	C(state_code)[T.30]	-8.291	0.515	-9.338	-7.245
W3	C(state_code)[T.31]	-1.631	0.794	-3.244	-0.017
W3	C(state_code)[T.32]	-8.172	0.163	-8.504	-7.840
W3	C(state_code)[T.33]	-15.271	0.254	-15.788	-14.755

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Model	Term	Beta pp	Cluster SE	95% low	95% high
W3	C(state_code)[T.34]	-13.878	1.440	-16.804	-10.952
W3	C(state_code)[T.35]	-14.410	0.199	-14.813	-14.006
W3	C(division)[T.06]	2.492	3.134	-3.877	8.861
W3	C(division)[T.07]	2.665	1.212	0.203	5.128
W3	C(division)[T.08]	9.085	1.202	6.641	11.529
W3	C(division)[T.09]	2.116	3.790	-5.586	9.819

W4 includes 550 observed context strata and two group indicators (rank 552). Its focal contrasts are above; every stratum coefficient is in `worker_regression_all_coefficients2011.csv`. Small cells and many absorbed terms warrant caution with the finite-sample covariance correction. Omitted reference levels are Remaining, Female, Rural, and the first observed state/division/stratum as applicable.

Table 21.8: Coal India employee counts and shares by snapshot and employment group

Date	Pay group	Social group	Count	Total	Share %
2024-01-01	A	SC	2,491	15,439	16.13
2024-01-01	A	ST	1,037	15,439	6.72
2024-01-01	A	OBC	3,361	15,439	21.77
2024-01-01	A	Remaining	8,550	15,439	55.38
2024-01-01	B	SC	2,229	16,368	13.62
2024-01-01	B	ST	1,627	16,368	9.94
2024-01-01	B	OBC	3,835	16,368	23.43
2024-01-01	B	Remaining	8,677	16,368	53.01
2024-01-01	C	SC	21,361	113,796	18.77
2024-01-01	C	ST	17,270	113,796	15.18
2024-01-01	C	OBC	26,750	113,796	23.51
2024-01-01	C	Remaining	48,415	113,796	42.55
2024-01-01	D	SC	18,520	85,455	21.67
2024-01-01	D	ST	13,209	85,455	15.46
2024-01-01	D	OBC	24,350	85,455	28.49
2024-01-01	D	Remaining	29,376	85,455	34.38
2026-01-01	A	SC	2,513	15,014	16.74
2026-01-01	A	ST	1,079	15,014	7.19
2026-01-01	A	OBC	3,765	15,014	25.08
2026-01-01	A	Remaining	7,657	15,014	51.00
2026-01-01	B	SC	1,617	12,648	12.78
2026-01-01	B	ST	1,042	12,648	8.24
2026-01-01	B	OBC	2,199	12,648	17.39
2026-01-01	B	Remaining	7,790	12,648	61.59
2026-01-01	C	SC	16,180	101,410	15.96
2026-01-01	C	ST	12,521	101,410	12.35
2026-01-01	C	OBC	19,124	101,410	18.86
2026-01-01	C	Remaining	53,585	101,410	52.84
2026-01-01	D	SC	15,041	85,261	17.64
2026-01-01	D	ST	12,113	85,261	14.21
2026-01-01	D	OBC	15,307	85,261	17.95
2026-01-01	D	Remaining	42,800	85,261	50.20

Table 21.9: Coal India full snapshot totals

Date	Group	Count	Total	Share %
2024-01-01	OBC	58,296	231,058	25.23
2024-01-01	Remaining	95,018	231,058	41.12
2024-01-01	SC	44,601	231,058	19.30
2024-01-01	ST	33,143	231,058	14.34
2026-01-01	OBC	40,395	214,333	18.85

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Date	Group	Count	Total	Share %
2026-01-01	Remaining	111,832	214,333	52.18
2026-01-01	SC	35,351	214,333	16.49
2026-01-01	ST	26,755	214,333	12.48

Table 21.10: SECL Mining Sirdar disclosed FY2024-25 vacancy filling

Vacancy category	Advertised	Filled	Unfilled	Fill %
UR	131	131	0	100.00
SC	48	48	0	100.00
ST	97	92	5	94.85
EWS	32	32	0	100.00
OBC	42	42	0	100.00

Source: Coal Ministry annual report 2025-26, chapter 16, PDF page 7. Counts sum to 350 advertised and 345 filled; ST non-joining is reported. No applicant denominator is disclosed here. UR and EWS are vacancy classifications, not caste identities.

Table 21.11: Published Arisa/CLRA local stone-worker composition

Site	Group	Published %	PDF page	Denominator note
Budhpura	SC	36	16	Composition paragraph does not explicitly resolve 37 main vs 47 including home-workers
Budhpura	ST	19	16	Composition paragraph does not explicitly resolve 37 main vs 47 including home-workers
Budhpura	OBC	45	16	Composition paragraph does not explicitly resolve 37 main vs 47 including home-workers
Makrana	SC	82	27	50 individual interviewees
Makrana	ST	0	27	50 individual interviewees; no ST interviewees reported
Makrana	OBC	18	27	50 individual interviewees

Table 21.12: ILO 2021 two-cluster stone-worker composition

Group	Published %	Sample N	PDF page
SC	34	115	55
ST	30	115	55
OBC	30	115	55
General	5	115	55

ILO percentages sum to 99 through publication rounding; they are not normalized and no individual group counts are reconstructed. Different local samples and vintages are never pooled into one representative survey.

Table 21.13: Makrana recruitment routes: individual-interview sample N=50

Route	Count	Share %	Count provenance
Relatives / co-workers	40	80	40 derived from reported 80% of 50
Approached employer	7	14	Published count 7
Contractor	3	6	Published count 3

Table 21.14: Other published stone-worker details with original denominators

Sample	Indicator	Count	Base	Percent	PDF page
Makrana	Male	45	50	90.00	27
Makrana	Illiterate	20	50	40.00	28
Makrana	Secondary education or higher	7	50	14.00	28
Makrana	Lives locally	48	50	96.00	28
Makrana	Uses any PPE	20	50	40.00	31
Makrana	Advance received	18	50	36.00	30
Makrana	Written employment contract	0	50	0.00	29
Makrana	Piece-rate paid	–	50	60.00	29
Budhpura main questionnaire	Piece-rate paid	31	37	83.78	18
Budhpura main questionnaire	Uses any PPE	21	37	56.76	23
ILO Rajasthan two clusters	Illiterate	–	115	40.00	55
ILO Rajasthan two clusters	Has bank account	–	115	30.00	55
ILO Rajasthan two clusters	Resides in work district	–	115	92.00	55
ILO mine-worker subgroup	Provident Fund access	–	–	21.00	62
ILO processing-worker subgroup	Provident Fund access	–	–	29.00	62

A blank count means a published percentage rather than an observed integer count. ILO PF percentages use separate mine/processing subgroups, whose counts are not reconstructed from N=115. Any PPE use is not adequate exposure protection. The Makrana advance paragraph gives 18/50 = 36 percent; the concluding 38-percent figure conflicts and is preserved in the CSV measurement note.

Table 21.15: ILO published average working year

Stage	Sex	Months/year	PDF page
Mines	Male	10.0	59
Mines	Female	4.5	59
Processing	Male	10.0	59
Processing	Female	8.5	59

Chapter 22

Full draft fieldwork questionnaire

The following 86 measures are a proposed instrument to pilot, translate and revise before any fieldwork. No responses were collected. Consent, private identifier handling and unknown/declined categories are mandatory parts of the protocol. Repeated outcome definitions and relationship event dates are needed to study growth and temporal order.

Table 22.1: Draft questionnaire measures and coding instructions

Module	ID	Question or measure	Coding instruction
Consent	C01	Consent to interview; voluntary participation and ability to skip questions	Yes/no; stop if no
Consent	C02	Separate permission for follow-up, administrative linkage and contact-based modules	Separate yes/no fields
Consent	C03	Preferred safe contact channel and interview language	Private key file only
Business	B01	Enterprise ID, site ID, operating address and historical location changes	Identifiers; separate private key
Business	B02	Mine operator, quarry owner, contractor, transport, trading or other supplier role	Multiple roles plus primary role
Business	B03	Mineral, grade, extraction/processing/service activity and NIC activity	Detailed activity, verified later
Business	B04	Start year, operating months, current operating status and closure episodes	Dates and reasons
Business	B05	Legal form, beneficial ownership, partners and owner changes	Shares/date; consent-dependent
Business	B06	CIN/LLPIN, lease/mine/permit IDs and registration identifiers	Optional; verify with consent
Business	B07	Other establishments, leases or businesses operated by same enterprise	Firm-site bridge
Owner	O01	Owner self-reported broad social category	SC/ST/OBC/other/ self-describe/decline
Owner	O02	Optional self-described jati/community and preferred wording	Voluntary free text; no inference
Owner	O03	Owner age, education, prior occupations and business experience	Baseline covariates
Owner	O04	Prior household business ownership, inherited capital and assets	Voluntary ranges/date
Owner	O05	Migration history and years in locality	Dates; sensitive details optional
Owner	O06	Business association membership and date of joining	Association ID/type
Contacts	N01	Who helped obtain your most recent buyer/ contract?	Up to named/coded alters; recall period
Contacts	N02	Who supplied useful price, permit or market information?	Functional name generator
Contacts	N03	Who helped obtain formal/informal finance or guarantees?	Functional name generator
Contacts	N04	Who helped hire labour or access equipment/ transport?	Functional name generator

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Module	ID	Question or measure	Coding instruction
Contacts	N05	Who helped resolve a business dispute?	Functional name generator
Contacts	N06	Which potentially helpful contacts did you approach without receiving help?	Include unsuccessful requests
Contacts	N07	For every alter, role, first-contact date, frequency and locality	Alter roster; private IDs
Contacts	N08	For every alter, how did the relationship begin?	Family/community/work/ association/other
Contacts	N09	Does respondent describe this relationship as linked to shared community identity?	Yes/no/uncertain/decline; do not infer
Contacts	N10	For every alter, what assistance occurred, when and with what result?	Event dates, resources, outcome
Contacts	N11	For every alter, reciprocal obligations, fees or exclusivity expectations	Optional amounts/ranges
Contacts	N12	Which contact classifications are unknown or declined?	Unknown/declined separate from different
Contacts	N13	Which contacts also know one another, if respondent voluntarily knows?	Alter-alter links; limit burden
Contacts	N14	Available local buyers, lenders and associations respondent could approach	Opportunity-set frame
Contacts	N15	Whether assistance predated entry, preceded growth or followed success	Temporal sequence
Sales	F01	Sales value in specified fiscal/reference periods	Rupees, period, record basis
Sales	F02	Physical output/dispatch by mineral/grade; contractor service quantity	Units and role-specific measures
Sales	F03	Prices, largest buyers and top-five revenue shares	Concentration and price verification
Sales	F04	Outstanding receivables, payment delays and buyer diversification	Same reference period
Sales	F05	Bids submitted, won, lost and disqualified; eligibility reasons	All attempts, not winners only
Employment	E01	Workers at baseline and each follow-up; paid/ unpaid, regular/casual	Same reference date/season
Employment	E02	Owners working in business, family workers and hired workers	Avoid double counting
Employment	E03	Operating days and seasonal employment peaks	Annual comparability
Capital	K01	Equipment, asset purchases and rented machinery	Values, dates, financing source
Capital	K02	Loan applications, approvals, terms, rejection and collateral	Formal and informal finance
Capital	K03	Working capital constraints and foregone orders	Recall period; documentary checks
Institutions	I01	Permit/lease applications, approval dates and operating commencement	Distinct event dates
Institutions	I02	Compliance costs, inspection dates and safety shutdowns	Document basis optional
Institutions	I03	Land access, displacement, local claims and dispute history	Voluntary, safe context
Institutions	I04	Perceived exclusion or unequal treatment in specific business events	Concrete events; neutral wording
Institutions	I05	Commodity, weather, infrastructure and local-policy shocks	Date and exposure
Follow-up	L01	Survival, temporary shutdown, permanent closure and ownership change	Separate states and dates
Follow-up	L02	Repeat sales, employment, output, capital and buyer modules	Same definitions and reference season
Follow-up	L03	Changes in contacts, activated help, new contacts and broken ties	Keep old alter IDs
Follow-up	L04	Treatment offered, delivered, used; exposure in neighboring clusters	Protocol-specific, blinded coding
Quality	Q01	Source of each numerical response and confidence in recall	Records/recall/range/unknown
Quality	Q02	Enumerator ID, duration, skips, corrections and recontact permissions	Audit trail; separate identifiers

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Module	ID	Question or measure	Coding instruction
Qualitative	V01	Narrative of entry, most consequential help and a failed attempt	Do not impose network mechanism
Qualitative	V02	How relationships facilitate opportunities or impose obligations	Concrete events and timelines
Qualitative	V03	Experiences crossing community boundaries and diversifying customers	Voluntary event-based narrative
Worker/job entry	W01	Separate worker consent and permission for anonymous site linkage	Consent yes/no; employer must not receive answers
Worker/job entry	W02	Anonymous worker, site and recruitment-event identifiers	Private identity key separate; dates recorded
Worker/job entry	W03	Voluntary broad social group and optional self-described jati	SC/ST/OBC/Other/unknown/ declined; never infer
Worker/job entry	W04	Age, sex and completed education before first mining job	Years and credential categories; relevant baseline date
Worker/job entry	W05	Origin, residence, migration and commuting history	District/state and dates; local does not imply no network
Worker/job entry	W06	Previous occupation and prior work/skill experience	Before recruitment; avoid post-job education proxy
Worker/job entry	W07	First information source about this vacancy	Multiple sources with sequence and earliest date
Worker/job entry	W08	Person who recommended or introduced the worker	Role and anonymous contact ID; no public name
Worker/job entry	W09	Relationship with referring contact before the vacancy	Relative/co-worker/neighbour/ broker/association/other; duration
Worker/job entry	W10	Voluntary shared-community connection with referrer	Self-report yes/no/unknown/ declined; separate broad category and jati
Worker/job entry	W11	Application, interview, selection and joining stages	Dates; include rejected and withdrawn applicants
Worker/job entry	W12	Recruitment method and vacancy category where relevant	Exam/direct/dependent/ land-related/referral; UR is a vacancy category
Worker/job entry	W13	Who selected, legally employs, pays and supervises	Separate actors; check site roster with consent
Worker/job entry	W14	Current task, occupation, worksite and production stage	Mine/processing/transport/ home-work; NIC and task
Worker/job entry	W15	Employer or contractor payroll and contract type	Verbal/written and length; inspect voluntarily
Worker/job entry	W16	Daily hours, days/month and months/year worked	Same reference period as earnings; seasonal interruptions
Worker/job entry	W17	Cash, piece-rate and in-kind pay with deductions	Units, reference period, payments due vs received
Worker/job entry	W18	Piece-rate task and quantities, rejects and tools	Avoid comparing rates across unlike tasks
Worker/job entry	W19	Advances, debt terms and ability to change employer	Amount, source, timing; distinguish referral from wage payer
Worker/job entry	W20	Payslips, attendance and proof of employment	Voluntary document check; absent not automatically illegal
Worker/job entry	W21	PF, insurance, leave and welfare access	Eligibility, enrolment, actual receipt separate
Worker/job entry	W22	Task exposure and PPE provision, training and use	Dust/noise/lifting; no medical diagnosis from questionnaire
Worker/job entry	W23	Injury/illness-related absence and voluntary care history	Self-report with consent; no diagnostic inference
Worker/job entry	W24	Promotion, skill training and changes of task	Dates and selection route; cohort before/after outcomes
Worker/job entry	W25	Other income and household employment alternatives	Do not treat secondary work as involuntary unemployment
Worker/job entry	W26	Union/association participation and grievance route	Voluntary; protect employer confidentiality

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Module	ID	Question or measure	Coding instruction
Worker/job entry	W27	Concrete discrimination or exclusion event if volunteered	Dates, setting and action; no leading assumption
Worker/job entry	W28	Job separation, reason and subsequent destination	Track quits, dismissals, season end and site closure
Worker/job entry	W29	Employer vacancy and applicant roster with consent	Anonymous stage denominators; refusals and unmatched cases retained
Worker/job entry	W30	Repeat follow-up and private safe recontact preference	Baseline/new-recruit/exit cohorts; anonymous matched firm outcomes

Chapter 23

Source inventory and acquisition priorities

The 65 entries are source documents, systems and source families. S02, S07, S09 and S56-S62 supply executed quantitative analysis or selected published empirical summaries. S63 provides acquired survey documentation; other entries are metadata, verified public documentation, conditional or licensed records, candidate families or data still to be collected. Source status is assessed for my research on 11 October 2026; access may change. A URL does not guarantee downloadable microdata.

23.1 S01: Sixth Economic Census unit records and metadata

Source link: https://microdata.gov.in/nada/index.php/catalog/47

Access status: Public metadata; microdata registration and terms; download not obtained

Unit and coverage: Establishment 2013-14

Caste or relationship information: SC ST OBC Others for eligible ownership

Growth information: No linked firm panel

Research use and limitations: Best broad frame for informal mining; obtain original microdata to estimate within-district differences and employment distributions. Social group is identity, not a network. Verify ownership skip rules in schedule 6A.

23.2 S02: Sixth Economic Census All India Report

Source link: https://ecensus.in/wp-content/uploads/2022/10/All-India-Report-of-Sixth-Economic-Census.pdf

Access status: Downloaded government-authored report from a third-party mirror

Unit and coverage: Published state and industry cells 2013-14

Caste or relationship information: Owner social-group aggregates

Growth information: Repeated census context only

Research use and limitations: Actual empirical input. Tables 2.5 and 4.5 give state mining establishments and workers; Tables 12 and 15 give proprietary establishments and employment by owner group. Official landing page is https://www.mospi.gov.in/all-india-report-sixth-economic-census. The mirror is an acquisition route, not a new statistical producer.

23.3 S03: Fifth Economic Census 2005

Source link: https://microdata.gov.in/nada/index.php/catalog/46

Access status: Metadata verified; unit data not acquired

Unit and coverage: Establishment 2005

Caste or relationship information: Owner social group

Growth information: Area and group changes if definitions harmonized

Research use and limitations: Pair with EC6 at stable geographic and industry cells. Do not treat independent censuses as a firm panel. Harmonize NIC2004 and NIC2008, ownership coverage, reporting rules and geography. Retrieve final rather than provisional tables.

23.4 S04: Fourth Economic Census and eSankhyiki dashboard

Source link: https://esankhyiki.mospi.gov.in/economic-census

Access status: Public dashboard; export route to verify

Unit and coverage: Establishment aggregates 1998 2005 2013

Caste or relationship information: Check individual releases

Growth information: Long-run area trends

Research use and limitations: Third cross-section can improve pre-trend context; a dashboard does not ensure a downloadable consistent panel. Preserve the release vintage and denominators.

23.5 S05: Seventh Economic Census release status

Source link: https://mospi.gov.in/faq

Access status: Official FAQ says finalization underway

Unit and coverage: Enumeration conducted 2019-21

Caste or relationship information: Check final release when available

Growth information: Potential third recent census

Research use and limitations: Do not assume an accessible final microdata release. The checked FAQ describes continuing finalization. Record the status on 10 October 2026; revisit before a dissertation data freeze.

23.6 S06: Economic Census establishment directory

Source link: https://mospi.gov.in/sites/default/files/6ec_dirEst/ec6_note_dir_esta.html

Access status: Public state directories

Unit and coverage: Named establishments with at least ten workers

Caste or relationship information: No direct caste ties

Growth information: Possible linkage to current registers

Research use and limitations: Potential fieldwork frame for larger units. The official note describes

undercoverage and an employment threshold. It cannot represent small quarries alone; collect a supplemental area listing.

23.7 S07: Maharashtra Sixth Economic Census report

Source link: https://www.nitiforstates.gov.in/public-assets/Policy/policy-repo/urbanization/RSS71M000226.pdf

Access status: Downloaded government report from NITI for States repository

Unit and coverage: 35 districts 2013

Caste or relationship information: Industry social-group tables; district context

Growth information: All-sector repeated counts, not mining firm growth

Research use and limitations: Actual input for district mining counts and employment, Tables 2.5 and 4.5. Matched to 2011 district boundaries; keep Thane before Palghar split and preserve historical district names.

23.8 S08: State Economic Census reports and statistical offices

Source link: https://des3.mahaonline.gov.in/economicReport.do

Access status: Public reports; state-specific archives vary

Unit and coverage: District and state tables

Caste or relationship information: Varies by table

Growth information: Repeat census area totals

Research use and limitations: Expand district replication to Rajasthan, Odisha, Jharkhand, Karnataka, Gujarat, Telangana and Andhra Pradesh. State report families are candidates, not all downloaded. Seek custom caste by mining by district tabulations from DES.

23.9 S09: Census 2011 Primary Census Abstract

Source link: https://censusindia.gov.in/nada/index.php/catalog/6191

Access status: Downloaded Excel workbook

Unit and coverage: State district rural urban 2011

Caste or relationship information: SC and ST population counts

Growth information: Population denominator and baseline controls

Research use and limitations: Actual demographic input. SC/ST population share measures local composition; it is not the caste of entrepreneurs or a tie intensity. Literacy uses population age seven and above. No comparable universal OBC denominator is present here.

23.10 S10: District Census Handbooks and village directories

Source link: https://censusindia.gov.in/census.website/en/data

Access status: Public tables and handbooks

Unit and coverage: Village town ward district 2001 and 2011

Caste or relationship information: SC ST and amenities

Growth information: Spatial contextual controls

Research use and limitations: Village infrastructure, accessibility, workforce structure and historical settlement context. Match mining coordinates to contemporaneous boundaries and calculate buffer-based measures only when resolution supports them.

23.11 S11: PLFS and earlier NSS employment surveys

Source link: https://microdata.gov.in/NADA/index.php/catalog/204/data-dictionary/F3?file_name=hh_per_rv_2017-18

Access status: Public metadata; register for unit data

Unit and coverage: Person household survey waves

Caste or relationship information: Household social group

Growth information: Labour trends; limited rotating panels

Research use and limitations: Estimate mining workers employment, wages and occupational segmentation using industry codes and survey weights. Workers social group does not identify employer caste. Inspect annual design changes and industry-code vintages before pooling.

23.12 S12: India Human Development Survey

Source link: https://ihds.umd.edu/ihds-1-codebook-topic-index

Access status: Public documentation; data via repository terms

Unit and coverage: Household person 2004-05 and 2011-12

Caste or relationship information: Jati broad group religion social capital

Growth information: Household panel business income

Research use and limitations: A useful source for mechanisms and survey-question development; assess whether mining businesses have sufficient observations. Household nonfarm business income may combine activities and cannot automatically be assigned to a mining firm.

23.13 S13: IHDS social group documentation

Source link: https://ihds.umd.edu/node/191

Access status: Public documentation

Unit and coverage: Household identification variables

Caste or relationship information: Reported jati and category coding

Growth information: No mining outcome on this page

Research use and limitations: Separate jati, broad social group and religion. The documentation offers coding guidance; do not treat its categories as interchangeable with the EC residual Others category.

23.14 S14: IHDS income and social capital questionnaire

Source link: https://ihds.umd.edu/sites/default/files/2022-08/ihds2isq.pdf

Access status: Public questionnaire

Unit and coverage: Household questions

Caste or relationship information: Association membership and network questions

Growth information: Business and income modules

Research use and limitations: Adapt neutral name-generator questions and distinguish family, jati, religious community, business association and political access. It is a questionnaire source, not evidence about mining ties in the current analysis.

23.15 S15: ASUSE and NSS unincorporated enterprise surveys

Source link: https://www.mospi.gov.in/node/current-surveys-0

Access status: Public survey documentation and registered microdata

Unit and coverage: Eligible manufacturing trade services establishments

Caste or relationship information: Some owner categories

Growth information: Cross-sections rather than firm panels

Research use and limitations: Excludes mining section B in the checked coverage. Can study eligible transport, repair, trade and other ancillary businesses with a mining exposure measure. Never label these records a mining operator sample.

23.16 S16: Annual Survey of Industries

Source link: https://nmds.mospi.gov.in/

Access status: Public metadata; registered unit data

Unit and coverage: Registered factory establishment year

Caste or relationship information: No general owner caste field

Growth information: Factory outcomes subject to identifier access

Research use and limitations: Downstream processing and eligible mineral-based manufacturing. Quarrying, mine operations and mineral manufacture have different coverage. Verify the factory frame and NIC before adding any plant.

23.17 S17: Historical mining establishment follow-up survey 1992-93

Source link: https://ecensus.in/india-fifth-economic-census-2005/

Access status: Existence described in census history; archive not acquired

Unit and coverage: Historical mining establishments

Caste or relationship information: Unknown until instrument inspected

Growth information: Potential historical benchmark

Research use and limitations: An obscure lead worth pursuing with MoSPI archives: the EC history identifies a mining and quarrying follow-up survey in 1992-93. No usable microdata, variables or complete report were located in this work; do not overstate availability.

23.18 S18: Indian Bureau of Mines Indian Minerals Yearbook

Source link: https://www.ibm.gov.in/IBMPortal/pages/indian-minerals-yearbook---2023

Access status: Public volumes and commodity chapters

Unit and coverage: Mineral state mine industry annual

Caste or relationship information: No caste ties

Growth information: Production and mine activity context

Research use and limitations: Mineral definitions, production, reserves, sector composition and state reviews. Coverage differs across fuels, atomic and minor minerals. Useful for geology and commodity-market confounders, with a mine-level availability audit.

23.19 S19: IBM monthly mineral production statistics

Source link: https://www.ibm.gov.in/IBMPortal/pages/monthly-statistics-of-mineral-production---2024

Access status: Public monthly reports

Unit and coverage: Mineral state month

Caste or relationship information: None

Growth information: Production cycles

Research use and limitations: Construct mineral-specific demand and output trends. Publication lag and revisions matter. A state-mineral series is contextual and does not reveal a particular firms growth.

23.20 S20: IBM Mining Tenement System returns

Source link: https://mts.ibm.gov.in/RMS/

Access status: Operational login; research access must be requested

Unit and coverage: Mine holder monthly and annual returns

Caste or relationship information: No verified caste variable

Growth information: Potential mine production employment sales panel

Research use and limitations: High-priority administrative request for deidentified mine identifiers and longitudinal quantities. The existence of a filing portal does not mean private returns are open. Request filing status and nonresponse flags with outcomes.

23.21 S21: IBM registration plans notices and ratings

Source link: https://ibm.gov.in/IBMPortal/pages/MINING_TENEMENT_SYSTEM_PROJECT

Access status: Public modules plus restricted sections

Unit and coverage: Mine lease registrant plan inspection

Caste or relationship information: Business links only

Growth information: Approvals compliance operating events

Research use and limitations: Link mine identifiers, operators and approval dates. Ratings are selected among reporting mines and are not exogenous treatments. Access and historical snapshots require verification.

23.22 S22: IBM average sale prices

Source link: https://www.ibm.gov.in/IBMPortal/pages/Average_Sale_Price

Access status: Public month and year series

Unit and coverage: Mineral grade month

Caste or relationship information: None

Growth information: Outcome deflators and price exposure

Research use and limitations: Deflate sales using relevant mineral and grade. Revenue may rise while tonnage falls. These prices are not an automatic valid instrument for caste ties because they directly affect demand and profitability.

23.23 S23: IBM mining lease bulletins and National Mineral Inventory

Source link: https://ibm.gov.in/IBMPortal/pages/DOWNLOADS

Access status: Public publications; detailed spatial access varies

Unit and coverage: Lease mineral state resource inventory

Caste or relationship information: None

Growth information: Lease frame and geological controls

Research use and limitations: Request vintage-consistent lease and resource measures. Geological endowment is crucial to regressions but is not a credible excluded instrument for caste networks because it directly affects mine growth.

23.24 S24: Ministry of Mines annual reports and auction summaries

Source link: https://mines.gov.in/webportal/annualreportflipbook

Access status: Public reports

Unit and coverage: Block state fiscal year

Caste or relationship information: None

Growth information: Policy and auction timing

Research use and limitations: Build a legal and auction timeline then verify the operative instrument for each event. The report alone cannot establish random assignment or a caste mechanism.

23.25 S25: MSTC major mineral block auctions

Source link: https://www.mstcecommerce.com/auctionhome/mlcln/index.jsp

Access status: Public notices; bidding logs restricted

Unit and coverage: Block auction bidder

Caste or relationship information: No caste variable

Growth information: Allocation and subsequent operations

Research use and limitations: Public tender and block summaries can define exposure. Full bids, participation, eligibility failures and unsuccessful bidders are essential for selection and close-auction analysis; availability is unverified.

23.26 S26: MSTC minor mineral auctions

Source link: https://www.mstcecommerce.com/auctionhome/mmb/welcome.jsp

Access status: Public state auction notices; fuller data conditional

Unit and coverage: Sand stone minor mineral block

Caste or relationship information: None

Growth information: Concession events

Research use and limitations: Important for small operators often missed in major-mineral records. State regimes vary. Separate auction award, permit approval, commencement and actual production dates.

23.27 S27: Coal Directory of India

Source link: https://www.coal.nic.in/index.php/major-statistics/coal-statistics

Access status: Public PDF and chapter Excel files

Unit and coverage: Coal lignite mine company state year

Caste or relationship information: None

Growth information: Production dispatch employment series

Research use and limitations: A verified source family with downloadable chapter spreadsheets listed for 2024-25. Relevant for coal operators and procurement ecosystems, but PSU production does not reveal supplier-owner caste.

23.28 S28: Coal monthly statistics

Source link: https://www.coal.nic.in/public-information/statistical-report

Access status: Public reports

Unit and coverage: Coal month company state

Caste or relationship information: None

Growth information: High-frequency operating context

Research use and limitations: Match dispatch and production cycles to supplier surveys and outcomes. Preserve fiscal-calendar mapping and distinguish mining from coal trading.

23.29 S29: Directorate General of Mines Safety

Source link: https://www.dgms.gov.in/UserView/index?mid=1651

Access status: Public annual reports; mine-level request needed

Unit and coverage: Mine employment accidents inspections year

Caste or relationship information: No verified caste of owner

Growth information: Safety and employment context

Research use and limitations: Triangulate mining coverage and incidents. Inspections are often targeted, so comparing inspected and uninspected mines is not a causal evaluation without an assignment design.

23.30 S30: Directorate General of Hydrocarbons

Source link: https://www.dghindia.gov.in/?page=home

Access status: Public outlook; repository terms vary

Unit and coverage: Hydrocarbon block operator basin year

Caste or relationship information: None

Growth information: Exploration and production

Research use and limitations: NIC mining includes hydrocarbons, but they require a separate institutional and data regime. Core small-quarry fieldwork should not be extrapolated to petroleum extraction.

23.31 S31: PARIVESH environmental and forest clearances

Source link: https://parivesh.nic.in/

Access status: Public project records and documents

Unit and coverage: Project proposal clearance event

Caste or relationship information: None

Growth information: Approval dates capacity and location

Research use and limitations: Potential project-to-firm linkage and permit delay outcomes. Approved capacity is not observed production. Delays can reflect project risk and application quality, so approval timing is endogenous.

23.32 S32: State Pollution Control Board consent records

Source link: https://parivesh.nic.in/

Access status: Candidate state-board portals and requests

Unit and coverage: Plant mine consent event

Caste or relationship information: None

Growth information: Entry operation and compliance dates

Research use and limitations: Follow the relevant board from project records and verify access to Consent to Establish and Consent to Operate. No unified national mine panel was acquired. Useful for entity validation and operating-status checks.

23.33 S33: Odisha mining lessee register

Source link: https://www.odishaminerals.gov.in/MIS/LesseeReport/LesseeReport

Access status: Public lessee report verified

Unit and coverage: Lessee lease mine

Caste or relationship information: No social-group field verified

Growth information: Linkable administrative frame

Research use and limitations: Columns include lessee code, area, IBM code, mine code, location and mineral. A strong starting point for fieldwork and linking returns to firms without inferring caste from names.

23.34 S34: Odisha i3MS transaction system

Source link: https://i3ms.odishaminerals.gov.in/i3MSStakeholder.aspx

Access status: Public descriptions and some MIS; transactions conditional

Unit and coverage: Lessee licensee production dispatch transport

Caste or relationship information: None

Growth information: Potential transaction panel

Research use and limitations: Administrative codes distinguish lessees and licensees. Seek deidentified production and dispatch histories and legal access to cross-firm transactions. A transporter, licensee and mine owner are different units.

23.35 S35: Rajasthan mines and geology records

Source link: https://mines.rajasthan.gov.in/

Access status: Candidate public directories and state requests

Unit and coverage: Lease quarry permit operator

Caste or relationship information: Not verified

Growth information: Renewals dispatch production where accessible

Research use and limitations: Priority fieldwork state for stone and other quarrying. Check Rajasthan-specific minor mineral rules and obtain a current operator frame; the source family was identified but no lease panel downloaded.

23.36 S36: Other state mining and geology departments

Source link: https://mines.gov.in/

Access status: Candidate state portals and administrative requests

Unit and coverage: Lease permit concession transaction

Caste or relationship information: Not verified

Growth information: State-dependent longitudinal records

Research use and limitations: Include Gujarat, Karnataka, Maharashtra, Jharkhand, Chhattisgarh, Madhya Pradesh, Tamil Nadu, Telangana, Andhra Pradesh, Kerala and Meghalaya. Inspect each departments source schema rather than presuming identical registries or public access.

23.37 S37: MCA company and LLP master data

Source link: https://www.mcacdm.nic.in/company-master-details

Access status: Public master-data portal verified

Unit and coverage: Legal entity CIN LLPIN

Caste or relationship information: No caste field

Growth information: Incorporation status and registration snapshots

Research use and limitations: Link mine operators to legal firms. Registered office state may differ from mine location. Incorporation does not establish operations; current active status does not prove historical survival.

23.38 S38: MCA company filings and research access

Source link: https://www.mca.gov.in/Ministry/pdf/GuidelinesMCA_final_12022018.pdf

Access status: Research guidelines; filings may need account and payment

Unit and coverage: Company fiscal year

Caste or relationship information: No caste tie measure

Growth information: Sales assets payroll borrowing

Research use and limitations: Build financial growth outcomes with standalone versus consolidated accounting clearly separated. Guidelines describe researcher access conditions; company-specific data access is not assumed here.

23.39 S39: CMIE Prowess

Source link: https://www.cmie.com/kommon/bin/sr.php?kall=wproducts&prd=prowess&tabno=7010&xportal_code=0300100100500000000000000000000000000000000000

Access status: Commercial institutional subscription

Unit and coverage: Company year financial panel

Caste or relationship information: No validated caste field

Growth information: Sales assets profits borrowing

Research use and limitations: Practical dissertation option where the university has access. Inspect coverage of unlisted mining operators, exit treatment, identifier history and business groups. Company panels still require a consented network module.

23.40 S40: NSE corporate annual reports and financial filings

Source link: https://www.nseindia.com/companies-listing/corporate-filings-annual-reports-xbrl

Access status: Public filings

Unit and coverage: Listed company year quarter

Caste or relationship information: No self-reported caste ties

Growth information: Audited financial outcomes

Research use and limitations: Cheap route for listed operators and parent groups. Provides a selected formal sample, not the quarry population. Ownership and director links measure corporate networks, not caste.

23.41 S41: Company investor relations and PSU reports

Source link: https://www.coal.nic.in/

Access status: Candidate firm and PSU websites

Unit and coverage: Operator subsidiary year

Caste or relationship information: No verified caste tie measure

Growth information: Production finance vendor context

Research use and limitations: Collect mining-specific segment outcomes from relevant operators such as coal, metal and state mineral enterprises. The catalogue flags a source family; it does not claim all company reports were inspected.

23.42 S42: Udyam registration and MSME dashboards

Source link: https://dashboard.msme.gov.in/dashboard.aspx/clcss_Cummulative.aspx

Access status: Public aggregates; restricted unit records

Unit and coverage: Registered enterprise plant

Caste or relationship information: Administrative owner social category

Growth information: Possible turnover employment histories by permission

Research use and limitations: A high-priority request for NIC mining by social group by district by year. Registration counts measure formalization as well as entry. Multi-activity enterprises and multiple plants complicate industry attribution.

23.43 S43: Udyam registration instrument

Source link: https://udyamregistration.gov.in/docs/UdyamApplication.pdf

Access status: Public application documentation

Unit and coverage: Registration form

Caste or relationship information: General SC ST OBC and ownership rules

Growth information: No outcomes on the form itself

Research use and limitations: Verified evidence that social category is recorded. Read entity ownership rules before assigning the entrepreneurs category to a multi-owner enterprise; administrative category is not a direct jati or tie measure.

23.44 S44: National SC ST Hub and NSIC

Source link: https://www.nsic.co.in/Schemes/NSSH

Access status: Public scheme information; unit monitoring by request

Unit and coverage: Beneficiary enterprise procurement assistance

Caste or relationship information: Eligibility social category

Growth information: Potential support and procurement outcomes

Research use and limitations: Possible partner for an ethical network-brokerage trial or eligibility-based evaluation. Application, selection and take-up are endogenous unless randomized or a defensible cutoff exists.

23.45 S45: CPPP procurement records

Source link: https://eprocure.gov.in/eprocure/app

Access status: Public notices; awards and bids need audit

Unit and coverage: Tender bidder award contract

Caste or relationship information: Category status only where documented

Growth information: Contract wins values and buyer links

Research use and limitations: Mine and PSU supplier networks, transport, overburden removal and equipment services. Request losing bids as well as awards; award-only data cannot estimate success probabilities.

23.46 S46: GeM and PSU vendor registration

Source link: https://gem.gov.in/

Access status: Candidate public portal; richer unit data conditional

Unit and coverage: Seller bid order buyer

Caste or relationship information: Verified status may be available by agreement

Growth information: Sales diversification procurement access

Research use and limitations: Potential supply-chain outcome source. No GeM mining seller panel was obtained in this work. Avoid treating broad portal aggregates as a dataset of mining contracts.

23.47 S47: RBI Database on Indian Economy

Source link: https://data.rbi.org.in/

Access status: Public macro and banking series

Unit and coverage: State district bank time

Caste or relationship information: No direct business caste ties

Growth information: Credit infrastructure and economic controls

Research use and limitations: Baseline financial access, state income and banking density. Aggregate credit does not reveal a firms loan access or who introduced the borrower.

23.48 S48: GST EPFO ESIC banking and credit records

Source link: https://www.mca.gov.in/

Access status: Restricted administrative sources; agreements required

Unit and coverage: Taxpayer employer borrower period

Caste or relationship information: Generally no validated caste field

Growth information: Sales payroll employment and credit

Research use and limitations: Candidate research partnerships only, not downloaded data. Require authorized linkage and stable identities. No individual tax returns, payroll histories or credit reports were accessed.

23.49 S49: DGCI and S trade data and port records

Source link: https://www.dgcis.gov.in/

Access status: Candidate public aggregates and licensed details

Unit and coverage: Commodity partner month exporter

Caste or relationship information: None

Growth information: Export exposure and prices

Research use and limitations: Demand context for stone and mineral exports. Exporter-level availability and license terms need checking. Commodity HS codes must be crosswalked to NIC and mineral categories.

23.50 S50: GSI Bhukosh and geological publications

Source link: https://ap.data.gov.in/catalog/bhukosh

Access status: Public catalogue; layer access varies

Unit and coverage: Geological spatial layers

Caste or relationship information: None

Growth information: Endowment and resource-quality controls

Research use and limitations: Geology is a central omitted variable in the completed regressions.

Record resolution, survey vintage and uncertainty; do not equate mineral occurrence with commercially viable reserves.

23.51 S51: District Mineral Foundations and PMKKKY records

Source link: https://bahraich.nic.in/district-mineral-foundation-trust-bahraich/

Access status: Public district records; other districts to audit

Unit and coverage: District project beneficiary payment

Caste or relationship information: Sometimes beneficiary categories; verify

Growth information: Local infrastructure and mining-affected development

Research use and limitations: Potential local-context and procurement sources. This verified district page describes project disclosure; no unified national panel was obtained. Spending assignment reflects mining revenues and need.

23.52 S52: NCRB crimes against SC and ST

Source link: https://www.data.gov.in/resource/stateut-wise-crimeatrocities-against-scheduled-castes-2021-2023

Access status: Public crime aggregates

Unit and coverage: State district recorded case year

Caste or relationship information: Victim category not owner network

Growth information: Contextual exclusion and conflict

Research use and limitations: Optional historical context. Recorded crime reflects reporting and enforcement; it is not a direct discrimination measure, and current crime must not be mechanically used as a pre-treatment control.

23.53 S53: Remote sensing imagery and night lights

Source link: https://www.usgs.gov/landsat-missions

Access status: Candidate open satellite sources

Unit and coverage: Pixel mine footprint month year

Caste or relationship information: None

Growth information: Land disturbance and operating proxies

Research use and limitations: Landsat, Sentinel and VIIRS can validate footprint changes or activity. Physical excavation is not firm profitability or employment; clouds, illegal activity and neighbouring plants complicate attribution. No imagery analysis was executed.

23.54 S54: Mining associations cooperatives unions and district industry centres

Source link: https://www.nsic.co.in/

Access status: Candidate institutional lists and interviews

Unit and coverage: Member business worker association

Caste or relationship information: Potential consented membership information

Growth information: Historical entry and relationship histories

Research use and limitations: Local associations can identify overlooked operators and contractors. Membership lists are selected. Do not use an association or chamber as the sole sampling frame, and do not infer individual caste from names.

23.55 S55: Consent-based primary firm and relationship panel

Source link: https://ihds.umd.edu/sites/default/files/2022-08/ihds2isq.pdf

Access status: To be collected through fieldwork

Unit and coverage: Firm owner alter relationship wave

Caste or relationship information: Self-reported identity and explicitly reported ties

Growth information: Employment sales production survival

Research use and limitations: The indispensable source for the actual research question. The URL is a questionnaire reference, not a dataset. Collect baseline, 12-month and 24-month outcomes and dated relationship histories; never fabricate respondents.

23.56 S56: Census 2011 B18: all mining-industry workers

Source link: https://censusindia.gov.in/nada/index.php/catalog/43234

Access status: Downloaded official Excel; executed worker analysis

Unit and coverage: Persons by industry, main/marginal duration, sex, residence, India and 35 historical states/UTs

Caste or relationship information: All workers; join published SC/ST tables by exact codes

Growth information: One historical cross-section, no firm panel

Research use and limitations: Select division 05-09 with group 000 and class 0000 to avoid hierarchical double counting. Counts include workers beyond employees. Official duration definitions: https://data.gov.in/catalog/main-workers-marginal-workers-non-workers-and-those-marginal-workers-non-workers .

23.57 S57: Census 2011 B18SC: SC mining workers

Source link: https://censusindia.gov.in/nada/index.php/catalog/43270

Access status: Downloaded official Excel; executed worker analysis

Unit and coverage: SC persons by NIC division, sex, rural/urban and main/marginal status

Caste or relationship information: Worker own SC identity, no jati or referral tie

Growth information: No linked growth

Research use and limitations: Matched with all/ST tabulations. Preserve zero and absent division rows and workbook row numbers. Does not identify employer group.

23.58 S58: Census 2011 B18ST: ST mining workers

Source link: https://censusindia.gov.in/nada/index.php/catalog/43302

Access status: Downloaded official Excel; executed worker analysis

Unit and coverage: ST persons by NIC division, sex, residence and duration

Caste or relationship information: Worker own ST identity, not a caste-network measure

Growth information: No linked growth

Research use and limitations: Actual worker composition and duration baseline. Remaining = All minus SC minus ST includes OBC. Workforce shares do not measure access probabilities among the ST population.

23.59 S59: Coal Ministry Annual Report 2023-24, welfare chapter 16

Source link: https://www.coal.nic.in/sites/default/files/2024-07/chap16AnnualReport2024en2.pdf

Access status: Downloaded official PDF; counts transcribed and reconciled

Unit and coverage: Coal India employment groups A-D as on 1 January 2024

Caste or relationship information: SC/ST/OBC administrative employee composition

Growth information: Corporate stock only

Research use and limitations: Full workforce composition table PDF page 5. Distinguish reservation quotas and PwBD tables from actual workforce shares; does not cover all Indian miners or all contractors.

23.60 S60: Coal Ministry Annual Report 2025-26, welfare chapter 16

Source link: https://www.coal.nic.in/sites/default/files/2026-02/chap16AnnualReport2026en.pdf

Access status: Downloaded official PDF; employee and selected recruitment tables used

Unit and coverage: Coal India snapshot 1 January 2026; subsidiary advertised and filled vacancies

Caste or relationship information: SC/ST/OBC/EWS in payroll; separate reserved/unreserved vacancy categories

Growth information: Two company stocks, no matched turnover or firm growth causal data

Research use and limitations: Workforce PDF page 4; SECL vacancy example page 7. Applicant denominators missing. UR vacancy category is not General worker caste. A separate SC/ST/OBC table concerns PwBD only and is not used as the full workforce numerator.

23.61 S61: Arisa and CLRA: Caste in stone, November 2024

Source link: https://arisa.nl/wp-content/uploads/Caste-in-stone_November-2024.pdf

Access status: Downloaded and inspected primary producer report; selected aggregate facts retained

Unit and coverage: 97 individual interviews in Budhpura/Makrana; local stone supply chain

Caste or relationship information: Broad groups and published recruitment-route margins; no joint individual records

Growth information: No owner-linked longitudinal growth

Research use and limitations: Composition pages 16/27; Makrana routes page 29. Non-probability local evidence. Ten Budhpura home-workers used a different instrument. Exact composition denominator unresolved; advance percentage discrepancy retained. Full publication is linked, not redistributed under the code licence.

23.62 S62: ILO: Rajasthan natural stone productive employment and decent work, 2021

Source link: https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40ed_dialogue/%40sector/documents/publication/wcms_781885.pdf

Access status: Downloaded and inspected primary ILO report; selected aggregates retained

Unit and coverage: 115 workers and 43 owners in two local clusters; primarily formal value-chain respondents

Caste or relationship information: Published SC/ST/OBC/General worker shares; qualitative job-entry routes

Growth information: No linked individual firm-worker growth panel

Research use and limitations: Methods page 18; profile page 55; work-year averages page 59. Rounded caste shares sum to 99; not normalized. No joint caste-route data acquired. Full publication is linked rather than redistributed.

23.63 S63: PLFS 2023-24 first-visit/revisit layout and instructions

Source link: https://microdata.gov.in/NADA/index.php/catalog/213

Access status: Downloaded layout Excel and instructions PDF; no unit records acquired

Unit and coverage: Household social group and person-level labour survey records

Caste or relationship information: Household sg; no employer caste or validated same-caste referral measure

Growth information: Labour outcomes and limited survey revisit design, not firm growth

Research use and limitations: Verified ind_pas/ind_sas, job_pas/job_sas, occupation, enterprise and earnings fields. Obtain unit records and exact survey design before controlled wage/formality analysis. Principal and subsidiary jobs and first/revisit records require explicit population definitions.

23.64 S64: Labour Bureau Occupational Wage Survey, mining

Source link: https://www.labourbureau.gov.in/occupational-wage-survey

Access status: Official survey description verified; specific seventh-round PDF identified, not downloaded

Unit and coverage: Occupation-level wages/employment in selected mining industries

Caste or relationship information: Caste fields not verified

Growth information: Repeated industry wage context, not worker-firm panel

Research use and limitations: Specific report: https://labourbureau.gov.in/assets/images/pdf/OWS_7th_MINING_Jul_Dec_2016.pdf . Verify sample, occupation definitions and unit-data access; do not assume this gives caste-specific pay.

23.65 S65: Labour Bureau SC/ST, unorganised and women-worker survey families

Source link: https://www.labourbureau.gov.in/sesdsl-1

Access status: Candidate official archive; mining-specific release/schema not acquired

Unit and coverage: Working/living conditions; industry and locality vary

Caste or relationship information: SC/ST-specific survey family; jati/referral fields unknown

Growth information: Mostly cross-sectional, vintage-specific

Research use and limitations: Search archives by coal, stone, quarrying, contractor and mineral; request schedules and mining subsamples. Survey-family availability does not establish a national modern mining-worker dataset.

Chapter 24

Input provenance and verification record

The raw files in the archive are retained with their original institutional authorship. The Ecensus-hosted national PDF is a third-party acquisition route; it is not a new statistical producer. Redistributed public reports and workbook remain subject to their original terms. Code and original report prose have separate licensing.

24.1 Downloaded input hashes

ec6_report.pdf: 15,378,638 bytes. `6ddb2d8b689257e4eaeef0d96b74e724e7ad28768d5db81417b0a3f1a67b2df0`

maharashtra_ec6.pdf: 15,911,003 bytes. `e1d10089c2662f20badfdd7c517287db0598bba25ba24d8c3dfbe760134500da`

pca_2011.xlsx: 1,376,414 bytes. `7a8f70d46c43b5dd30eefe920f251752f69f6af8589594c21ac06499074646f5`

24.2 Additional acquired worker inputs

ilo_rajasthan_stone.pdf: 4,595,859 bytes. https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40ed_dialogue/%40sector/documents/publication/wcms_781885.pdf
`3064f41e22a01421746e491a2f64955fb52c6e7b40a6ae7f9a22b6fa60b774b3`

Primary producer report inspected; selected aggregate facts archived, full publication linked rather than redistributed

plfs_layout_2023_24.xlsx: 56,677 bytes. https://microdata.gov.in/NADA/index.php/catalog/213/download/2463
`3de65c3dcaaadfd90b78579b8eaa65557e32dc7074f178dcc1e11195e8cf3836`

Acquired survey documentation; no individual data analyzed

plfs_instructions_2023_24.pdf: 1,848,924 bytes. https://microdata.gov.in/NADA/index.php/catalog/213/download/5155
`038994ccb43a1d11aa5248dc463cab074984562aa131f749e49a58e06c3841ed`

Acquired survey documentation; no individual data analyzed

coal__welfare__2024.pdf: 210,004 bytes. https://www.coal.nic.in/sites/default/files/2024-07/chap16AnnualReport2024en2.pdf
2d42d95df760e4252967a8356784e2ca
a233881e75d0e6b43fd6e2238a8be1ff

Official acquired worker empirical input; raw file bundled

coal__welfare__2026.pdf: 339,446 bytes. https://www.coal.nic.in/sites/default/files/2026-02/chap16AnnualReport2026en.pdf
91924643bdbcfcce32c075312f014d65
544d60f8d2ce31686280669b98fdcf26

Official acquired worker empirical input; raw file bundled

caste__in__stone__2024.pdf: 1,250,372 bytes. https://www.arisa.nl/wp-content/uploads/Caste-in-stone_November-2024.pdf
e39d44947cae00c4488dd38b65096603
f4f42bdc7a33c7f35d3e10461b777a0a

Primary producer report inspected; selected aggregate facts archived, full publication linked rather than redistributed

census__b18__all__2011.xlsx: 3,583,367 bytes. https://censusindia.gov.in/nada/index.php/catalog/43234/download/46902/DDW-B18-0000-nic-final.xlsx
844666b5e2d45941112271703f44b03c
02fb72adec811efc1e2d1dc6555b2598

Official acquired worker empirical input; raw file bundled

census__b18__sc__2011.xlsx: 2,626,194 bytes. https://censusindia.gov.in/nada/index.php/catalog/43270/download/46938/DDW-B18SC-0000-nic-final.xlsx
06705fc7db8a4223f270a512a352a959
9f7c9008741794b7a72530a1516f0f3b

Official acquired worker empirical input; raw file bundled

census__b18__st__2011.xlsx: 2,507,054 bytes. https://censusindia.gov.in/nada/index.php/catalog/43302/download/46970/DDW-B18ST-0000-nic-final.xlsx
fa9997214e78982ad1723decce68d670
6e834e0f697ceb1b662eae55d5ae57bc

Official acquired worker empirical input; raw file bundled

Worker checks additionally reconcile sexes, residence, group residuals and every historical state sum to India. The four weighted models are reproduced from 1,275 observed nonempty cells. Local-study percentages remain published margins; no recruitment-by-caste joint distribution is invented. The ILO rounded 99-percent total and the conflicting Arisa advance percentage are explicitly retained.

24.3 Meaningful checks performed

Extraction checks verify own-account plus hired-worker establishments equals all establishments, and rural plus urban equals combined totals for every region and column. National mining sums are 85,176 establishments and 557,474 workers; Maharashtra sums are 6,229 and 29,527. Proprietary owner-group sums are 73,170 and 319,425. Join cardinality is enforced, Maharashtra names are compared, residual AP/Telangana counts are pooled, and zero denominators are retained as missing size rather than invented observations. Source hashes, exact coefficients, all figures and archive integrity are checked. Model uncertainty and unavailable causal data remain substantive limitations.

24.4 Reporting conventions

Data vintages include 2011, EC6 2013–14, ILO 2021, Arisa/CLRA 2024 and Coal India snapshots 2024/2026; source entries state the exact dates. Historical series are not automatically comparable. Broad industry code 05 in the EC tables means Mining and quarrying; it is not NIC division 05 alone. Establishment is the observed EC unit; enterprise, company, mine, lease and household are different units. Population shares are geographic context. Others is residual. Every proposed design and hypothetical power parameter is marked as such. Reproducibility is the ability to regenerate the observed-data outputs, not proof that causal identification is valid.